

Solutions Manual for

Heat and Mass Transfer: Fundamentals & Applications

6th Edition in SI Units

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Chapter 10

BOILING AND CONDENSATION

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Boiling Heat Transfer

10-1C The different boiling regimes that occur in a vertical tube during flow boiling are forced convection of liquid, bubbly flow, slug flow, annular flow, transition flow, mist flow, and forced convection of vapor.

10-2C Both boiling and evaporation are liquid-to-vapor phase change processes, but evaporation occurs at the *liquid-vapor interface* when the vapor pressure is less than the saturation pressure of the liquid at a given temperature, and it involves no bubble formation or bubble motion. Boiling, on the other hand, occurs at the *solid-liquid interface* when a liquid is brought into contact with a surface maintained at a temperature T_s sufficiently above the saturation temperature T_{sat} of the liquid.

10-3C Boiling is called *pool boiling* in the absence of bulk fluid flow, and *flow boiling* (or *forced convection boiling*) in the presence of it. In pool boiling, the fluid is stationary, and any motion of the fluid is due to natural convection currents and the motion of the bubbles due to the influence of buoyancy.

10-4C The boiling curve is given in Figure 10-6 in the text. In the *natural convection boiling* regime, the fluid motion is governed by natural convection currents, and heat transfer from the heating surface to the fluid is by natural convection. In the *nucleate boiling* regime, bubbles form at various preferential sites on the heating surface, and rise to the top. In the *transition boiling* regime, part of the surface is covered by a vapor film. In the *film boiling* regime, the heater surface is completely covered by a continuous stable vapor film, and heat transfer is by combined convection and radiation.

10-5C In the *film boiling* regime, the heater surface is completely covered by a continuous stable vapor film, and heat transfer is by combined convection and radiation. In the *nucleate boiling* regime, the heater surface is covered by the liquid. The boiling heat flux in the stable film boiling regime can be higher or lower than that in the nucleate boiling regime, as can be seen from the boiling curve.

10-6C The boiling curve is given in Figure 10-6 in the text. The burnout point in the curve is point C. The *burnout* during boiling is caused by the heater surface being blanketed by a continuous layer of vapor film at increased heat fluxes, and the resulting rise in heater surface temperature in order to maintain the same heat transfer rate across a low-conducting vapor film. Any attempt to increase the heat flux beyond $\dot{q}_{\text{max}}$ will cause the operation point on the boiling curve to jump suddenly from point C to point E. However, the surface temperature that corresponds to point E is beyond the melting point of most heater materials, and burnout occurs. The burnout point is avoided in the design of boilers in order to avoid the disastrous explosions of the boilers.

10-7C Pool boiling heat transfer can be increased *permanently* by increasing the number of nucleation sites on the heater surface by *coating* the surface with a thin layer (much less than 1 mm) of very porous material, or by *forming cavities* on the surface mechanically to facilitate continuous vapor formation. Such surfaces are reported to enhance heat transfer in the nucleate boiling regime by a factor of up to 10, and the critical heat flux by a factor of 3. The use of finned surfaces is also known to enhance nucleate boiling heat transfer and the critical heat flux.

10-8C Yes. Otherwise we can create energy by alternately vaporizing and condensing a substance.

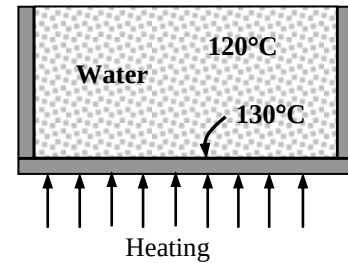
10-9 Water is boiled at $T_{\text{sat}} = 120^\circ\text{C}$ in a mechanically polished stainless steel pressure cooker whose inner surface temperature is maintained at $T_s = 130^\circ\text{C}$. The heat flux on the surface is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the boiler are negligible.

Properties The properties of water at the saturation temperature of 120°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 943.4 \text{ kg/m}^3 & h_{fg} &= 2203 \times 10^3 \text{ J/kg} \\ \rho_v &= 1.121 \text{ kg/m}^3 & \mu_l &= 0.232 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \sigma &= 0.0550 \text{ N/m} & c_{pl} &= 4244 \text{ J/kg} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.44\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.



Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 130 - 120 = 10^\circ\text{C}$ which is relatively low (less than 30°C). Therefore, nucleate boiling will occur. The heat flux in this case can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.232 \times 10^{-3})(2203 \times 10^3) \left[\frac{9.81(943.4 - 1.121)}{0.0550} \right]^{1/2} \left(\frac{4244(130 - 120)}{0.0130(2203 \times 10^3)1.44} \right)^3 \\ &= 228,400 \text{ W/m}^2 = \mathbf{228.4 \text{ kW/m}^2}\end{aligned}$$

10-10 The nucleate pool boiling heat transfer rate per unit length and the rate of evaporation per unit length of water being boiled by a rod that is maintained at 10°C above the saturation temperature are to be determined.

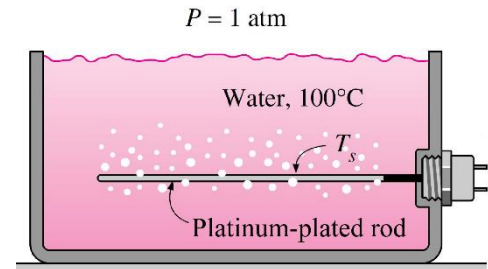
Assumptions 1 Steady operating condition exists. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are $\sigma = 0.0589 \text{ N/m}$ (Table 10-1) and, from Table A-9,

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.5978 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \text{Pr}_l &= 1.75 & c_{pl} &= 4217 \text{ J/kg}\cdot\text{K}\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on platinum surface (Table 10-3).

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 10^\circ\text{C}$, which is less than 30°C for water from Fig. 10-6. Therefore, nucleate boiling will occur. The heat flux in this case can be determined from the Rohsenow relation to be



$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3 \\ &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} \left[\frac{4217(10)}{(0.013)(2257 \times 10^3)(1.75)} \right]^3 \\ &= 1.408 \times 10^5 \text{ W/m}^2\end{aligned}$$

Finally, the nucleate pool boiling heat transfer rate per unit length is

$$\dot{Q}_{\text{boiling}} / L = \pi D \dot{q}_{\text{nucleate}} = \pi(0.010 \text{ m})(1.408 \times 10^5 \text{ W/m}^2) = \mathbf{4420 \text{ W/m}}$$

The rate of evaporation per unit length is

$$\frac{\dot{m}_{\text{evaporation}}}{L} = \frac{(\dot{Q}_{\text{boiling}} / L)}{h_{fg}} = \frac{4420 \text{ J/s}\cdot\text{m}}{2257 \times 10^3 \text{ J/kg}} = \mathbf{1.96 \times 10^{-3} \text{ kg/s}\cdot\text{m}}$$

Discussion The value for the rate of evaporation per unit length indicates that 1 m of the platinum-plated rod would boil water at a rate of about 2 grams per second.

10-11  Water is boiled at 361 kPa by a horizontal ASTM B371 brass rod. The rod is supplied with a heat transfer rate of 20 kW. The supplied heat transfer rate is to be determined whether it would heat the rod surface temperature above the maximum use temperature.

Assumptions 1 Steady state conditions. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 140°C (at 361 kPa) are from Table A-9,

$$c_{pl} = 4286 \text{ J/kg}\cdot\text{K}, \quad h_{fg} = 2145 \text{ kJ/kg}, \quad \text{Pr}_l = 1.24$$

$$\rho_l = 921.7 \text{ kg/m}^3, \quad \rho_v = 1.965 \text{ kg/m}^3, \quad \mu_l = 0.197 \times 10^{-3} \text{ kg/m}\cdot\text{s}$$

The surface tension is from Table 10-1 at 140°C, $\sigma = 0.0509 \text{ N/m}$.

Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a brass surface (Table 10-3).

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 149 - 140 = 9^\circ\text{C}$, which is less than 30°C for water. Therefore, it is in the nucleate boiling regime. The nucleate boiling heat flux is

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3$$

$$\dot{q}_{\text{nucleate}} = (0.197 \times 10^{-3})(2145000) \left[\frac{(9.81)(921.7 - 1.965)}{0.0509} \right]^{1/2} \left[\frac{(4286)(9)}{(0.0060)(2145000)(1.24)^{1.0}} \right]^3$$

$$\dot{q}_{\text{nucleate}} = 2.512 \times 10^6 \text{ W/m}^2$$

The highest heat transfer rate that can be supplied from the ASTM B371 brass rod to boil the water at the excess temperature of $\Delta T = T_s - T_{\text{sat}} = 9^\circ\text{C}$, without heating the rod surface above 149°C is

$$\dot{Q} = A \dot{q}_{\text{total}} = (\pi DL) \dot{q}_{\text{total}} = \pi(0.005 \text{ m})(0.15 \text{ m})(2.512 \times 10^6 \text{ W/m}^2) = 5919 \text{ W} < 20,000 \text{ W}$$

Discussion The highest heat transfer rate that can be supplied to the ASTM B371 rod to boil water is 5919 W. If the ASTM B371 rod is to be supplied with a heat transfer rate of 20 kW, this would require the excess temperature to increase above 9°C . Thus, the rod surface temperature would be higher than the maximum use temperature established by the ASME Code for Process Piping. For a nucleate boiling heat transfer rate of 20 kW, the rod surface temperature would be at 153.5°C .

10-12 Water is boiled at a saturation (or boiling) temperature of $T_{\text{sat}} = 120^\circ\text{C}$ by a brass heating element whose temperature is not to exceed $T_s = 125^\circ\text{C}$. The highest rate of steam production is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling since $\Delta T = T_s - T_{\text{sat}} = 125 - 120 = 5^\circ\text{C}$ which is in the nucleate boiling range of 5 to 30°C for water.

Properties The properties of water at the saturation temperature of 120°C are (Tables 10-1 and A-9)

$$\rho_l = 943.4 \text{ kg/m}^3$$

$$\rho_v = 1.12 \text{ kg/m}^3$$

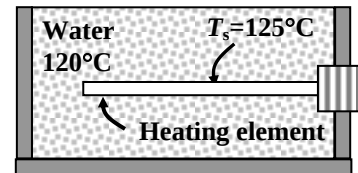
$$\sigma = 0.0550 \text{ N/m}$$

$$\text{Pr}_l = 1.44$$

$$h_{fg} = 2203 \times 10^3 \text{ J/kg}$$

$$\mu_l = 0.232 \times 10^{-3} \text{ kg} \cdot \text{m/s}$$

$$c_{pl} = 4244 \text{ J/kg} \cdot ^\circ\text{C}$$



Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a brass surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis Assuming nucleate boiling, the heat flux in this case can be determined from Rohsenow relation to be

$$\begin{aligned} \dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.232 \times 10^{-3})(2203 \times 10^3) \left[\frac{9.81(943.4 - 1.12)}{0.0550} \right]^{1/2} \left(\frac{4244(125 - 120)}{0.0060(2203 \times 10^3)1.44} \right)^3 \\ &= 290,300 \text{ W/m}^2 \end{aligned}$$

The surface area of the heater is

$$A_s = \pi DL = \pi(0.02 \text{ m})(0.65 \text{ m}) = 0.04084 \text{ m}^2$$

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (0.04084 \text{ m}^2)(290,300 \text{ W/m}^2) = 11,856 \text{ W}$$

(b) The rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{11,856 \text{ J/s}}{2203 \times 10^3 \text{ J/kg}} \left(\frac{3600 \text{ s}}{1 \text{ h}} \right) = \mathbf{19.4 \text{ kg/h}}$$

Therefore, steam can be produced at a rate of about 20 kg/h by this heater.

10-13  Water is boiled at 1254 kPa by a heating element sheathed in an ASTM B75 copper tube. The tube is immersed in the water horizontally. The highest evaporation rate of water that can be achieved by the heater is to be determined.

Assumptions 1 Steady state conditions. 2 Heat losses from the boiler are negligible.

Properties The saturation temperature of water at 1254 kPa is 190°C (Table A-9), and properties of water at the saturation temperature are from Table A-9,

$$c_{pl} = 4460 \text{ J/kg}\cdot\text{K}, \quad h_{fg} = 1979 \text{ kJ/kg}, \quad \text{Pr}_l = 0.947$$

$$\rho_l = 876.4 \text{ kg/m}^3, \quad \rho_v = 6.388 \text{ kg/m}^3, \quad \mu_l = 0.142 \times 10^{-3} \text{ kg/m}\cdot\text{s}$$

The surface tension is from Table 10-1 at 190°C, $\sigma = 0.03995 \text{ N/m}$.

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a polished copper surface (Table 10-3).

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 204 - 190 = 14^\circ\text{C}$, which is less than 30°C for water. Therefore, it is in the nucleate boiling regime. The Rohsenow equation for nucleate boiling heat flux is

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3$$

$$\dot{q}_{\text{nucleate}} = (0.142 \times 10^{-3})(1979000) \left[\frac{(9.81)(876.4 - 6.388)}{0.03995} \right]^{1/2} \left[\frac{(4460)(14)}{(0.0130)(1979000)(0.947)^{1.0}} \right]^3$$

$$\dot{q}_{\text{nucleate}} = 2.186 \times 10^6 \text{ W/m}^2$$

Thus, the highest evaporation rate of water that can be achieved by the heater, without heating the ASTM B75 tube above 204°C , is

$$\dot{m}_{\text{evap}} = \frac{\dot{Q}_{\text{boil}}}{h_{fg}} = \frac{(\pi DL) \dot{q}_{\text{nucleate}}}{h_{fg}} = \frac{\pi(0.005 \text{ m})(0.10 \text{ m})(2.186 \times 10^6 \text{ W/m}^2)}{1979000 \text{ J/kg}} = 0.00174 \text{ kg/s} = 1.74 \text{ g/s}$$

Discussion The highest evaporation rate that can be achieved by the heater, with ASTM B75 copper tube as the sheath, is about 1.74 g/s. If the water evaporation rate is higher than 1.74 g/s, the surface temperature of the ASTM B75 tube would heat above the maximum use temperature established by the ASME Code for Process Piping.

10-14 Water is boiled at 1 atm pressure and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ in a mechanically polished stainless steel pan whose inner surface temperature is maintained at $T_s = 110^\circ\text{C}$. The rate of heat transfer to the water and the rate of evaporation of water are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\rho_l = 957.9 \text{ kg/m}^3$$

$$\rho_v = 0.60 \text{ kg/m}^3$$

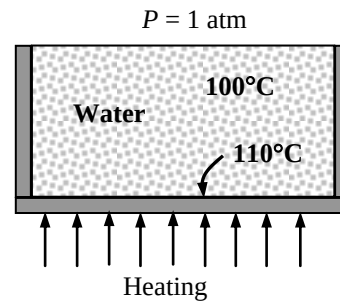
$$\sigma = 0.0589 \text{ N/m}$$

$$\text{Pr}_l = 1.75$$

$$h_{fg} = 2257 \times 10^3 \text{ J/kg}$$

$$\mu_l = 0.282 \times 10^{-3} \text{ kg} \cdot \text{m/s}$$

$$c_{pl} = 4217 \text{ J/kg} \cdot ^\circ\text{C}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 110 - 100 = 10^\circ\text{C}$ which is relatively low (less than 30°C). Therefore, nucleate boiling will occur. The heat flux in this case can be determined from Rohsenow relation to be

$$\begin{aligned} \dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(110 - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3 \\ &= 140,700 \text{ W/m}^2 \end{aligned}$$

The surface area of the bottom of the pan is

$$A_s = \pi D^2 / 4 = \pi (0.30 \text{ m})^2 / 4 = 0.07069 \text{ m}^2$$

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (0.07069 \text{ m}^2)(140,700 \text{ W/m}^2) = \mathbf{9945 \text{ W}}$$

(b) The rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{9945 \text{ J/s}}{2257 \times 10^3 \text{ J/kg}} = \mathbf{0.00441 \text{ kg/s}}$$

That is, water in the pan will boil at a rate of 4.4 grams per second.

10-15  Water is boiled at 3344 kPa by an ASME SA-240 stainless steel plate. The plate surface temperature is to be determined whether it complies with the ASME Boiler and Pressure Vessel Code if it is used to boil the water at an evaporation rate of 4 kg/s.

Assumptions 1 Steady state conditions. 2 Heat losses from the boiler are negligible. 3 Boiling regime is nucleate.

Properties The saturation temperature of water at 3344 kPa is 240°C (Table A-9), and properties of water at the saturation temperature are from Table A-9,

$$c_{pl} = 4760 \text{ J/kg}\cdot\text{K}, \quad h_{fg} = 1767 \text{ kJ/kg}, \quad \text{Pr}_l = 0.836 \\ \rho_l = 813.7 \text{ kg/m}^3, \quad \rho_v = 16.73 \text{ kg/m}^3, \quad \mu_l = 0.111 \times 10^{-3} \text{ kg/m}\cdot\text{s}$$

The surface tension is from Table 10-1 at 240°C, $\sigma = 0.0284 \text{ N/m}$.

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3).

Analysis The evaporation rate of water is

$$\dot{m}_{\text{evap}} = \frac{\dot{Q}_{\text{boil}}}{h_{fg}} = \frac{(\pi D^2/4)\dot{q}_{\text{nucleate}}}{h_{fg}} = 4 \text{ kg/s}$$

Thus,

$$\dot{q}_{\text{nucleate}} = \frac{\dot{m}_{\text{evap}} h_{fg}}{\pi D^2/4} = \frac{(4 \text{ kg/s})(1767000 \text{ J/kg})}{\pi (0.5 \text{ m})^2/4} = 3.5997 \times 10^7 \text{ W/m}^2$$

The Rohsenow equation for nucleate boiling heat flux is

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3 \\ 3.5997 \times 10^7 = (0.111 \times 10^{-3})(1767000) \left[\frac{(9.81)(813.7 - 16.73)}{0.0284} \right]^{1/2} \left[\frac{(4760)(T_s - 240)}{(0.0130)(1767000)(0.836)^{1.0}} \right]^3$$

Solving for the surface temperature yields,

$$T_s = 268.4^\circ\text{C} > 260^\circ\text{C}$$

Note: The nucleate boiling assumption is appropriate. With $T_s = 268.4^\circ\text{C}$, the excess temperature is $\Delta T = T_s - T_{\text{sat}} = 268.4 - 240 = 28.4^\circ\text{C}$, which is less than 30°C for water.

Discussion To boil the water at an evaporation rate of 4 kg/s would require the ASME SA-240 plate surface temperature to be 268.4°C . This is above the maximum use temperature limited by the ASME Boiler and Pressure Vessel Code. To comply with the code, the evaporation rate of water will need to be reduced to 1.39 kg/s or lower.

10-16 A polished copper tube is used to generate 1.5 kg/s of steam at 270 kPa. The surface temperature of the tube, with the interest to minimize the excess temperature, is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The boiling regime is nucleate boiling since ΔT_{excess} is to be minimized (this assumption will be verified).

Properties At 270 kPa, the saturation temperature of water is $T_{\text{sat}} = 130^\circ\text{C}$. The properties of water at $T_{\text{sat}} = 130^\circ\text{C}$ are $\sigma = 0.05295 \text{ N/m}$ (Tables 10-1) and, from Table A-9,

$$\begin{aligned}\rho_l &= 934.6 \text{ kg/m}^3 & h_{fg} &= 2174 \times 10^3 \text{ J/kg} \\ \rho_v &= 1.496 \text{ kg/m}^3 & \mu_l &= 0.213 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \text{Pr}_l &= 1.33 & c_{pl} &= 4263 \text{ J/kg} \cdot \text{K}\end{aligned}$$

Also, $C_{sf} = 0.013$ and $n = 1$ for the boiling of water on a mechanically polished copper surface (Table 10-3).

Analysis The heat flux can be determined from the rate of vaporization to be

$$\dot{q}_{\text{nucleate}} = \frac{\dot{m}_{\text{vapor}} h_{fg}}{A_s} = \frac{\dot{m}_{\text{vapor}} h_{fg}}{\pi DL} = \frac{(1.5 \text{ kg/s})(2174 \times 10^3 \text{ J/kg})}{\pi(0.05 \text{ m})(15 \text{ m})} = 1.384 \times 10^6 \text{ W/m}^2$$

In the interest of minimizing the excess temperature, the boiling regime would be nucleate boiling. The heat flux can be expressed using the Rohsenow relation,

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 1.384 \times 10^6 \text{ W/m}^2 &= (0.213 \times 10^{-3})(2174 \times 10^3) \left[\frac{9.81(934.6 - 1.496)}{0.05295} \right]^{1/2} \left[\frac{4263(T_s - 130)}{0.013(2174 \times 10^3)1.33} \right]^3\end{aligned}$$

Solving for T_s yields

$$T_s = 147^\circ\text{C}$$

Discussion With $T_s = 147^\circ\text{C}$, the excess temperature would be since $\Delta T_{\text{excess}} = T_s - T_{\text{sat}} = 17^\circ\text{C}$, which is in the nucleate boiling range of 5 to 30°C for water. Thus the assumption of nucleate boiling regime is verified.

10-17  Water is boiled at 3344 kPa by a heating element sheathed in an ASTM B725 nickel-copper alloy tube. The tube is immersed in the water horizontally. Determine whether the tube complies with the ASME Code for Process Piping if it is used to boil the water at an evaporation rate of 1 kg/s. If not, the highest evaporation rate of water that can be achieved by the heater is to be determined.

Assumptions 1 Steady state conditions. 2 Heat losses from the boiler are negligible. 3 Boiling regime is nucleate.

Properties The saturation temperature of water at 3344 kPa is 240°C (Table A-9), and properties of water at the saturation temperature are from Table A-9,

$$c_{pl} = 4760 \text{ J/kg}\cdot\text{K}, \quad h_{fg} = 1767 \text{ kJ/kg}, \quad \text{Pr}_l = 0.836$$

$$\rho_l = 813.7 \text{ kg/m}^3, \quad \rho_v = 16.73 \text{ kg/m}^3, \quad \mu_l = 0.111 \times 10^{-3} \text{ kg/m}\cdot\text{s}$$

The surface tension is from Table 10-1 at 240°C, $\sigma = 0.0284 \text{ N/m}$.

Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a nickel surface (Table 10-3). ASTM B725 nickel-copper alloy is majority nickel in material composition.

Analysis The evaporation rate of water is

$$\dot{m}_{\text{evap}} = \frac{\dot{Q}_{\text{boil}}}{h_{fg}} = \frac{(\pi DL)\dot{q}_{\text{nucleate}}}{h_{fg}} = 1 \text{ kg/s}$$

Thus,

$$\dot{q}_{\text{nucleate}} = \frac{\dot{m}_{\text{evap}} h_{fg}}{\pi DL} = \frac{(1 \text{ kg/s})(1767000 \text{ J/kg})}{\pi(0.005 \text{ m})(0.2 \text{ m})} = 5.6245 \times 10^8 \text{ W/m}^2$$

The Rohsenow equation for nucleate boiling heat flux is

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3$$

$$5.6245 \times 10^8 = (0.111 \times 10^{-3})(1767000) \left[\frac{(9.81)(813.7 - 16.73)}{0.0284} \right]^{1/2} \left[\frac{(4760)(T_s - 240)}{(0.0060)(1767000)(0.836)^{1.0}} \right]^3$$

Solving for the surface temperature yields,

$$T_s = 272.8^\circ\text{C} > 260^\circ\text{C}$$

The highest boiling heat flux that can be achieved by the heater, without heating the ASTM B725 tube above 260°C, is

$$\dot{q}_{\text{nucleate}} = (0.111 \times 10^{-3})(1767000) \left[\frac{(9.81)(813.7 - 16.73)}{0.0284} \right]^{1/2} \left[\frac{(4760)(260 - 240)}{(0.0060)(1767000)(0.836)^{1.0}} \right]^3$$

$$\dot{q}_{\text{nucleate}} = 1.2752 \times 10^8 \text{ W/m}^2$$

Note: The nucleate boiling assumption is appropriate. With $T_s = 260^\circ\text{C}$, the excess temperature is $\Delta T = T_s - T_{\text{sat}} = 260 - 240 = 20^\circ\text{C}$, which is less than 30°C for water.

Thus, the highest evaporation rate of water that can be achieved by the heater, without heating the ASTM B725 tube above 260°C, is

$$\dot{m}_{\text{evap}} = \frac{\dot{Q}_{\text{boil}}}{h_{fg}} = \frac{(\pi DL)\dot{q}_{\text{nucleate}}}{h_{fg}} = \frac{\pi(0.005 \text{ m})(0.20 \text{ m})(1.2752 \times 10^8 \text{ W/m}^2)}{1767000 \text{ J/kg}} = 0.2267 \text{ kg/s}$$

Discussion To boil the water at an evaporation rate of 1 kg/s would require the ASTM B725 tube surface temperature to be 272.8°C. This is above the maximum use temperature set by the ASME Code for Process Piping. To comply with the code, the evaporation rate of water will need to be reduced to 0.2267 kg/s or lower.

10-18 The nucleate pool boiling heat transfer coefficient of water being boiled by a horizontal platinum-plated rod is to be determined.

Assumptions 1 Steady operating condition exists. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are $\sigma = 0.0589$ N/m (Table 10-1) and, from Table A-9,

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.5978 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \text{Pr}_l &= 1.75 & c_{pl} &= 4217 \text{ J/kg}\cdot\text{K}\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on platinum surface (Table 10-3).

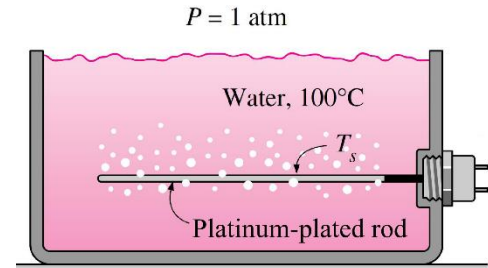
Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 10^\circ\text{C}$, which is less than 30°C for water from Fig. 10-6. Therefore, nucleate boiling will occur. The heat flux in this case can be determined from the Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3 \\ &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} \left[\frac{4217(10)}{(0.013)(2257 \times 10^3)(1.75)} \right]^3 \\ &= 1.408 \times 10^5 \text{ W/m}^2\end{aligned}$$

Using the Newton's law of cooling, the boiling heat transfer coefficient is

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= h(T_s - T_{\text{sat}}) \quad \rightarrow \quad h = \frac{\dot{q}_{\text{nucleate}}}{T_s - T_{\text{sat}}} \\ h &= \frac{1.408 \times 10^5 \text{ W/m}^2}{(110 - 100) \text{ K}} = \mathbf{14,100 \text{ W/m}^2 \cdot \text{K}}\end{aligned}$$

Discussion Heat transfer coefficient on the order of $10^4 \text{ W/m}^2 \cdot \text{K}$ can be obtained in nucleate boiling with a temperature difference of just 10°C .



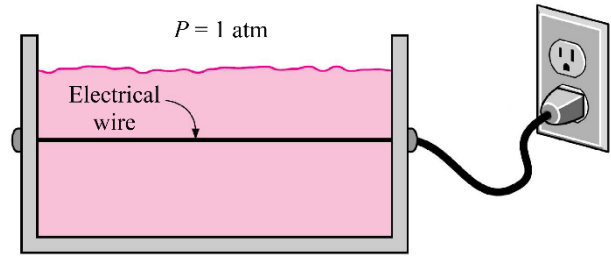
10-19 The nucleate boiling heat transfer coefficient and the value of C_{sf} for water being boiled by a long electrical wire are to be determined.

Assumptions 1 Steady operating condition exists. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are $\sigma = 0.0589 \text{ N/m}$ (Table 10-1) and, from Table A-9,

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.5978 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \text{Pr}_l &= 1.75 & c_{pl} &= 4217 \text{ J/kg}\cdot\text{K}\end{aligned}$$

Also, $n = 1.0$ is given.



Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 28^\circ\text{C}$, which is less than 30°C for water from Fig. 10-6. Therefore, nucleate boiling will occur. The nucleate boiling heat transfer coefficient can be determined using

$$\dot{q}_{\text{boiling}} = h(T_s - T_{\text{sat}}) \quad \rightarrow \quad h = \frac{\dot{q}_{\text{boiling}}}{T_s - T_{\text{sat}}}$$

Also, we know

$$\begin{aligned}\dot{Q}_{\text{boiling}} / L &= \pi D \dot{q}_{\text{boiling}} = 4100 \text{ W/m} \\ \dot{q}_{\text{boiling}} &= \frac{4100 \text{ W/m}}{\pi D} = \frac{4100 \text{ W/m}}{\pi(0.001 \text{ m})} = 1.305 \times 10^6 \text{ W/m}^2\end{aligned}$$

Hence, the nucleate boiling heat transfer coefficient is

$$h = \frac{1.305 \times 10^6 \text{ W/m}^2}{(128 - 100) \text{ K}} = \mathbf{46,600 \text{ W/m}^2 \cdot \text{K}}$$

The value of the experimental constant C_{sf} can be determined from the Rohsenow relation to be

$$\dot{q}_{\text{boiling}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3$$

or

$$\begin{aligned}C_{sf} &= \left\{ \frac{\mu_l h_{fg}}{\dot{q}_{\text{boiling}}} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{h_{fg} \text{Pr}_l^n} \right]^3 \right\}^{1/3} \\ C_{sf} &= \left\{ \frac{(0.282 \times 10^{-3})(2257 \times 10^3)}{1.305 \times 10^6} \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} \left[\frac{4217(128 - 100)}{(2257 \times 10^3)(1.75)} \right]^3 \right\}^{1/3} \\ &= \mathbf{0.0173}\end{aligned}$$

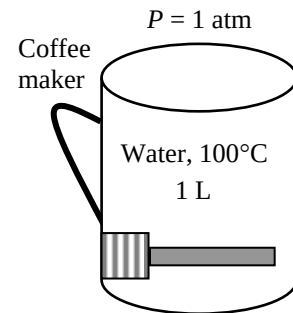
Discussion The boiling heat transfer coefficient of $46,600 \text{ W/m}^2 \cdot \text{K}$ is within the range suggested by Table 1-5 for boiling and condensation (2500 to $100,000 \text{ W/m}^2 \cdot \text{K}$).

10-20 Water is boiled at sea level (1 atm pressure) and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by a stainless steel heating element. The surface temperature of the heating element and its power rating are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the coffee maker are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later).

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 \\ \rho_v &= 0.60 \text{ kg/m}^3 \\ \sigma &= 0.0589 \text{ N/m} \\ \text{Pr}_l &= 1.75 \\ h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \mu_l &= 0.282 \times 10^{-3} \text{ kg} \cdot \text{m/s} \\ c_{pl} &= 4217 \text{ J/kg} \cdot ^\circ\text{C}\end{aligned}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 connection with their definitions in order to avoid unit manipulations.

Analysis The density of water at room temperature is very nearly 1 kg/L , and thus the mass of 1 L water at 18°C is nearly 1 kg . The rate of energy transfer needed to evaporate half of this water in 32 min and the heat flux are

$$\begin{aligned}Q &= \dot{Q}\Delta t = m h_{fg} \rightarrow \dot{Q} = \frac{m h_{fg}}{\Delta t} = \frac{(0.5 \text{ kg})(2257 \text{ kJ/kg})}{(32 \times 60 \text{ s})} = 0.5878 \text{ kW} \\ A_s &= \pi D L = \pi(0.004 \text{ m})(0.30 \text{ m}) = 0.003770 \text{ m}^2 \\ \dot{q} &= \dot{Q} / A_s = (0.5878 \text{ kW}) / (0.003770 \text{ m}^2) = 155.92 \text{ kW/m}^2 = 155,920 \text{ W/m}^2\end{aligned}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given.

Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 155,920 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives

$$T_s = \mathbf{110.3^\circ\text{C}}$$

which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

The specific heat of water at the average temperature of $(14 + 100)/2 = 57^\circ\text{C}$ is $c_p = 4.184 \text{ kJ/kg} \cdot ^\circ\text{C}$. Then the time it takes for the entire water to be heated from 14°C to 100°C is determined to be

$$Q = \dot{Q}\Delta t = mc_p \Delta T \rightarrow \Delta t = \frac{mc_p \Delta T}{\dot{Q}} = \frac{(1 \text{ kg})(4.184 \text{ kJ/kg} \cdot ^\circ\text{C})(100 - 14)^\circ\text{C}}{0.5878 \text{ kJ/s}} = 612 \text{ s} = \mathbf{10.2 \text{ min}}$$

10-21 Water is boiled at sea level (1 atm pressure) and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by a copper heating element. The surface temperature of the heating element and its power rating are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the coffee maker are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later).

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\rho_l = 957.9 \text{ kg/m}^3$$

$$\rho_v = 0.60 \text{ kg/m}^3$$

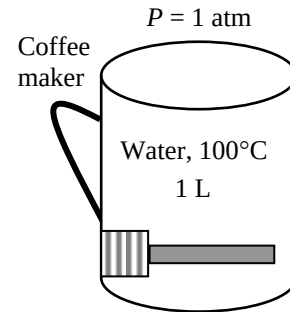
$$\sigma = 0.0589 \text{ N/m}$$

$$\text{Pr}_l = 1.75$$

$$h_{fg} = 2257 \times 10^3 \text{ J/kg}$$

$$\mu_l = 0.282 \times 10^{-3} \text{ kg} \cdot \text{m/s}$$

$$c_{pl} = 4217 \text{ J/kg} \cdot ^\circ\text{C}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a copper surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 connection with their definitions in order to avoid unit manipulations.

Analysis The density of water at room temperature is very nearly 1 kg/L, and thus the mass of 1 L water at 18°C is nearly 1 kg. The rate of energy transfer needed to evaporate half of this water in 32 min and the heat flux are

$$Q = \dot{Q}\Delta t = m h_{fg} \rightarrow \dot{Q} = \frac{m h_{fg}}{\Delta t} = \frac{(0.5 \text{ kg})(2257 \text{ kJ/kg})}{(32 \times 60 \text{ s})} = 0.5878 \text{ kW}$$

$$A_s = \pi D L = \pi (0.004 \text{ m})(0.30 \text{ m}) = 0.003770 \text{ m}^2$$

$$\dot{q} = \dot{Q} / A_s = (0.5878 \text{ kW}) / (0.003770 \text{ m}^2) = 155.92 \text{ kW/m}^2 = 155,920 \text{ W/m}^2$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given.

Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3$$

$$155,920 = (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3$$

It gives

$$T_s = 110.3^\circ\text{C}$$

which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

The specific heat of water at the average temperature of $(14 + 100)/2 = 57^\circ\text{C}$ is $c_p = 4.184 \text{ kJ/kg} \cdot ^\circ\text{C}$. Then the time it takes for the entire water to be heated from 14°C to 100°C is determined to be

$$Q = \dot{Q}\Delta t = mc_p \Delta T \rightarrow \Delta t = \frac{mc_p \Delta T}{\dot{Q}} = \frac{(1 \text{ kg})(4.184 \text{ kJ/kg} \cdot ^\circ\text{C})(100 - 14)^\circ\text{C}}{0.5878 \text{ kJ/s}} = 612 \text{ s} = 10.2 \text{ min}$$

10-22 Water is boiled at sea level (1 atm pressure) and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ in a teflon-pitted stainless steel pan placed on an electric burner. The water level drops by 10 cm in 30 min during boiling. The inner surface temperature of the pan is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the pan are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later).

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\rho_l = 957.9 \text{ kg/m}^3$$

$$\rho_v = 0.60 \text{ kg/m}^3$$

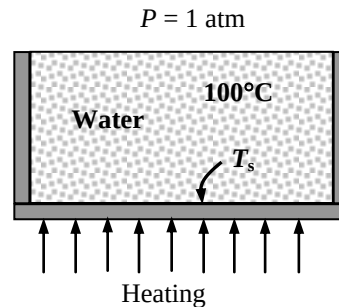
$$\sigma = 0.0589 \text{ N/m}$$

$$\text{Pr}_l = 1.75$$

$$h_{fg} = 2257 \times 10^3 \text{ J/kg}$$

$$\mu_l = 0.282 \times 10^{-3} \text{ kg} \cdot \text{m/s}$$

$$c_{pl} = 4217 \text{ J/kg} \cdot ^\circ\text{C}$$



Also, $C_{sf} = 0.0058$ and $n = 1.0$ for the boiling of water on a teflon-pitted stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 connection with their definitions in order to avoid unit manipulations.

Analysis The rate of heat transfer to the water and the heat flux are

$$\dot{m}_{\text{evap}} = \frac{m_{\text{evap}}}{\Delta t} = \frac{\rho \Delta V}{\Delta t} = \frac{(957.9 \text{ kg/m}^3)(\pi \times (0.2 \text{ m})^2 / 4 \times 0.10 \text{ m})}{15 \times 60 \text{ s}} = 0.003344 \text{ kg/s}$$

$$\dot{Q} = \dot{m}_{\text{evap}} h_{fg} = (0.003344 \text{ kg/s})(2257 \text{ kJ/kg}) = 7.547 \text{ kW}$$

$$A_s = \pi D^2 / 4 = \pi (0.20 \text{ m})^2 / 4 = 0.03142 \text{ m}^2$$

$$\dot{q} = \dot{Q} / A_s = (7547 \text{ W}) / (0.03142 \text{ m}^2) = 240,200 \text{ W/m}^2$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given. Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3$$

$$240,200 = (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.60)}{0.0589} \right]^{1/2} \left\{ \frac{4217(T_s - 100)}{0.0058(2257 \times 10^3)1.75} \right\}^3$$

It gives

$$T_s = 105.3^\circ\text{C}$$

which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

10-23 Water is boiled at sea level (1 atm pressure) and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ in a polished copper pan placed on an electric burner. The water level drops by 10 cm in 30 min during boiling. The inner surface temperature of the pan is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the pan are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later).

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\rho_l = 957.9 \text{ kg/m}^3$$

$$\rho_v = 0.60 \text{ kg/m}^3$$

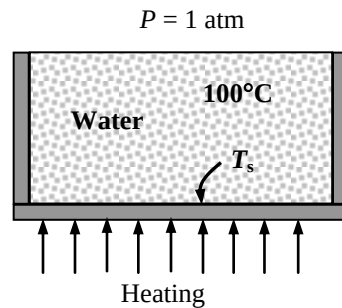
$$\sigma = 0.0589 \text{ N/m}$$

$$\text{Pr}_l = 1.75$$

$$h_{fg} = 2257 \times 10^3 \text{ J/kg}$$

$$\mu_l = 0.282 \times 10^{-3} \text{ kg} \cdot \text{m/s}$$

$$c_{pl} = 4217 \text{ J/kg} \cdot ^\circ\text{C}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a copper surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 connection with their definitions in order to avoid unit manipulations.

Analysis The rate of heat transfer to the water and the heat flux are

$$\dot{m}_{\text{evap}} = \frac{m_{\text{evap}}}{\Delta t} = \frac{\rho \Delta V}{\Delta t} = \frac{(957.9 \text{ kg/m}^3)(\pi \times (0.2 \text{ m})^2 / 4 \times 0.10 \text{ m})}{15 \times 60 \text{ s}} = 0.003344 \text{ kg/s}$$

$$\dot{Q} = \dot{m}_{\text{evap}} h_{fg} = (0.003344 \text{ kg/s})(2257 \text{ kJ/kg}) = 7.547 \text{ kW}$$

$$A_s = \pi D^2 / 4 = \pi (0.20 \text{ m})^2 / 4 = 0.03142 \text{ m}^2$$

$$\dot{q} = \dot{Q} / A_s = (7547 \text{ W}) / (0.03142 \text{ m}^2) = 240,200 \text{ W/m}^2$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given. Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3$$

$$240,200 = (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3$$

It gives

$$T_s = 111.9^\circ\text{C}$$

which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

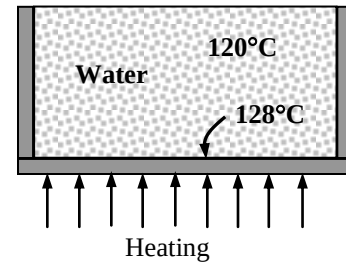
10-24 Water is boiled at $T_{\text{sat}} = 120^\circ\text{C}$ in a mechanically polished stainless steel pressure cooker whose inner surface temperature is maintained at $T_s = 128^\circ\text{C}$. The boiling heat transfer coefficient is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the boiler are negligible.

Properties The properties of water at the saturation temperature of 120°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 943.4 \text{ kg/m}^3 & h_{fg} &= 2203 \times 10^3 \text{ J/kg} \\ \rho_v &= 1.121 \text{ kg/m}^3 & \mu_l &= 0.232 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \sigma &= 0.0550 \text{ N/m} & c_{pl} &= 4244 \text{ J/kg} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.44\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.



Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 128 - 120 = 8^\circ\text{C}$ which is relatively low (less than 30°C). Therefore, nucleate boiling will occur. The heat flux in this case can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.232 \times 10^{-3})(2203 \times 10^3) \left[\frac{9.81(943.4 - 1.121)}{0.0550} \right]^{1/2} \left(\frac{4244(128 - 120)}{0.0130(2203 \times 10^3)1.44} \right)^3 \\ &= 116,900 \text{ W/m}^2\end{aligned}$$

The boiling heat transfer coefficient is

$$\dot{q}_{\text{nucleate}} = h(T_s - T_{\text{sat}}) \longrightarrow h = \frac{\dot{q}_{\text{nucleate}}}{T_s - T_{\text{sat}}} = \frac{116,900 \text{ W/m}^2}{(128 - 120)^\circ\text{C}} = 14,610 \text{ W/m}^2 \cdot ^\circ\text{C} = \mathbf{14.6 \text{ kW/m}^2 \cdot ^\circ\text{C}}$$

10-25 Water is boiled at a temperature of $T_{\text{sat}} = 120^\circ\text{C}$ by a nickel-plated heating element whose surface temperature is maintained at $T_s = 140^\circ\text{C}$. The boiling heat transfer coefficient, the electric power consumed, and the rate of evaporation of water are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling since $\Delta T = T_s - T_{\text{sat}} = 140 - 120 = 20^\circ\text{C}$ which is in the nucleate boiling range of 5 to 30°C for water.

Properties The properties of water at the saturation temperature of 120°C are (Tables 10-1 and A-9)

$$\rho_l = 943.4 \text{ kg/m}^3$$

$$\rho_v = 1.121 \text{ kg/m}^3$$

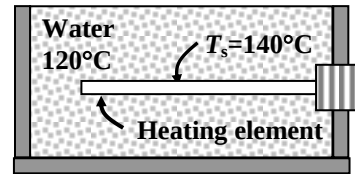
$$\sigma = 0.0550 \text{ N/m}$$

$$\text{Pr}_l = 1.44$$

$$h_{fg} = 2203 \text{ kJ/kg}$$

$$\mu_l = 0.232 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

$$c_{pl} = 4244 \text{ J/kg} \cdot ^\circ\text{C}$$



Also, $g = 9.81 \text{ m/s}^2$ and $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a nickel plated surface (Table 10-3). Note that we expressed the properties in units that will cancel each other in boiling heat transfer relations.

Analysis (a) Assuming nucleate boiling, the heat flux can be determined from Rohsenow relation to be

$$\begin{aligned} \dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.232 \times 10^{-3})(2203 \times 10^3) \left[\frac{9.81(943.4 - 1.12)}{0.0550} \right]^{1/2} \left(\frac{4244(140 - 120)}{0.0060(2203 \times 10^3)1.44} \right)^3 \\ &= 1.858 \times 10^7 \text{ W/m}^2 \end{aligned}$$

Then the convection heat transfer coefficient becomes

$$\dot{q} = h(T_s - T_{\text{sat}}) \rightarrow h = \frac{\dot{q}}{T_s - T_{\text{sat}}} = \frac{1.858 \times 10^7 \text{ W/m}^2}{(140 - 120)^\circ\text{C}} = \mathbf{929,000 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

(b) The electric power consumed is equal to the rate of heat transfer to the water, and is determined from

$$\dot{W}_e = \dot{Q} = \dot{Q}_s = (\pi DL)\dot{q} = (\pi \times 0.007 \text{ m} \times 0.6 \text{ m})(1.858 \times 10^7 \text{ W/m}^2) = 245,157 \text{ W} = \mathbf{245.2 \text{ kW}}$$

(c) Finally, the rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{245.2 \text{ kJ/s}}{2203 \text{ kJ/kg}} = \mathbf{0.111 \text{ kg/s}}$$

10-26 Water is boiled at a temperature of $T_{\text{sat}} = 120^\circ\text{C}$ by a platinum-plated heating element whose surface temperature is maintained at $T_s = 140^\circ\text{C}$. The boiling heat transfer coefficient, the electric power consumed, and the rate of evaporation of water are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling since $\Delta T = T_s - T_{\text{sat}} = 140 - 120 = 20^\circ\text{C}$ which is in the nucleate boiling range of 5 to 30°C for water.

Properties The properties of water at the saturation temperature of 120°C are (Tables 10-1 and A-9)

$$\rho_l = 943.4 \text{ kg/m}^3$$

$$\rho_v = 1.121 \text{ kg/m}^3$$

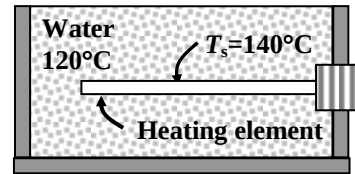
$$\sigma = 0.0550 \text{ N/m}$$

$$\text{Pr}_l = 1.44$$

$$h_{fg} = 2203 \text{ kJ/kg}$$

$$\mu_l = 0.232 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

$$c_{pl} = 4244 \text{ J/kg} \cdot ^\circ\text{C}$$



Also, $g = 9.81 \text{ m/s}^2$ and $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a platinum plated surface (Table 10-3). Note that we expressed the properties in units that will cancel each other in boiling heat transfer relations.

Analysis (a) Assuming nucleate boiling, the heat flux can be determined from Rohsenow relation to be

$$\begin{aligned} \dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.232 \times 10^{-3})(2203 \times 10^3) \left[\frac{9.81(943.4 - 1.12)}{0.0550} \right]^{1/2} \left(\frac{4244(140 - 120)}{0.0130(2203 \times 10^3)1.44} \right)^3 \\ &= 1.827 \times 10^6 \text{ W/m}^2 \end{aligned}$$

Then the convection heat transfer coefficient becomes

$$\dot{q} = h(T_s - T_{\text{sat}}) \rightarrow h = \frac{\dot{q}}{T_s - T_{\text{sat}}} = \frac{1.827 \times 10^6 \text{ W/m}^2}{(140 - 120)^\circ\text{C}} = \mathbf{91,350 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

(b) The electric power consumed is equal to the rate of heat transfer to the water, and is determined from

$$\dot{W}_e = \dot{Q} = \dot{q} A_s = (\pi D L) \dot{q} = (\pi \times 0.007 \text{ m} \times 0.6 \text{ m})(1.827 \times 10^6 \text{ W/m}^2) = 24,107 \text{ W} = \mathbf{24.11 \text{ kW}}$$

(c) Finally, the rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{24.11 \text{ kJ/s}}{2203 \text{ kJ/kg}} = \mathbf{0.0109 \text{ kg/s}}$$

10-27  Prob. 10-26 is reconsidered. The effect of surface temperature of the heating element on the boiling heat transfer coefficient, the electric power, and the rate of evaporation of water is to be investigated.

Analysis The problem is solved using EES, and the solution is given below.

"GIVEN"

T_sat=120 [C]
L=0.6 [m]
D=0.007 [m]
T_s=140 [C]

"PROPERTIES"

Fluid\$='steam_iapws'
P_sat=pressure(Fluid\$, T=T_sat, x=1)
rho_l=density(Fluid\$, T=T_sat, x=0)
rho_v=density(Fluid\$, T=T_sat, x=1)
sigma=SurfaceTension(Fluid\$, T=T_sat)
mu_l=Viscosity(Fluid\$, T=T_sat, x=0)
Pr_l=Prandtl(Fluid\$, T=T_sat, P=P_sat+1[kPa]) "P=P_sat+1 is used to get liquid state"
C_l=CP(Fluid\$, T=T_sat, x=0)
h_f=enthalpy(Fluid\$, T=T_sat, x=0)
h_g=enthalpy(Fluid\$, T=T_sat, x=1)
h_fg=h_g-h_f
C_sf=0.0060 "from Table 10-3 of the text"
n=1 "from Table 10-3 of the text"
g=9.81 [m/s^2] "gravitational acceleration"

"ANALYSIS"

"(a)"

q_dot_nucleate=mu_l*h_fg*(((g*(rho_l-rho_v))/sigma)^0.5)*((C_l*(T_s-T_sat))/(C_sf*h_fg*Pr_l^n))^3
q_dot_nucleate=h*(T_s-T_sat)

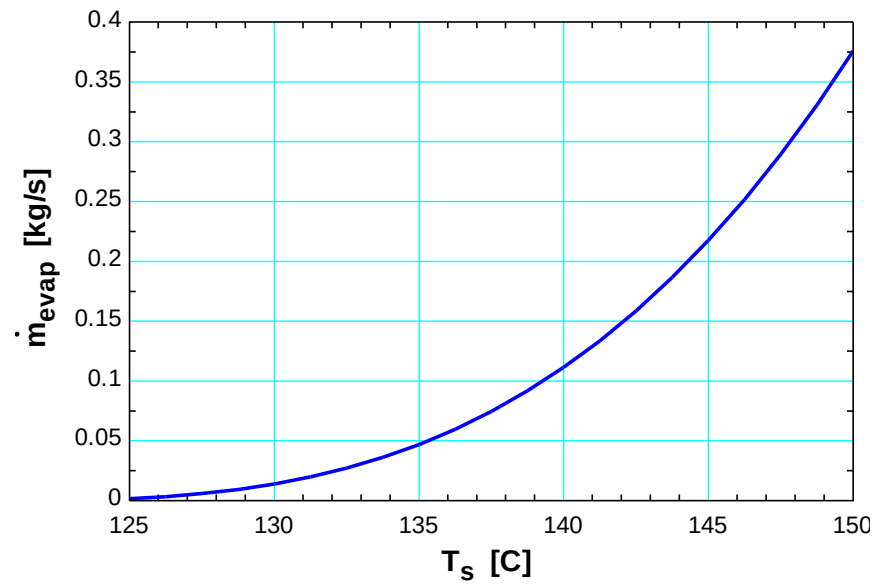
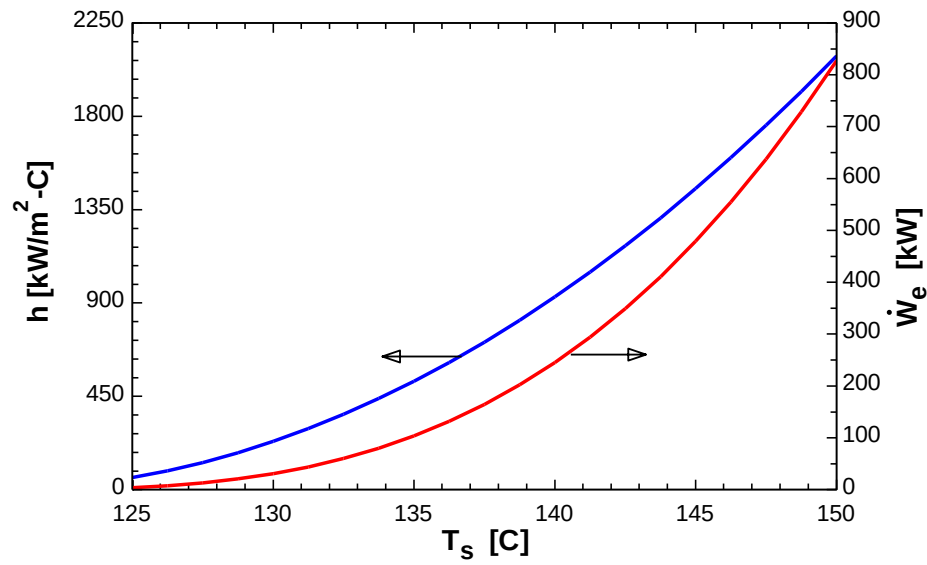
"(b)"

W_dot_e=q_dot_nucleate*A
A=pi*D*L

"(c)"

m_dot_evap=Q_dot_boiling/h_fg
Q_dot_boiling=W_dot_e

T _s [C]	h [kW/m ² -C]	W _e [kW]	m _{evap} [kg/s]
125.00	58.06	3.831	0.001739
126.25	90.72	7.481	0.003397
127.50	130.6	12.93	0.005871
128.75	177.8	20.53	0.009322
130.00	232.2	30.64	0.01392
131.25	293.9	43.63	0.01981
132.50	362.9	59.85	0.02718
133.75	439.1	79.66	0.03618
135.00	522.6	103.4	0.04697
136.25	613.3	131.5	0.05971
137.50	711.3	164.2	0.07458
138.75	816.5	202	0.09173
140.00	929	245.2	0.1113
141.25	1049	294.1	0.1335
142.50	1176	349.1	0.1585
143.75	1310	410.5	0.1864
145.00	1452	478.8	0.2174
146.25	1600	554.3	0.2517
147.50	1756	637.3	0.2894
148.75	1920	728.2	0.3307
150.00	2090	827.4	0.3757



10-28 Hot mechanically polished stainless steel ball bearings are cooled by submerging them in water at 1 atm. The rate of heat removed from a ball bearing at the instant it is submerged in the water is to be determined.

Assumptions **1** Steady operating conditions exist at the instant of submersion. **2** Surface temperature is uniform. **3** The boiling regime is nucleate boiling since $\Delta T_{\text{excess}} = T_s - T_{\text{sat}} = 25^\circ\text{C}$, which is in the nucleate boiling range of 5 to 30°C for water.

Properties At 1 atm, the saturation temperature of water is $T_{\text{sat}} = 100^\circ\text{C}$. The properties of water at $T_{\text{sat}} = 100^\circ\text{C}$ are $\sigma = 0.0589 \text{ N/m}$ (Tables 10-1) and, from Table A-9,

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.5978 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \text{Pr}_l &= 1.75 & c_{pl} &= 4217 \text{ J/kg} \cdot \text{K}\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3).

Analysis The instant a ball bearing is submerged in the water, with $\Delta T_{\text{excess}} = 25^\circ\text{C}$, nucleate boiling would occur. The heat flux can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} \left[\frac{4217(125 - 100)}{0.0130(2257 \times 10^3)1.75} \right]^3 \\ &= 2.1998 \times 10^6 \text{ W/m}^2\end{aligned}$$

The heat transfer surface area is

$$A_s = \pi D^2 = \pi(0.05 \text{ m})^2 = 0.007854 \text{ m}^2$$

The rate of heat removed from a ball bearing at the instant it is submerged in the water is

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (0.007854 \text{ m}^2)(2.1998 \times 10^6 \text{ W/m}^2) = \mathbf{17.3 \text{ kW}}$$

Discussion Note that a 5-cm-diameter stainless steel ball can release 17.3 kW of heat at the instant it is submerged in water. This high value of heat rate removal, even with a temperature difference of only 25°C , is a result from nucleate boiling.

10-29 PtD A long hot mechanically polished stainless steel sheet is being cooled in a water bath. The temperature of the stainless steel sheet leaving the water bath is to be determined whether or not it has the risk of thermal burn hazard.

Assumptions 1 Steady operating conditions exist. 2 Surface temperature is uniform. 3 The boiling regime is nucleate boiling since $\Delta T_{\text{excess}} = T_s - T_{\text{sat}} = 25^\circ\text{C}$, which is in the nucleate boiling range of 5 to 30°C for water.

Properties The specific heat and the density of stainless steel are given as $c_{p,ss} = 450 \text{ J/kg}\cdot\text{K}$ and $\rho_{ss} = 7900 \text{ kg/m}^3$, respectively.

At 1 atm, the saturation temperature of water is $T_{\text{sat}} = 100^\circ\text{C}$. The properties of water at $T_{\text{sat}} = 100^\circ\text{C}$ are $\sigma = 0.0589 \text{ N/m}$ (Tables 10-1) and, from Table A-9,

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.5978 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \text{Pr}_l &= 1.75 & c_{pl} &= 4217 \text{ J/kg}\cdot\text{K}\end{aligned}$$

Also, $C_{sf} = 0.013$ and $n = 1$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3).

Analysis The mass of the stainless steel sheet being conveyed enters and exits the water bath at a rate of

$$\dot{m} = \rho_{ss} V w t$$

The rate of heat that needs to be removed from the sheet so that it leaves the water bath below 45°C is

$$\dot{Q}_{\text{removed}} = \dot{m} c_{p,ss} (T_{\text{in}} - T_{\text{out}})$$

Then,

$$\begin{aligned}\dot{Q}_{\text{removed}} &= \rho_{ss} V w t c_{p,ss} (T_{\text{in}} - T_{\text{out}}) \\ &= (7900 \text{ kg/m}^3)(2 \text{ m/s})(0.5 \text{ m})(0.005 \text{ m})(450 \text{ J/kg}\cdot\text{K})(125 - 45) \text{ K} \\ &= 1.422 \times 10^6 \text{ W}\end{aligned}$$

With $\Delta T_{\text{excess}} = 25^\circ\text{C}$, nucleate boiling would occur in the water bath. The heat flux can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} \left[\frac{4217(125 - 100)}{0.013(2257 \times 10^3)1.75} \right]^3 \\ &= 2.1998 \times 10^6 \text{ W/m}^2\end{aligned}$$

The heat transfer surface area of the sheet submerged in the water bath is

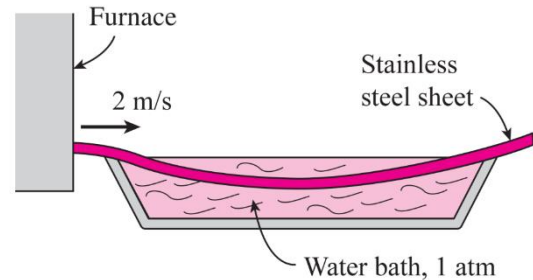
$$A_s = 2(1 \text{ m})(0.5 \text{ m}) + 2(1 \text{ m})(0.005 \text{ m}) = 1.01 \text{ m}^2$$

The rate of heat that could be removed from the sheet in the water bath is

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (1.01 \text{ m}^2)(2.1998 \times 10^6 \text{ W/m}^2) = 2.222 \times 10^6 \text{ W} > \dot{Q}_{\text{removed}} = 1.422 \times 10^6 \text{ W}$$

Discussion The rate of heat that could be removed from the stainless steel sheet in the water bath via nucleate boiling is greater than the heat that needs to be removed from the sheet so that it leaves the water bath below 45°C . Thus, there is no risk of thermal burn on the stainless steel sheet as it leaves the water bath.

Note that this analysis is simplified to steady state conditions, but the actual cooling process is transient.

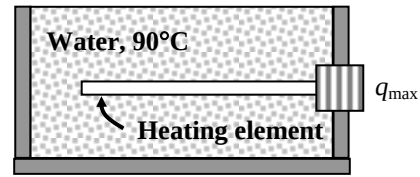


10-30 Water is boiled at the saturation (or boiling) temperature of $T_{\text{sat}} = 90^\circ\text{C}$ by a horizontal brass heating element. The maximum heat flux in the nucleate boiling regime is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 90°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 965.3 \text{ kg/m}^3 & h_{fg} &= 2283 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.4235 \text{ kg/m}^3 & \mu_l &= 0.315 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \sigma &= 0.0608 \text{ N/m} & c_{pl} &= 4206 \text{ J/kg} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.96\end{aligned}$$



Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a brass heating (Table 10-3). Note that we expressed the properties in units specified under Eqs. 10-2 and 10-3 in connection with their definitions in order to avoid unit manipulations.

Analysis For a large horizontal heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned}L^* &= L [g (\rho_l - \rho_v) / \sigma]^2 = (0.008/2) [9.81(965.3 - 0.4235) / (0.0608)]^2 = 1.58 > 1.2 \\ C_{cr} &= 0.12 \text{ (since } L^* = 1.58 > 1.2 \text{ and thus large cylinder).}\end{aligned}$$

The maximum or critical heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.12(2283 \times 10^3) [0.0608 \times 9.81 \times (0.4235)^2 (965.3 - 0.4235)]^{1/4} \\ &= 873,200 \text{ W/m}^2 = \mathbf{873.2 \text{ kW/m}^2}\end{aligned}$$

10-31 The electrical current at which a nickel wire would be in danger of burnout in nucleate boiling is to be determined.

Assumptions 1 Steady operating condition exists. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are $\sigma = 0.0589$ N/m (Table 10-1) and, from Table A-9,

$$\rho_l = 957.9 \text{ kg/m}^3$$

$$h_{fg} = 2257 \times 10^3 \text{ J/kg}$$

$$\rho_v = 0.5978 \text{ kg/m}^3$$

Analysis The danger of burnout occurs when the heat flux is at maximum in nucleate boiling, which can be determined using

$$\dot{q}_{\max} = C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4}$$

Using Table 10-4, the parameter L^* and the constant C_{cr} are determined to be

$$L^* = L \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} = (0.0005) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} = 0.1997$$

which correspond to

$$C_{cr} = 0.12 L^{*-0.25} = 0.12(0.1997)^{-0.25} = 0.1795$$

Hence, the maximum heat flux is

$$\begin{aligned} \dot{q}_{\max} &= (0.1795)(2257 \times 10^3) [(0.0589)(9.81)(0.5978)^2 (957.9 - 0.5978)]^{1/4} \\ &= 1.519 \times 10^6 \text{ W/m}^2 \end{aligned}$$

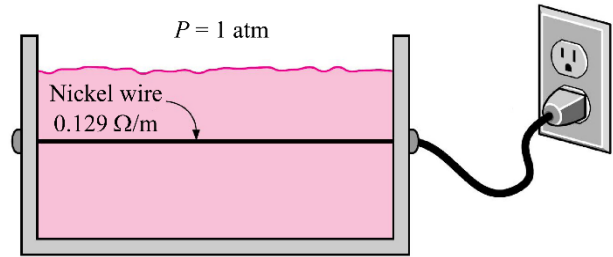
We know that

$$\dot{q} = \frac{I^2 R}{A_s} = \frac{I^2 R}{\pi D L}$$

Thus, the electrical current at which the wire would be in danger of burnout is

$$I = \left[\frac{\dot{q}_{\max} \pi D}{(R/L)} \right]^{1/2} = \left[\frac{(1.519 \times 10^6 \text{ W/m}^2) \pi (0.001 \text{ m})}{0.129 \Omega/\text{m}} \right]^{1/2} = \mathbf{192 \text{ A}}$$

Discussion The electrical current at which burnout could occur will decrease if the resistance of the wire increases.



10-32 Water is boiled at a temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by hot gases flowing through a tube submerged in water. The maximum rate of vaporization is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible.

Properties The properties of water at $T_{\text{sat}} = 100^\circ\text{C}$ are $\sigma = 0.0589 \text{ N/m}$ (Tables 10-1) and, from Table A-9, $\rho_l = 957.9 \text{ kg/m}^3$, $\rho_v = 0.5978 \text{ kg/m}^3$, $h_{fg} = 2257 \times 10^3 \text{ J/kg}$.

Analysis The maximum rate of vaporization occurs at the maximum heat flux.

For a horizontal cylindrical heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$L^* = \frac{D}{2} \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.050/2) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} = 9.98 > 1.2$$

$$C_{cr} = 0.12 \quad (\text{since } L^* > 1.2 \text{ and thus large cylinder})$$

Then the maximum heat flux is determined from

$$\dot{q}_{\text{max}} = C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4}$$

$$= 0.12(2257 \times 10^3) [0.0589 \times 9.81 \times (0.5978)^2 (957.9 - 0.5978)]^{1/4} = 1.0155 \times 10^6 \text{ W/m}^2$$

The heat transfer surface area is

$$A_s = \pi DL = \pi(0.05 \text{ m})(10 \text{ m}) = 1.5708 \text{ m}^2$$

Then, the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{max}} = (1.5708 \text{ m}^2)(1.0155 \times 10^6 \text{ W/m}^2) = 1.5952 \times 10^6 \text{ W}$$

The maximum rate of vaporization of water is determined from

$$\dot{m}_{\text{vapor}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{1.5952 \times 10^6 \text{ J/s}}{2257 \times 10^3 \text{ J/kg}} = \mathbf{0.707 \text{ kg/s}}$$

Discussion The rate of vaporization can be increased by increasing the tube diameter, thereby increasing the heat transfer surface area.

10-33 Water is boiled at a temperature of $T_{\text{sat}} = 160^\circ\text{C}$ by a $3\text{ m} \times 3\text{ m}$ horizontal flat heater heated by hot gases flowing through an array of tubes embedded in it. The maximum rate of vaporization is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater are negligible.

Properties The properties of water at $T_{\text{sat}} = 160^\circ\text{C}$ are $\sigma = 0.0466\text{ N/m}$ (Tables 10-1) and, from Table A-9, $\rho_l = 907.4\text{ kg/m}^3$, $\rho_v = 3.256\text{ kg/m}^3$, $h_{fg} = 2083 \times 10^3\text{ J/kg}$.

Analysis The maximum rate of vaporization occurs at the maximum heat flux.

For a horizontal flat heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$L^* = L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (3) \left[\frac{9.81(907.4 - 3.256)}{0.0466} \right]^{1/2} = 1309 > 27$$

$$C_{cr} = 0.149 \quad (\text{since } L^* > 27 \text{ and thus large flat heater})$$

Then the maximum heat flux is determined from

$$\begin{aligned} \dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.149(2083 \times 10^3) [0.0466 \times 9.81 \times (3.256)^2 (907.4 - 3.256)]^{1/4} \\ &= 2.5252 \times 10^6 \text{ W/m}^2 \end{aligned}$$

The heat transfer surface area is

$$A_s = L \times L = 3\text{ m} \times 3\text{ m} = 9\text{ m}^2$$

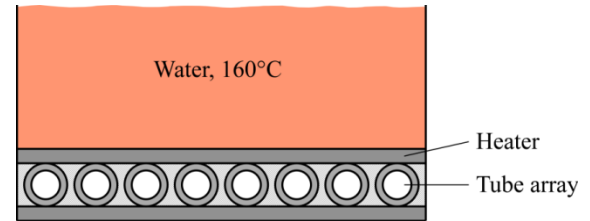
Then, the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{max}} = (9\text{ m}^2)(2.5252 \times 10^6 \text{ W/m}^2) = 2.2727 \times 10^7 \text{ W}$$

The maximum rate of vaporization of water is determined from

$$\dot{m}_{\text{vapor}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{2.2727 \times 10^7 \text{ J/s}}{2083 \times 10^3 \text{ J/kg}} = \mathbf{10.9 \text{ kg/s}}$$

Discussion The rate of vaporization can be increased by increasing the surface area of the plate.



10-34 Water is to be boiled at 1 atm by a spherical heater and a square horizontal flat heater of the same surface area, and which heater geometry would produce higher maximum rate of vaporization is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heaters are negligible.

Properties At 1 atm, the saturation temperature of water is $T_{\text{sat}} = 100^\circ\text{C}$. The properties of water at $T_{\text{sat}} = 100^\circ\text{C}$ are $\sigma = 0.0589 \text{ N/m}$ (Tables 10-1) and, from Table A-9, $\rho_l = 957.9 \text{ kg/m}^3$, $\rho_v = 0.5978 \text{ kg/m}^3$, $h_{fg} = 2257 \times 10^3 \text{ J/kg}$.

Analysis The maximum rate of vaporization occurs at the maximum heat flux.

For a spherical heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$L^* = \frac{D}{2} \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (1/2) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} = 199.7 > 4.26$$

$$C_{cr} = 0.11 \quad (\text{since } L^* > 4.26 \text{ and thus large sphere})$$

Then the maximum heat flux is determined from

$$\dot{q}_{\text{max}} = C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4}$$

$$= 0.11(2257 \times 10^3) [0.0589 \times 9.81 \times (0.5978)^2 (957.9 - 0.5978)]^{1/4} = 9.3092 \times 10^5 \text{ W/m}^2$$

The heat transfer surface area for a sphere is

$$A_s = \pi D^2 = \pi (1 \text{ m})^2 = 3.142 \text{ m}^2$$

The maximum rate of vaporization of water is determined from

$$\dot{m}_{\text{vapor}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{A_s \dot{q}_{\text{max}}}{h_{fg}} = \frac{(3.142 \text{ m}^2)(9.3092 \times 10^5 \text{ W/m}^2)}{2257 \times 10^3 \text{ J/kg}} = \mathbf{1.296 \text{ kg/s}} \quad (\text{spherical heater})$$

The width of a square with the same surface area as the spherical heater is

$$A_s = L^2 = 3.142 \text{ m}^2 \rightarrow L = 1.7726 \text{ m}$$

For a horizontal flat heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$L^* = L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (1.7726) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} = 707.8 > 27$$

$$C_{cr} = 0.149 \quad (\text{since } L^* > 27 \text{ and thus large flat heater})$$

Then the maximum heat flux is determined from

$$\dot{q}_{\text{max}} = C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4}$$

$$= 0.149(2257 \times 10^3) [0.0589 \times 9.81 \times (0.5978)^2 (957.9 - 0.5978)]^{1/4} = 1.261 \times 10^6 \text{ W/m}^2$$

The maximum rate of vaporization of water is determined from

$$\dot{m}_{\text{vapor}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{A_s \dot{q}_{\text{max}}}{h_{fg}} = \frac{(3.142 \text{ m}^2)(1.261 \times 10^6 \text{ W/m}^2)}{2257 \times 10^3 \text{ J/kg}} = \mathbf{1.755 \text{ kg/s}} \quad (\text{square heater})$$

Discussion For the same surface area, the square heater would produce about 35% higher in the maximum rate of vaporization than the spherical heater. This is because, for the same surface area, the square heater has a C_{cr} coefficient that is about 35% higher than that of the spherical heater.

10-35 Water is boiled at a temperature of $T_{\text{sat}} = 180^\circ\text{C}$ by a $3\text{ m} \times 3\text{ m}$ nickel plated flat heater that is heated by hot gases flowing through an array of tubes embedded in it. The surface temperature that produced the maximum rate of steam generation is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater are negligible. 3 The boiling regime is nucleate boiling.

Properties The properties of water at $T_{\text{sat}} = 180^\circ\text{C}$ are $\sigma = 0.0422\text{ N/m}$ (Tables 10-1) and, from Table A-9,

$$\begin{aligned}\rho_l &= 887.3\text{ kg/m}^3 & h_{fg} &= 2015 \times 10^3\text{ J/kg} \\ \rho_v &= 5.153\text{ kg/m}^3 & \mu_l &= 0.150 \times 10^{-3}\text{ kg/m} \cdot \text{s} \\ \text{Pr}_l &= 0.983 & c_{pl} &= 4410\text{ J/kg} \cdot \text{K}\end{aligned}$$

Also, $C_{sf} = 0.0060$ and $n = 1$ for the boiling of water on a nickel surface (Table 10-3).

Analysis The maximum rate of steam generation occurs at the maximum heat flux.

For a horizontal flat heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$L^* = L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (3) \left[\frac{9.81(887.3 - 5.153)}{0.0422} \right]^{1/2} = 1359 > 27$$

$$C_{cr} = 0.149 \quad (\text{since } L^* > 27 \text{ and thus large flat heater})$$

Then the maximum heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.149(2015 \times 10^3) [0.0422 \times 9.81 \times (5.153)^2 (887.3 - 5.153)]^{1/4} = 2.9794 \times 10^6\text{ W/m}^2\end{aligned}$$

The heat flux for nucleate boiling can be expressed using the Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{max}} = \dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 2.9794 \times 10^6\text{ W/m}^2 &= (0.150 \times 10^{-3})(2015 \times 10^3) \left[\frac{9.81(887.3 - 5.153)}{0.0422} \right]^{1/2} \left[\frac{4410(T_s - 180)}{0.0060(2015 \times 10^3)0.983} \right]^3\end{aligned}$$

Solving for T_s yields

$$T_s = 187.5^\circ\text{C}$$

The convection heat transfer coefficient can be determined as

$$\dot{q} = h(T_s - T_{\text{sat}}) \rightarrow h = \frac{2.9794 \times 10^6\text{ W/m}^2}{(187.5 - 180)\text{ K}} = 3.97 \times 10^5\text{ W/m}^2 \cdot \text{K}$$

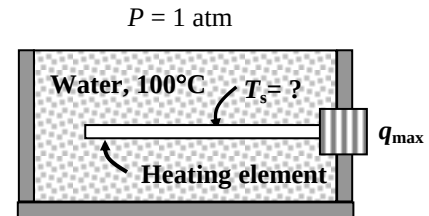
Discussion Note that a heat transfer coefficient of about $400\text{ kW/m}^2 \cdot \text{K}$ can be achieved in nucleate boiling with a temperature difference of less than 10°C .

10-36 Water is boiled at 1 atm pressure and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by a mechanically polished stainless steel heating element. The maximum heat flux in the nucleate boiling regime and the surface temperature of the heater for that case are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg} \cdot \text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & c_{pl} &= 4217 \text{ J/kg} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eqs. 10-2 and 10-3 in connection with their definitions in order to avoid unit manipulations.

Analysis For a large horizontal heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$L^* = L [g(\rho_l - \rho_v) / \sigma]^2 = (0.02/2) [9.81(957.9 - 0.60) / (0.0589)]^2 = 3.99 > 1.2$$

$$C_{cr} = 0.12 \text{ (since } L^* = 3.99 > 1.2 \text{ and thus large cylinder).}$$

The maximum or critical heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.12 (2257 \times 10^3) [0.0589 \times 9.81 \times (0.6)^2 (957.9 - 0.60)]^{1/4} \\ &= \mathbf{1,017,000 \text{ W/m}^2}\end{aligned}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given. Substituting the maximum heat flux into the Rohsenow relation together with other properties gives

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 1,017,000 &= (0.282 \times 10^{-3}) (2257 \times 10^3) \left[\frac{9.81(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives

$$T_s = \mathbf{119.3^\circ\text{C}}$$

Therefore, the temperature of the heater surface will be only 19.3°C above the boiling temperature of water when burnout occurs.

10-37  Prob. 10-36 is reconsidered. The effect of local atmospheric pressure on the maximum heat flux and the temperature difference $T_s - T_{\text{sat}}$ is to be investigated.

Analysis The problem is solved using EES, and the solution is given below.

"GIVEN"

D=0.02 [m]
P_sat=101.3 [kPa]

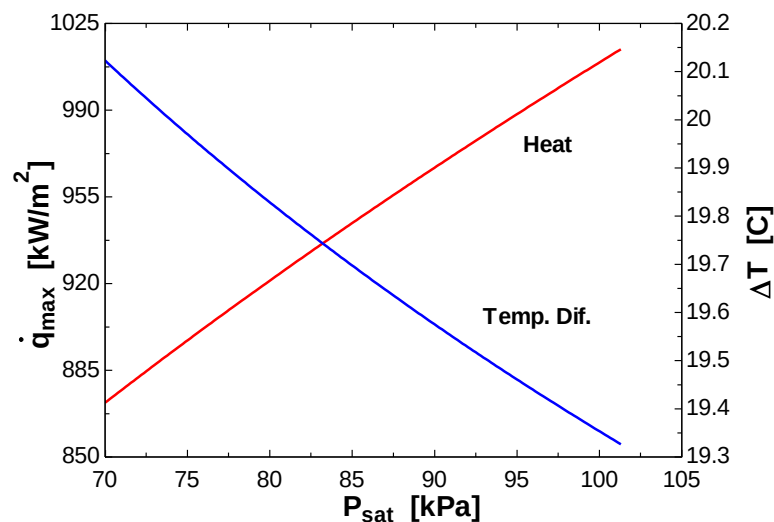
"PROPERTIES"

Fluid\$='steam_IAPWS'
T_sat=temperature(Fluid\$, P=P_sat, x=1)
rho_l=density(Fluid\$, T=T_sat, x=0)
rho_v=density(Fluid\$, T=T_sat, x=1)
sigma=SurfaceTension(Fluid\$, T=T_sat)
mu_l=Viscosity(Fluid\$, T=T_sat, x=0)
Pr_l=Prandtl(Fluid\$, T=T_sat, P=P_sat+1[kPa])
c_l=CP(Fluid\$, T=T_sat, x=0)
h_f=enthalpy(Fluid\$, T=T_sat, x=0)
h_g=enthalpy(Fluid\$, T=T_sat, x=1)
h_fg=h_g-h_f
C_sf=0.0130 "from Table 10-3 of the text"
n=1 "from Table 10-3 of the text"
C_cr=0.12 "from Table 10-4 of the text"
g=9.81 [m/s^2] "gravitational acceleration"

"ANALYSIS"

q_dot_max=C_cr*h_fg*(sigma*g*rho_v^2*(rho_l-rho_v))^0.25
q_dot_nucleate=q_dot_max
q_dot_nucleate=mu_l*h_fg*(((g*(rho_l-rho_v))/sigma)^0.5)*((c_l*(T_s-T_sat))/(C_sf*h_fg*Pr_l^n))^3
DELTA T=T_s-T_sat

P _{sat} [kPa]	$\dot{q}_{\text{max}}$ [kW/m ²]	ΔT [C]
70	871.9	20.12
71.65	880.3	20.07
73.29	888.6	20.02
74.94	896.8	19.97
76.59	904.9	19.92
78.24	912.8	19.88
79.88	920.7	19.83
81.53	928.4	19.79
83.18	936.1	19.74
84.83	943.6	19.7
86.47	951.1	19.66
88.12	958.5	19.62
89.77	965.8	19.58
91.42	973	19.54
93.06	980.1	19.5
94.71	987.2	19.47
96.36	994.1	19.43
98.01	1001	19.4
99.65	1008	19.36
101.3	1015	19.33



10-38 A 10 cm × 10 cm flat heater is used for vaporizing refrigerant-134a at 350 kPa. The surface temperature of the heater is given as 25°C and the heater is subjected to a heat flux of 0.35 MW/m². The coefficient C_{sf} is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater are negligible. 3 The boiling regime is nucleate boiling since $\Delta T_{\text{excess}} = T_s - T_{\text{sat}} = 20^\circ\text{C}$.

Properties At 350 kPa, the saturation temperature of R134a is 5°C (Table A-10). The properties of R134a at $T_{\text{sat}} = 5^\circ\text{C}$ are from Table A-10,

$$\begin{aligned}\rho_l &= 1278 \text{ kg/m}^3 & h_{fg} &= 194.8 \times 10^3 \text{ J/kg} \\ \rho_v &= 17.12 \text{ kg/m}^3 & \mu_l &= 2.589 \times 10^{-4} \text{ kg/m} \cdot \text{s} \\ \text{Pr}_l &= 3.802 & c_{pl} &= 1358 \text{ J/kg} \cdot \text{K} \\ & & \sigma &= 0.01084 \text{ N/m}\end{aligned}$$

Also, $n = 1.7$ for the boiling of R134a is given.

Analysis The heat flux for nucleate boiling can be expressed using the Rohsenow relation to be

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3$$

Using the Rohsenow relation to solve for C_{sf} yields

$$\begin{aligned}C_{sf} &= \frac{c_{p,l}(T_s - T_{\text{sat}})}{h_{fg} \text{Pr}_l^n} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/6} \left(\frac{\mu_l h_{fg}}{\dot{q}_{\text{nucleate}}} \right)^{1/3} \\ &= \frac{1358(25 - 5)}{(194.8 \times 10^3)(3.802^{1.7})} \left[\frac{9.81(1278 - 17.12)}{0.01084} \right]^{1/6} \left(\frac{(2.589 \times 10^{-4})(194.8 \times 10^3)}{0.35 \times 10^6} \right)^{1/3} = \mathbf{0.00772}\end{aligned}$$

For a horizontal flat heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned}L^* &= L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.1) \left[\frac{9.81(1278 - 17.12)}{0.01084} \right]^{1/2} = 106.8 > 27 \\ C_{cr} &= 0.149 \quad (\text{since } L^* > 27 \text{ and thus large flat heater})\end{aligned}$$

The maximum heat flux in the nucleate boiling regime can be determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.149(194.8 \times 10^3) [0.01084 \times 9.81 \times (17.12)^2 (1278 - 17.12)]^{1/4} \\ &= 4.087 \times 10^5 \text{ W/m}^2 > 0.35 \text{ MW/m}^2\end{aligned}$$

Discussion Since $\dot{q}_{\text{max}} > 0.35 \text{ MW/m}^2$, the Rohsenow relation for nucleate boiling is appropriate for this analysis.

10-39 Water is boiled at a temperature of $T_{\text{sat}} = 150^\circ\text{C}$ by hot gases flowing through a mechanically polished stainless steel pipe submerged in water whose outer surface temperature is maintained at $T_s = 160^\circ\text{C}$. The rate of heat transfer to the water, the rate of evaporation, the ratio of critical heat flux to current heat flux, and the pipe surface temperature at critical heat flux conditions are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling since $\Delta T = T_s - T_{\text{sat}} = 160 - 150 = 10^\circ\text{C}$ which is in the nucleate boiling range of 5 to 30°C for water.

Properties The properties of water at the saturation temperature of 150°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 916.6 \text{ kg/m}^3 & h_{fg} &= 2114 \times 10^3 \text{ J/kg} \\ \rho_v &= 2.55 \text{ kg/m}^3 & \mu_l &= 0.183 \times 10^{-3} \text{ kg} \cdot \text{m/s} \\ \sigma &= 0.0488 \text{ N/m} & c_{pl} &= 4311 \text{ J/kg} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.16\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis (a) Assuming nucleate boiling, the heat flux can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.183 \times 10^{-3})(2114 \times 10^3) \left[\frac{9.81(916.6 - 2.55)}{0.0488} \right]^{1/2} \left(\frac{4311(160 - 150)}{0.0130(2114 \times 10^3)1.16} \right)^3 \\ &= 410,090 \text{ W/m}^2\end{aligned}$$

The heat transfer surface area is

$$A_s = \pi DL = \pi(0.05 \text{ m})(25 \text{ m}) = 3.927 \text{ m}^2$$

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (3.927 \text{ m}^2)(410,090 \text{ W/m}^2) = 1,610,400 \text{ W} = \mathbf{1610 \text{ kW}}$$

(b) The rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{1610 \text{ kJ/s}}{2114 \text{ kJ/kg}} = \mathbf{0.762 \text{ kg/s}}$$

(c) For a horizontal cylindrical heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$L^* = L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.025) \left(\frac{9.8(916.6 - 2.55)}{0.0488} \right)^{1/2} = 10.7 > 1.2$$

$$C_{cr} = 0.12 \quad (\text{since } L^* > 1.2 \text{ and thus large cylinder})$$

Then the maximum or critical heat flux is determined from

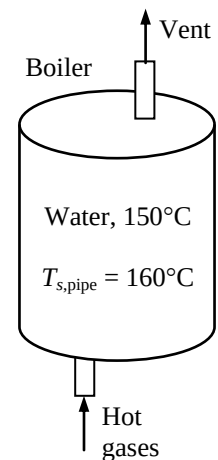
$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.12(2114 \times 10^3) [0.0488 \times 9.81 \times (2.55)^2 (916.6 - 2.55)]^{1/4} = 1,852,000 \text{ W/m}^2\end{aligned}$$

Therefore,

$$\frac{\dot{q}_{\text{max}}}{\dot{q}_{\text{current}}} = \frac{1,852,000}{410,090} = \mathbf{4.52}$$

(d) The surface temperature of the pipe at the burnout point is determined from Rohsenow relation at the critical heat flux value to be

$$\begin{aligned}\dot{q}_{\text{nucleate,cr}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_{s,cr} - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 1,852,000 &= (0.183 \times 10^{-3})(2114 \times 10^3) \left[\frac{9.81(916.6 - 2.55)}{0.0488} \right]^{1/2} \left(\frac{4311(T_{s,cr} - 150)}{0.0130(2114 \times 10^3)1.16} \right)^3 \\ T_{s,cr} &= \mathbf{166.5^\circ\text{C}}\end{aligned}$$



10-40 Water is boiled at a temperature of $T_{\text{sat}} = 155^\circ\text{C}$ by hot gases flowing through a mechanically polished stainless steel pipe submerged in water whose outer surface temperature is maintained at $T_s = 160^\circ\text{C}$. The rate of heat transfer to the water, the rate of evaporation, the ratio of critical heat flux to current heat flux, and the pipe surface temperature at critical heat flux conditions are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling since $\Delta T = T_s - T_{\text{sat}} = 160 - 155 = 5^\circ\text{C}$ which is in the nucleate boiling range of 5 to 30°C for water.

Properties The properties of water at the saturation temperature of 155°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 912.0 \text{ kg/m}^3 & h_{fg} &= 2099 \times 10^3 \text{ J/kg} \\ \rho_v &= 2.901 \text{ kg/m}^3 & \mu_l &= 0.177 \times 10^{-3} \text{ kg} \cdot \text{m/s} \\ \sigma &= 0.0477 \text{ N/m} & c_{pl} &= 4326 \text{ J/kg} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.125\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis (a) Assuming nucleate boiling, the heat flux can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.177 \times 10^{-3})(2099 \times 10^3) \left[\frac{9.81(912.0 - 2.901)}{0.0477} \right]^{1/2} \left(\frac{4326(160 - 155)}{0.0130(2099 \times 10^3)1.125} \right)^3 \\ &= 56,197 \text{ W/m}^2\end{aligned}$$

The heat transfer surface area is

$$A_s = \pi DL = \pi(0.05 \text{ m})(25 \text{ m}) = 3.927 \text{ m}^2$$

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (3.927 \text{ m}^2)(56,197 \text{ W/m}^2) = \mathbf{220,700 \text{ W}}$$

(b) The rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{220.7 \text{ kJ/s}}{2099 \text{ kJ/kg}} = \mathbf{0.105 \text{ kg/s}}$$

(c) For a horizontal cylindrical heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned}L^* &= L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.025) \left(\frac{9.81(912.0 - 2.901)}{0.0477} \right)^{1/2} = 10.8 > 1.2 \\ C_{cr} &= 0.12 \quad (\text{since } L^* > 1.2 \text{ and thus large cylinder})\end{aligned}$$

Then the maximum or critical heat flux is determined from

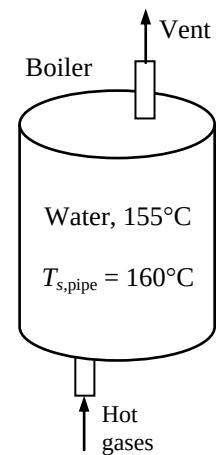
$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.12(2099 \times 10^3) [0.0477 \times 9.81 \times (2.901)^2 (912.0 - 2.901)]^{1/4} = 1.948 \times 10^6 \text{ W/m}^2\end{aligned}$$

Therefore,

$$\frac{\dot{q}_{\text{max}}}{\dot{q}_{\text{current}}} = \frac{1.948 \times 10^6 \text{ W/m}^2}{56,197 \text{ W/m}^2} = \mathbf{34.7}$$

(d) The surface temperature of the pipe at the burnout point is determined from Rohsenow relation at the critical heat flux value to be

$$\begin{aligned}1.948 \times 10^6 &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_{s,cr} - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.177 \times 10^{-3})(2099 \times 10^3) \left[\frac{9.81(912.0 - 2.901)}{0.0477} \right]^{1/2} \left(\frac{4326(T_{s,cr} - 155)}{0.0130(2099 \times 10^3)1.125} \right)^3 \\ T_{s,cr} &= \mathbf{171.3^\circ\text{C}}\end{aligned}$$



10-41 Steam is generated by a $1\text{ m} \times 1\text{ m}$ flat heater boiling water at 1 atm with an excess temperature above 300°C . The range of the steam generation rate that can be achieved by the heater is to be determined.

Assumptions 1 Steady operating conditions exists. 2 Heat losses from the heater are negligible. 3 The boiling regime is film boiling since $\Delta T_{\text{excess}} > 300^\circ\text{C}$, which is much larger than 30°C .

Properties At 1 atm, the saturation temperature of water is $T_{\text{sat}} = 100^\circ\text{C}$. The properties of water at $T_{\text{sat}} = 100^\circ\text{C}$ are $\sigma = 0.0589\text{ N/m}$ (Tables 10-1) and, from Table A-9, $\rho_l = 957.9\text{ kg/m}^3$, $\rho_v = 0.5978\text{ kg/m}^3$, $h_{fg} = 2257 \times 10^3\text{ J/kg}$.

Analysis For a horizontal flat heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$L^* = L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (1) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} = 399.4 > 27$$

$$C_{cr} = 0.149 \quad (\text{since } L^* > 27 \text{ and thus large flat heater})$$

The range of steam generation rate not exceeding the burnout point can be determined from the minimum and maximum boiling heat fluxes.

The minimum rate of vaporization occurs at the minimum heat flux, which can be determined from

$$\dot{q}_{\min} = 0.09 \rho_v h_{fg} \left[\frac{\sigma g (\rho_l - \rho_v)}{(\rho_l + \rho_v)^2} \right]^{1/4}$$

$$= 0.09(0.5978)(2257 \times 10^3) \left[\frac{(0.0589)(9.81)(957.9 - 0.5978)}{(957.9 + 0.5978)^2} \right]^{1/4} = 19021\text{ W/m}^2$$

The maximum rate of vaporization occurs at the maximum heat flux, which can be determined from

$$\dot{q}_{\max} = C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4}$$

$$= 0.149(2257 \times 10^3) [0.0589 \times 9.81 \times (0.5978)^2 (957.9 - 0.5978)]^{1/4} = 1.261 \times 10^6\text{ W/m}^2$$

The heat transfer surface area is

$$A_s = L \times L = 1\text{ m} \times 1\text{ m} = 1\text{ m}^2$$

Then, the rate of heat transfer during boiling is

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}$$

Thus, the range of the steam generation rate in the film boiling regime is

$$\frac{\dot{Q}_{\text{boiling, min}}}{h_{fg}} \leq \dot{m}_{\text{vapor}} \leq \frac{\dot{Q}_{\text{boiling, max}}}{h_{fg}}$$

$$\frac{A_s \dot{q}_{\min}}{h_{fg}} \leq \dot{m}_{\text{vapor}} \leq \frac{A_s \dot{q}_{\max}}{h_{fg}}$$

$$\frac{(1\text{ m}^2)(19021\text{ W/m}^2)}{2257 \times 10^3\text{ J/kg}} \leq \dot{m}_{\text{vapor}} \leq \frac{(1\text{ m}^2)(1.261 \times 10^6\text{ W/m}^2)}{2257 \times 10^3\text{ J/kg}}$$

$$\text{or} \quad \mathbf{0.00843\text{ kg/s} \leq \dot{m}_{\text{vapor}} \leq 0.559\text{ kg/s}}$$

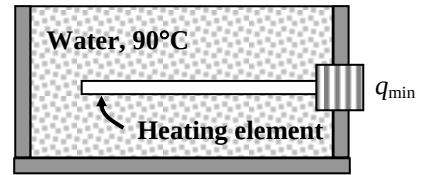
Discussion The maximum rate of steam generation is more than 66 times larger than the minimum rate of steam generation.

10-42 Water is boiled at $T_{\text{sat}} = 90^\circ\text{C}$ in a brass heating element. The surface temperature of the heater is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the boiler are negligible.

Properties The properties of water at the saturation temperature of 90°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 965.3 \text{ kg/m}^3 & h_{fg} &= 2283 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.4235 \text{ kg/m}^3 & \mu_l &= 0.315 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \sigma &= 0.0608 \text{ N/m} & c_{p,l} &= 4206 \text{ J/kg} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.96\end{aligned}$$



Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a brass heating (Table 10-3).

Analysis The minimum heat flux is determined from

$$\begin{aligned}\dot{q}_{\min} &= 0.09 \rho_v h_{fg} \left[\frac{\sigma g (\rho_l - \rho_v)}{(\rho_l + \rho_v)^2} \right]^{1/4} \\ &= 0.09 (0.4235) (2283 \times 10^3) \left[\frac{(0.0608)(9.81)(965.3 - 0.4235)}{(965.3 + 0.4235)^2} \right]^{1/4} = 13,715 \text{ W/m}^2\end{aligned}$$

The surface temperature can be determined from Rohsenow equation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 13,715 \text{ W/m}^2 &= (0.315 \times 10^{-3}) (2283 \times 10^3) \left[\frac{9.81(965.3 - 0.4235)}{0.0608} \right]^{1/2} \left(\frac{4206(T_s - 90)}{0.0060(2283 \times 10^3)1.96} \right)^3 \\ T_s &= \mathbf{92.3^\circ\text{C}}\end{aligned}$$

10-43  Water is boiled at 1 atm by a heating element sheathed in an ASTM B165 tube. The tube is immersed in the water horizontally. The highest evaporation rate of water that can be achieved by the heater is to be determined.

Assumptions 1 Steady state conditions. 2 Heat losses from the boiler are negligible. 3 The water is boiled at 1 atm.

Properties The properties of water at the saturation temperature of 100°C are (Table A-9) $h_{fg} = 2257 \text{ kJ/kg}$ and $\rho_l = 957.9 \text{ kg/m}^3$. The properties of vapor at the film temperature of $T_f = (100 + 260)^\circ\text{C}/2 = 180^\circ\text{C}$ are (Table A-16)

$$c_{pv} = 1924.2 \text{ J/kg}\cdot\text{K}, \quad k_v = 0.0314 \text{ W/m}\cdot\text{K}, \quad \rho_v = 0.48596 \text{ kg/m}^3, \quad \mu_v = 1.5724 \times 10^{-5} \text{ kg/m}\cdot\text{s}$$

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 260 - 100 = 160^\circ\text{C}$, which is larger than 30°C for water. Therefore, it is in the film boiling regime. The film boiling heat flux is

$$\dot{q}_{\text{film}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}})$$

where $C_{\text{film}} = 0.62$ for horizontal cylinders

$$\dot{q}_{\text{film}} = 0.62 \left[\frac{(9.81)(0.0314)^3(0.48596)(957.9 - 0.48596)[2257000 + 0.4(1924.2)(160)]}{(1.5724 \times 10^{-5})(0.005)(160)} \right]^{1/4} \quad (160)$$

$$\dot{q}_{\text{film}} = 40113 \text{ W/m}^2$$

The highest heat transfer rate for the film boiling that can be achieved by the nickel-copper tube, without heating the tube surface above 260°C , is

$$\dot{Q}_{\text{boil}} = A \dot{q}_{\text{film}} = (\pi D L) \dot{q}_{\text{film}} = \pi(0.005 \text{ m})(0.15 \text{ m})(40113 \text{ W/m}^2) = 94.514 \text{ W}$$

Thus, the highest evaporation rate of water that can be achieved by the heater is

$$\dot{m}_{\text{evap}} = \frac{\dot{Q}_{\text{boil}}}{h_{fg}} = \frac{94.514 \text{ J/s}}{2257000 \text{ J/kg}} = 4.188 \times 10^{-5} \text{ kg/s} = 0.04188 \text{ g/s}$$

Discussion The highest evaporation rate that can be achieved by the heater in the film boiling regime, with ASTM B165 nickel-copper alloy tube as the sheath, is about 0.042 g/s. The heat transfer rate to boil water at this evaporation rate is 94.5 W. If the heat transfer rate is higher than 94.5 W, the surface temperature of the ASTM B165 tube would heat above the maximum use temperature established by the ASME Code for Process Piping. Note that in this problem since $T_s < 300^\circ\text{C}$ radiation is negligible.

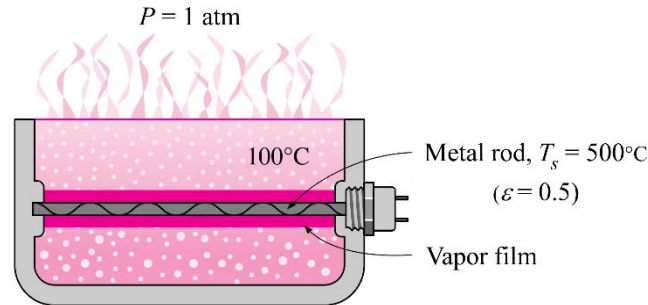
10-44 The power dissipation per unit length of a metal rod submerged horizontally in water, when electric current is passed through it, is to be determined.

Assumptions 1 Steady operating condition exists. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \text{ kJ/kg}$ (Table A-2) and $\rho_l = 957.9 \text{ kg/m}^3$ (Table A-9).

The properties of vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 300^\circ\text{C}$ are, from Table A-16,

$$\begin{aligned}\rho_v &= 0.3831 \text{ kg/m}^3 & c_{pv} &= 1997 \text{ J/kg}\cdot\text{K} \\ \mu_v &= 2.045 \times 10^{-5} \text{ kg/m}\cdot\text{s} & k_v &= 0.04345 \text{ W/m}\cdot\text{K}\end{aligned}$$



Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 400^\circ\text{C}$, which is much larger than 30°C for water from Fig. 10-6. Therefore, film boiling will occur. The film boiling heat flux in this case can be determined from

$$\begin{aligned}\dot{q}_{\text{film}} &= C_{\text{film}} \left[\frac{gk_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4c_{pv}(T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.62 \left[\frac{9.81(0.04345)^3 (0.3831)(957.9 - 0.3831) [2257 \times 10^3 + 0.4(1997)(400)]}{(2.045 \times 10^{-5})(0.002)(400)} \right]^{1/4} (400) \\ &= 1.152 \times 10^5 \text{ W/m}^2\end{aligned}$$

The radiation heat flux is determined from

$$\dot{q}_{\text{rad}} = \epsilon \sigma (T_s^4 - T_{\text{sat}}^4) = (0.5)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(773^4 - 373^4) \text{ K}^4 = 9573 \text{ W/m}^2$$

Then the total heat flux becomes

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 1.152 \times 10^5 \text{ W/m}^2 + \frac{3}{4} (9573 \text{ W/m}^2) = 1.224 \times 10^5 \text{ W/m}^2$$

Finally, the power dissipation per unit length of the metal rod is

$$\dot{Q}_{\text{total}} / L = \pi D \dot{q}_{\text{total}} = \pi (0.002 \text{ m})(1.224 \times 10^5 \text{ W/m}^2) = \mathbf{769 \text{ W/m}}$$

Discussion The contribution of radiation to the total heat flux is about 8%.

10-45  Water is boiled at 1 atm by a horizontal ASTM B335 nickel alloy rod. The highest heat transfer rate that can be supplied from the rod to the water, without heating the rod surface above the maximum use temperature, is to be determined.

Assumptions 1 Steady state conditions. 2 Heat losses from the boiler are negligible. 3 The water is boiled at 1 atm.

Properties The properties of water at the saturation temperature of 100°C are (Table A-9) $h_{fg} = 2257 \text{ kJ/kg}$ and $\rho_l = 957.9 \text{ kg/m}^3$. The properties of vapor at the film temperature of $T_f = (100 + 427)^\circ\text{C}/2 = 263.5^\circ\text{C}$ are (Table A-16)
 $c_{pv} = 1974.4 \text{ J/kg}\cdot\text{K}$, $k_v = 0.03973 \text{ W/m}\cdot\text{K}$, $\rho_v = 0.41263 \text{ kg/m}^3$, $\mu_v = 1.9008 \times 10^{-5} \text{ kg/m}\cdot\text{s}$

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 427 - 100 = 327^\circ\text{C}$, which is much larger than 30°C for water from Fig. 10-6. Therefore, it is in the film boiling regime. The film boiling heat flux is

$$\dot{q}_{\text{film}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}})$$

where $C_{\text{film}} = 0.62$ for horizontal cylinders

$$\dot{q}_{\text{film}} = 0.62 \left[\frac{(9.81)(0.03973)^3(0.41263)(957.9 - 0.41263)[2257000 + 0.4(1974.4)(327)]}{(1.9008 \times 10^{-5})(0.005)(327)} \right]^{1/4} (327) \quad (327)$$

$$\dot{q}_{\text{film}} = 75928 \text{ W/m}^2$$

Since $T_s > 300^\circ\text{C}$ radiation becomes significant. The heat flux from radiation heat transfer is

$$\dot{q}_{\text{rad}} = \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) = (0.3)(5.67 \times 10^{-8} \text{ W/m}^2)(427 + 273\text{K})^4 - (100 + 273\text{K})^4 = 3754.8 \text{ W/m}^2$$

The total heat flux is

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 75928 \text{ W/m}^2 + \frac{3}{4} (3754.8 \text{ W/m}^2) = 78744 \text{ W/m}^2$$

The highest heat transfer rate that can be supplied from the nickel alloy rod to the water, without heating the rod surface above 427°C , is

$$\dot{Q} = A \dot{q}_{\text{total}} = (\pi D L) \dot{q}_{\text{total}} = \pi(0.005 \text{ m})(0.10 \text{ m})(78744 \text{ W/m}^2) = 123.7 \text{ W}$$

Discussion The highest heat transfer rate that can be supplied to the ASTM B335 rod to boil water in the film boiling regime is 123.7 W. If the heat transfer rate is higher than 123.7 W, the surface temperature of the ASTM B335 rod would heat above 427°C , the maximum use temperature established by the ASME Code for Process Piping.

10-46 The initial heat transfer rate from a hot metal sphere that is suddenly submerged in a water bath is to be determined.

Assumptions 1 Steady operating condition exists. 2 The metal sphere has uniform initial surface temperature.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \text{ kJ/kg}$ (Table A-2) and $\rho_l = 957.9 \text{ kg/m}^3$ (Table A-9). The properties of vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 400^\circ\text{C}$ are, from Table A-16,

$$\begin{aligned}\rho_v &= 0.3262 \text{ kg/m}^3 & c_{pv} &= 2066 \text{ J/kg}\cdot\text{K} \\ \mu_v &= 2.446 \times 10^{-5} \text{ kg/m}\cdot\text{s} & k_v &= 0.05467 \text{ W/m}\cdot\text{K}\end{aligned}$$

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 600^\circ\text{C}$, which is much larger than 30°C for water from Fig. 10-6. Therefore, film boiling will occur. The film boiling heat flux in this case can be determined from

$$\begin{aligned}\dot{q}_{\text{film}} &= C_{\text{film}} \left[\frac{gk_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4c_{pv}(T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.67 \left[\frac{9.81(0.05467)^3 (0.3262)(957.9 - 0.3262)[2257 \times 10^3 + 0.4(2066)(600)]}{(2.446 \times 10^{-5})(0.02)(600)} \right]^{1/4} (600) \\ &= 1.052 \times 10^5 \text{ W/m}^2\end{aligned}$$

The radiation heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{rad}} &= \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) \\ &= (0.75)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(973^4 - 373^4) \text{ K}^4 \\ &= 3.729 \times 10^4 \text{ W/m}^2\end{aligned}$$

Then the total heat flux becomes

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 1.052 \times 10^5 \text{ W/m}^2 + \frac{3}{4} (3.729 \times 10^4 \text{ W/m}^2) = 1.332 \times 10^5 \text{ W/m}^2$$

Finally, the initial heat transfer rate from the submerged metal sphere is

$$\dot{Q}_{\text{total}} = \dot{q}_{\text{total}} \pi D^2 = (1.332 \times 10^5 \text{ W/m}^2) \pi (0.02 \text{ m})^2 = \mathbf{167 \text{ W}}$$

Discussion The contribution of radiation to the total heat flux is about 21%, which is significant and cannot be neglected.

10-47 The initial heat transfer rate from a hot steel rod that is suddenly submerged in a water bath is to be determined.

Assumptions 1 Steady operating condition exists. 2 The steel rod has uniform initial surface temperature.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \text{ kJ/kg}$ (Table A-2) and $\rho_l = 957.9 \text{ kg/m}^3$ (Table A-9). The properties of vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 300^\circ\text{C}$ are, from Table A-16,

$$\begin{aligned}\rho_v &= 0.3831 \text{ kg/m}^3 & c_{pv} &= 1997 \text{ J/kg}\cdot\text{K} \\ \mu_v &= 2.045 \times 10^{-5} \text{ kg/m}\cdot\text{s} & k_v &= 0.04345 \text{ W/m}\cdot\text{K}\end{aligned}$$

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 400^\circ\text{C}$, which is much larger than 30°C for water from Fig. 10-6. Therefore, film boiling will occur. The film boiling heat flux in this case can be determined from

$$\begin{aligned}\dot{q}_{\text{film}} &= C_{\text{film}} \left[\frac{gk_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4c_{pv}(T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.62 \left[\frac{9.81(0.04345)^3 (0.3831)(957.9 - 0.3831)[2257 \times 10^3 + 0.4(1997)(400)]}{(2.045 \times 10^{-5})(0.02)(400)} \right]^{1/4} (400) \\ &= 6.476 \times 10^4 \text{ W/m}^2\end{aligned}$$

The radiation heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{rad}} &= \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) \\ &= (0.9)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(773^4 - 373^4) \text{ K}^4 \\ &= 1.723 \times 10^4 \text{ W/m}^2\end{aligned}$$

Then the total heat flux becomes

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 6.476 \times 10^4 \text{ W/m}^2 + \frac{3}{4} (1.723 \times 10^4 \text{ W/m}^2) = 7.768 \times 10^4 \text{ W/m}^2$$

Finally, the initial heat transfer rate from the submerged steel rod is

$$\dot{Q}_{\text{total}} = \dot{q}_{\text{total}} \pi DL = (7.768 \times 10^4 \text{ W/m}^2) \pi (0.02 \text{ m})(0.2 \text{ m}) = \mathbf{976 \text{ W}}$$

Discussion The contribution of radiation to the total heat flux is about 17%, which is significant and cannot be neglected.

10-48 Water is boiled at $T_{\text{sat}} = 100^\circ\text{C}$ by a spherical platinum heating element immersed in water. The surface temperature is $T_s = 350^\circ\text{C}$. The rate of heat transfer is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Table A-9)

$$h_{fg} = 2257 \times 10^3 \text{ J/kg}$$

$$\rho_l = 957.9 \text{ kg/m}^3$$

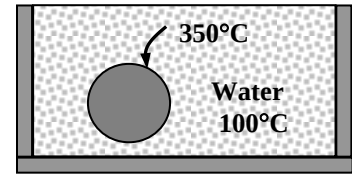
The properties of water vapor at $(350+100)/2 = 225^\circ\text{C}$ are (Table A-16)

$$\rho_v = 0.444 \text{ kg/m}^3$$

$$\mu_v = 1.749 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$c_{pv} = 1951 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_v = 0.03581 \text{ W/m} \cdot ^\circ\text{C}$$



Analysis The film boiling occurs since the temperature difference between the surface and the fluid. The heat flux in this case can be determined from

$$\begin{aligned} \dot{q}_{\text{film}} &= 0.67 \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.67 \left[\frac{(9.81)(0.03581)^3 (0.444)(957.9 - 0.444) [2257 \times 10^3 + 0.4(1951)(350 - 100)]}{(1.749 \times 10^{-5})(0.15)(350 - 100)} \right]^{1/4} (350 - 100) \\ &= 25,207 \text{ W/m}^2 \end{aligned}$$

The radiation heat transfer is

$$\dot{q}_{\text{rad}} = \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) = (0.10)(5.67 \times 10^{-8}) [(350 + 273)^4 - (100 + 273)^4] = 745 \text{ W/m}^2$$

The total heat flux is

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 25,207 + \frac{3}{4} (745) = 25,766 \text{ W/m}^2$$

Then the total rate of heat transfer becomes

$$\dot{Q}_{\text{total}} = A \dot{q}_{\text{total}} = \pi (0.15)^2 (25,766 \text{ W/m}^2) = \mathbf{1821 \text{ W}}$$

10-49 Cylindrical stainless steel rods are heated to 700°C and then suddenly quenched in water at 1 atm. The convection heat transfer coefficient and the total rate of heat removed from a rod at the instant it is submerged in the water are to be determined.

Assumptions 1 Steady operating conditions exist at the instant of submersion. 2 Surface temperature is uniform. 3 The boiling regime is film boiling since $\Delta T_{\text{excess}} = T_s - T_{\text{sat}} = 600^\circ\text{C}$, which is much larger than 30°C .

Properties At 1 atm, the saturation temperature of water is $T_{\text{sat}} = 100^\circ\text{C}$. The properties of water at $T_{\text{sat}} = 100^\circ\text{C}$ are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_l = 957.9 \text{ kg/m}^3$ (Table A-9). The properties of vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (100 + 700)/2 = 400^\circ\text{C}$ are, from Table A-16,

$$\begin{aligned}\rho_v &= 0.3262 \text{ kg/m}^3 & c_{pv} &= 2066 \text{ J/kg} \cdot \text{K} \\ \mu_v &= 2.446 \times 10^{-5} \text{ kg/m} \cdot \text{s} & k_v &= 0.05467 \text{ W/m} \cdot \text{K}\end{aligned}$$

Analysis The film boiling heat flux can be determined from

$$\dot{q}_{\text{film}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}})$$

where $C_{\text{film}} = 0.62$ (horizontal cylinders)

$$\begin{aligned}&= 0.62 \left[\frac{9.81 (0.05467)^3 (0.3262) (957.9 - 0.3262) [2257 \times 10^3 + 0.4 (2066) (700 - 100)]}{(2.446 \times 10^{-5}) (0.025) (700 - 100)} \right]^{1/4} (700 - 100) \\ &= 92097 \text{ W/m}^2\end{aligned}$$

Thus, the convection heat transfer coefficient is

$$\dot{q}_{\text{film}} = h(T_s - T_{\text{sat}}) \rightarrow h = \frac{92097 \text{ W/m}^2}{(700 - 100) \text{ K}} = 153.5 \text{ W/m}^2 \cdot \text{K}$$

The radiation heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{rad}} &= \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) \\ &= (0.3) (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) (973^4 - 373^4) \text{ K}^4 \\ &= 14917 \text{ W/m}^2\end{aligned}$$

Then the total heat flux becomes

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 92097 \text{ W/m}^2 + \frac{3}{4} (14917 \text{ W/m}^2) = 1.0328 \times 10^5 \text{ W/m}^2$$

The total rate of heat removed from a rod at the instant it is submerged in the water is

$$\dot{Q}_{\text{total}} = (\pi D L) \dot{q}_{\text{total}} = \pi (0.025 \text{ m}) (0.25 \text{ m}) (1.0328 \times 10^5 \text{ W/m}^2) = 2028 \text{ W}$$

Discussion Convection heat transfer coefficient in film boiling is generally lower than that of nucleate boiling, because the excess temperature of film boiling is much larger than that of nucleate boiling.

10-50  A long cylindrical stainless steel rod with mechanically polished surface is being quenched in a water bath. The temperature of the rod leaving the water bath is to be determined whether or not it has the risk of thermal burn hazard.

Assumptions 1 Steady operating conditions exist. 2 Surface temperature is uniform. 3 The boiling regime is film boiling since $\Delta T_{\text{excess}} = T_s - T_{\text{sat}} = 700^\circ\text{C} - 100^\circ\text{C} = 600^\circ\text{C}$, which is much larger than 30°C .

Properties The specific heat and the density of stainless steel are given as $c_{p,ss} = 450 \text{ J/kg}\cdot\text{K}$ and $\rho_{ss} = 7900 \text{ kg/m}^3$, respectively.

At 1 atm, the saturation temperature of water is $T_{\text{sat}} = 100^\circ\text{C}$. The properties of water at $T_{\text{sat}} = 100^\circ\text{C}$ are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_l = 957.9 \text{ kg/m}^3$ (Table A-9). The properties of vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (100 + 700) = 400^\circ\text{C}$ are, from Table A-16,

$$\begin{aligned}\rho_v &= 0.3262 \text{ kg/m}^3 & c_{pv} &= 2066 \text{ J/kg}\cdot\text{K} \\ \mu_v &= 2.446 \times 10^{-5} \text{ kg/m}\cdot\text{s} & k_v &= 0.05467 \text{ W/m}\cdot\text{K}\end{aligned}$$

Analysis With $\Delta T_{\text{excess}} = 600^\circ\text{C}$, film boiling would occur in the water bath. The heat flux can be determined from

$$\dot{q}_{\text{film}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}})$$

where $C_{\text{film}} = 0.62$ (horizontal cylinders)

$$\begin{aligned}&= 0.62 \left[\frac{9.81 (0.05467)^3 (0.3262) (957.9 - 0.3262) [2257 \times 10^3 + 0.4 (2066) (700 - 100)]}{(2.446 \times 10^{-5}) (0.025) (700 - 100)} \right]^{1/4} (700 - 100) \\ &= 92097 \text{ W/m}^2\end{aligned}$$

The radiation heat flux is determined from

$$\dot{q}_{\text{rad}} = \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) = (0.3) (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) (973^4 - 373^4) \text{ K}^4 = 14917 \text{ W/m}^2$$

Then the total heat flux becomes

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 92097 \text{ W/m}^2 + \frac{3}{4} (14917 \text{ W/m}^2) = 1.0328 \times 10^5 \text{ W/m}^2$$

The rate of heat that could be removed from the rod in the water bath is

$$\dot{Q}_{\text{total}} = A_s \dot{q}_{\text{total}} = (\pi D L) \dot{q}_{\text{total}} = \pi (0.025 \text{ m}) (3 \text{ m}) (1.0328 \times 10^5 \text{ W/m}^2) = 24335 \text{ W}$$

The mass of the stainless steel rod being conveyed enters and exits the water bath at a rate of

$$\dot{m} = \rho_{ss} V (\pi D^2 / 4)$$

The rate of heat that needs to be removed from the rod so that it leaves the water bath below 45°C can be determined using

$$\dot{Q}_{\text{total}} = \dot{m} c_{p,ss} (T_{\text{in}} - T_{\text{out}}) = \rho_{ss} V (\pi D^2 / 4) c_{p,ss} (T_{\text{in}} - T_{\text{out}})$$

Thus, the speed of the rod conveying through the water bath is

$$\begin{aligned}V &= \frac{\dot{Q}_{\text{total}}}{\rho_{ss} (\pi D^2 / 4) c_{p,ss} (T_{\text{in}} - T_{\text{out}})} \\ &= \frac{24335 \text{ W}}{(7900 \text{ kg/m}^3) [\pi (0.025 \text{ m})^2 / 4] (450 \text{ J/kg}\cdot\text{K}) (700 - 45) \text{ K}} = 0.0213 \text{ m/s} = \mathbf{76.7 \text{ m/hr}}\end{aligned}$$

Discussion To ensure that the stainless steel rod leaves the water bath below 45°C , in order to prevent thermal burn hazard, the speed of the rod conveying through the water bath should be about 77 m/hr or slower.

Note that this analysis is simplified to steady state conditions, but the actual quenching process is transient.

10-51 A cylindrical heater is used for boiling water at 1 atm. The film boiling convection heat transfer coefficient at the burnout point is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater are negligible. 3 The boiling regime is film boiling.

Properties At 1 atm, the saturation temperature of water is $T_{\text{sat}} = 100^\circ\text{C}$. The properties of water at $T_{\text{sat}} = 100^\circ\text{C}$ are $\sigma = 0.0589 \text{ N/m}$ (Tables 10-1) and, from Table A-9, $\rho_l = 957.9 \text{ kg/m}^3$, $\rho_{v,\text{sat}} = 0.5978 \text{ kg/m}^3$, $h_{fg} = 2257 \times 10^3 \text{ J/kg}$.

The properties of vapor at the film temperature of $T_f = 1150^\circ\text{C}$ are, from Table A-16,

$$\begin{aligned}\rho_v &= 0.1543 \text{ kg/m}^3 & c_{pv} &= 2571 \text{ J/kg} \cdot \text{K} \\ \mu_v &= 5.283 \times 10^{-5} \text{ kg/m} \cdot \text{s} & k_v &= 0.1588 \text{ W/m} \cdot \text{K}\end{aligned}$$

Analysis For a cylindrical heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$L^* = \frac{D}{2} \left(\frac{g(\rho_l - \rho_{v,\text{sat}})}{\sigma} \right)^{1/2} = (0.01/2) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} = 1.997 > 1.2$$

$$C_{cr} = 0.12 \quad (\text{since } L^* > 1.2 \text{ and thus large cylinder})$$

The burnout point occurs at the maximum heat flux, which is

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_{v,\text{sat}}^2 (\rho_l - \rho_{v,\text{sat}})]^{1/4} \\ &= 0.12(2257 \times 10^3) [0.0589 \times 9.81 \times (0.5978)^2 (957.9 - 0.5978)]^{1/4} = 1.0155 \times 10^6 \text{ W/m}^2\end{aligned}$$

To determine the film boiling convection heat transfer coefficient, the knowledge of T_s is needed, which can be determined from the heat transfer in the film boiling region:

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) + \frac{3}{4} \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4)$$

where $C_{\text{film}} = 0.62$ (horizontal cylinders)

Substituting the values,

$$\begin{aligned}1.0155 \times 10^6 &= 0.62 \left[\frac{9.81(0.1588)^3 (0.1543)(957.9 - 0.1543)[2257 \times 10^3 + 0.4(2571)(T_s - 100)]}{(5.283 \times 10^{-5})(0.01)(T_s - 100)} \right]^{1/4} (T_s - 100) \\ &\quad + \frac{3}{4} (0.3)(5.67 \times 10^{-8}) [(T_s + 273)^4 - (373)^4]\end{aligned}$$

Solving for the surface temperature yield $T_s = 2231^\circ\text{C}$

The film boiling heat flux is

$$\begin{aligned}\dot{q}_{\text{film}} &= C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.62 \left[\frac{9.81(0.1588)^3 (0.1543)(957.9 - 0.1543)[2257 \times 10^3 + 0.4(2571)(2231 - 100)]}{(5.283 \times 10^{-5})(0.01)(2231 - 100)} \right]^{1/4} (2231 - 100) \\ &= 5.1419 \times 10^5 \text{ W/m}^2\end{aligned}$$

Thus, the film boiling convection heat transfer coefficient is

$$h = \frac{\dot{q}_{\text{film}}}{T_s - T_{\text{sat}}} = \frac{5.1419 \times 10^5 \text{ W/m}^2}{(2231 - 100) \text{ K}} = 241.3 \text{ W/m}^2 \cdot \text{K}$$

Discussion Note that the film temperature $T_f = (2231 + 100)/2 = 1166^\circ\text{C}$, is close to the assumed value of 1150°C for the evaluation of vapor properties. Therefore, 1150°C is a reasonable film temperature for the vapor properties.

10-52 A cylindrical heater is used for boiling water at 100°C. The boiling convection heat transfer coefficients at the maximum heat flux for nucleate boiling and film boiling are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater are negligible.

Properties The properties of water at $T_{\text{sat}} = 100^\circ\text{C}$ are $\sigma = 0.0589 \text{ N/m}$ (Tables 10-1) and, from Table A-9,

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_{v,\text{sat}} &= 0.5978 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \text{Pr}_l &= 1.75 & c_{pl} &= 4217 \text{ J/kg} \cdot \text{K}\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3).

The properties of vapor at the film temperature of $T_f = 1150^\circ\text{C}$ are, from Table A-16,

$$\begin{aligned}\rho_v &= 0.1543 \text{ kg/m}^3 & c_{pv} &= 2571 \text{ J/kg} \cdot \text{K} \\ \mu_v &= 5.283 \times 10^{-5} \text{ kg/m} \cdot \text{s} & k_v &= 0.1588 \text{ W/m} \cdot \text{K}\end{aligned}$$

Analysis For a cylindrical heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned}L^* &= \frac{D}{2} \left(\frac{g(\rho_l - \rho_{v,\text{sat}})}{\sigma} \right)^{1/2} = (0.003/2) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} = 0.599 < 1.2 \\ C_{cr} &= 0.12L^{*-0.25} = 0.1364 \quad (\text{since } L^* < 1.2 \text{ and thus small cylinder})\end{aligned}$$

The maximum heat flux can be determined as

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_{v,\text{sat}}^2 (\rho_l - \rho_{v,\text{sat}})]^{1/4} \\ &= 0.1364 (2257 \times 10^3) [0.0589 \times 9.81 \times (0.5978)^2 (957.9 - 0.5978)]^{1/4} = 1.1543 \times 10^6 \text{ W/m}^2\end{aligned}$$

(a) The surface temperature T_s for nucleate boiling at $\dot{q}_{\text{max}}$ can be solved as

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_{v,\text{sat}})}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3$$

Substituting the values,

$$\begin{aligned}1.1543 \times 10^6 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} \left[\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right]^3 \\ \therefore T_s &= 120.2^\circ\text{C}\end{aligned}$$

Thus, the nucleate boiling convection heat transfer coefficient is

$$h_{\text{nucleate}} = \frac{\dot{q}_{\text{nucleate}}}{T_s - T_{\text{sat}}} = \frac{1.1543 \times 10^6 \text{ W/m}^2}{(120.2 - 100) \text{ K}} = 57,144 \text{ W/m}^2 \cdot \text{K}$$

(b) The surface temperature T_s for film boiling at $\dot{q}_{\text{max}}$ can be solved as

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv}(T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) + \frac{3}{4} \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4)$$

Substituting the values,

$$\begin{aligned}1.1543 \times 10^6 &= 0.62 \left[\frac{9.81(0.1588)^3 (0.1543)(957.9 - 0.1543)[2257 \times 10^3 + 0.4(2571)(T_s - 100)]}{(5.283 \times 10^{-5})(0.003)(T_s - 100)} \right]^{1/4} \\ &\quad \times (T_s - 100) + \frac{3}{4} (0.3)(5.67 \times 10^{-8}) [(T_s + 273)^4 - (373)^4] \\ \therefore T_s &= 2192^\circ\text{C}\end{aligned}$$

The film boiling heat flux is

$$\dot{q}_{\text{film}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{\text{fg}} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}})$$

where $C_{\text{film}} = 0.62$ (horizontal cylinders)

$$\begin{aligned} &= 0.62 \left[\frac{9.81(0.1588)^3(0.1543)(957.9 - 0.1543)[2257 \times 10^3 + 0.4(2571)(2192 - 100)]}{(5.283 \times 10^{-5})(0.003)(2192 - 100)} \right]^{1/4} (2192 - 100) \\ &= 6.8366 \times 10^5 \text{ W/m}^2 \end{aligned}$$

Thus, the film boiling convection heat transfer coefficient is

$$h = \frac{\dot{q}_{\text{film}}}{T_s - T_{\text{sat}}} = \frac{6.8366 \times 10^5 \text{ W/m}^2}{(2192 - 100) \text{ K}} = \mathbf{327 \text{ W/m}^2 \cdot \text{K}}$$

Discussion The nucleate boiling convection heat transfer coefficient is about 175 times higher than that of film boiling. This is because the vapor film surrounding the heater surface during film boiling impedes convection heat transfer.

Note that the film temperature $T_f = (2192 + 100)/2 = 1146^\circ\text{C}$, is close to the assumed value of 1150°C used in film boiling for the evaluation of vapor properties.

10-53 Water is boiled at 1 atm pressure and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by a horizontal nickel plated copper heating element. The maximum (critical) heat flux and the temperature jump of the wire when the operating point jumps from nucleate boiling to film boiling regime are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.5978 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg} \cdot \text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & c_{pl} &= 4217 \text{ J/kg} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.75\end{aligned}$$

Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a nickel plated surface (Table 10-3). Note that we expressed the properties in units specified under Eqs. 10-2 and 10-3 in connection with their definitions in order to avoid unit manipulations. The vapor properties at the anticipated film temperature of $T_f = (T_s + T_{\text{sat}})/2$ of 1000°C (will be checked) (Table A-16)

$$\begin{aligned}\rho_v &= 0.1725 \text{ kg/m}^3 & c_{pv} &= 2471 \text{ J/kg} \cdot ^\circ\text{C} \\ k_v &= 0.1362 \text{ W/m} \cdot ^\circ\text{C} & \mu_v &= 4.762 \times 10^{-5} \text{ kg/m} \cdot \text{s}\end{aligned}$$

Analysis (a) For a horizontal heating element, the coefficient C_{cr} is determined from Table 10-4 to be

$$L^* = L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.002) \left(\frac{9.81(957.9 - 0.5978)}{0.0589} \right)^{1/2} = 0.7986 < 1.2$$

$$C_{cr} = 0.12 L^{*-0.25} = 0.12(0.7986)^{-0.25} = 0.1269$$

Then the maximum or critical heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.1269(2257 \times 10^3) [0.0589 \times 9.81 \times (0.5978)^2 (957.9 - 0.5978)]^{1/4} = \mathbf{1,074,000 \text{ W/m}^2}\end{aligned}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given. Substituting the maximum heat flux into the Rohsenow relation together with other properties gives

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 1,074,000 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.5978)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0060(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives $T_s = 109.1^\circ\text{C}$

(b) Heat transfer in the film boiling region can be expressed as

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 0.62 \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv}(T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) + \frac{3}{4} \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4)$$

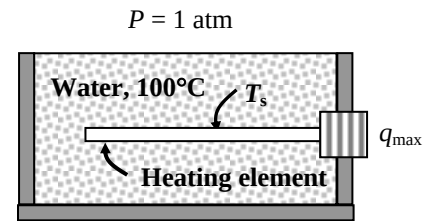
Substituting,

$$\begin{aligned}1,074,000 &= 0.62 \left[\frac{9.81(0.1362)^3(0.1725)(957.9 - 0.1725)[2257 \times 10^3 + 0.4 \times 2471(T_s - 100)]}{(4.762 \times 10^{-5})(0.004)(T_s - 100)} \right]^{1/4} \\ &\quad \times (T_s - 100) + \frac{3}{4} (0.3)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(T_s + 273)^4 - (100 + 273)^4]\end{aligned}$$

Solving for the surface temperature gives $T_s = 2200^\circ\text{C}$. Therefore, the temperature jump of the wire when the operating point jumps from nucleate boiling to film boiling is

$$\text{Temperature jump: } \Delta T = T_{s, \text{film}} - T_{s, \text{crit}} = 2200 - 109 = \mathbf{2091^\circ\text{C}}$$

Note that the film temperature is $(2200 + 100)/2 = 1150^\circ\text{C}$, which is close enough to the assumed value of 1000°C for the evaluation of vapor properties.





10-54 Prob. 10-53 is reconsidered. The effects of the local atmospheric pressure and the emissivity of the wire on the critical heat flux and the temperature rise of wire are to be investigated.

Analysis The problem is solved using EES, and the solution is given below.

"GIVEN"

L=0.3 [m]
D=0.004 [m]
epsilon=0.3
P=101.3 [kPa]

"PROPERTIES"

Fluid\$='steam_IAPWS'
T_sat=temperature(Fluid\$, P=P, x=1)
rho_l=density(Fluid\$, T=T_sat, x=0)
rho_v=density(Fluid\$, T=T_sat, x=1)
sigma=SurfaceTension(Fluid\$, T=T_sat)
mu_l=Viscosity(Fluid\$, T=T_sat, x=0)
Pr_l=Prandtl(Fluid\$, T=T_sat, P=P+1)
c_l=CP(Fluid\$, T=T_sat, x=0)*Convert(kJ/kg-C, J/kg-C)
h_f=enthalpy(Fluid\$, T=T_sat, x=0)
h_g=enthalpy(Fluid\$, T=T_sat, x=1)
h_fg=(h_g-h_f)*Convert(kJ/kg, J/kg)
C_sf=0.0060 "from Table 10-3 of the text"
n=1 "from Table 10-3 of the text"

T_vapor=1000-273 "[C], assumed vapor temperature in the film boiling region"
rho_v_f=density(Fluid\$, T=T_vapor, P=P) "f stands for film"
c_v_f=CP(Fluid\$, T=T_vapor, P=P)*Convert(kJ/kg-C, J/kg-C)
k_v_f=Conductivity(Fluid\$, T=T_vapor, P=P)
mu_v_f=Viscosity(Fluid\$, T=T_vapor, P=P)

g=9.81 [m/s^2] "gravitational acceleration"
sigma_rad=5.67E-8 [W/m^2-K^4] "Stefan-Boltzmann constant"

"ANALYSIS"

"(a)"

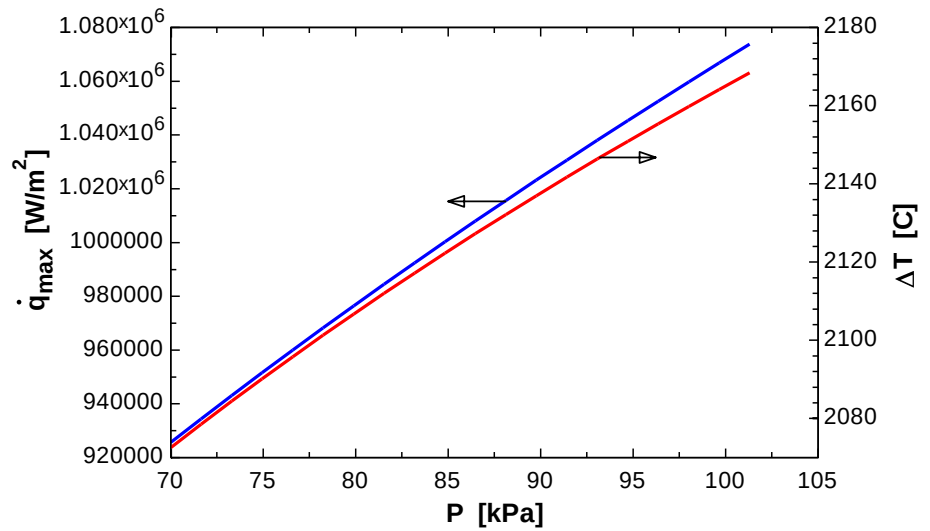
"C_cr is to be determined from Table 10-4 of the text"

C_cr=0.12*L_star^(-0.25)
L_star=D/2*((g*(rho_l-rho_v))/sigma)^0.5
q_dot_max=C_cr*h_fg*(sigma*g*rho_v^2*(rho_l-rho_v))^0.25
q_dot_nucleate=q_dot_max
q_dot_nucleate=mu_l*h_fg*(((g*(rho_l-rho_v))/sigma)^0.5)*((c_l*(T_s_crit-T_sat))/(C_sf*h_fg*Pr_l^n))^3

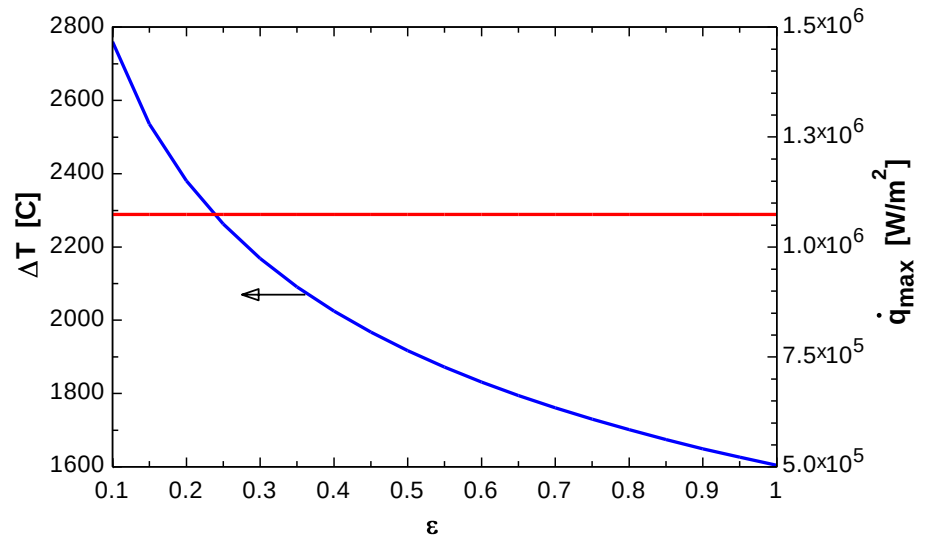
"(b)"

q_dot_total=q_dot_film+3/4*q_dot_rad "Heat transfer in the film boiling region"
q_dot_total=q_dot_nucleate
q_dot_film=0.62*((g*k_v_f^3*rho_v_f*(rho_l-rho_v_f)*(h_fg+0.4*c_v_f*(T_s_film-T_sat)))/(mu_v_f*D*(T_s_film-T_sat)))^0.25*(T_s_film-T_sat)
q_dot_rad=epsilon*sigma_rad*((T_s_film+273)^4-(T_sat+273)^4)
DELTAT=T_s_film-T_s_crit

P [kPa]	$\dot{q}_{\max}$ [kW/m ²]	ΔT [C]
70	925656	2073
71.65	934417	2079
73.29	943050	2084
74.94	951559	2090
76.59	959948	2096
78.24	968222	2101
79.88	976385	2107
81.53	984439	2112
83.18	992389	2117
84.83	1000237	2122
86.47	1007987	2127
88.12	1015642	2132
89.77	1023205	2137
91.42	1030677	2142
93.06	1038062	2146
94.71	1045363	2151
96.36	1052581	2155
98.01	1059719	2160
99.65	1066778	2164
101.3	1073762	2168



ε	$\dot{q}_{\max}$ [kW/m ²]	ΔT [C]
0.1	1073762	2760
0.15	1073762	2535
0.2	1073762	2380
0.25	1073762	2262
0.3	1073762	2168
0.35	1073762	2091
0.4	1073762	2025
0.45	1073762	1967
0.5	1073762	1917
0.55	1073762	1872
0.6	1073762	1831
0.65	1073762	1794
0.7	1073762	1761
0.75	1073762	1730
0.8	1073762	1701
0.85	1073762	1674
0.9	1073762	1649
0.95	1073762	1626
1	1073762	1604



Condensation Heat Transfer

10-55C Condensation is a vapor-to-liquid phase change process. It occurs when the temperature of a vapor is reduced *below* its saturation temperature T_{sat} . This is usually done by bringing the vapor into contact with a solid surface whose temperature T_s is *below* the saturation temperature T_{sat} of the vapor.

10-56C In *film condensation*, the condensate wets the surface and forms a liquid film on the surface which slides down under the influence of gravity. The thickness of the liquid film increases in the flow direction as more vapor condenses on the film. This is how condensation normally occurs in practice. In *dropwise condensation*, the condensed vapor forms droplets on the surface instead of a continuous film, and the surface is covered by countless droplets of varying diameters. Dropwise condensation is a much more effective mechanism of heat transfer.

10-57C The presence of noncondensable gases in the vapor has a detrimental effect on condensation heat transfer. Even small amounts of a noncondensable gas in the vapor cause significant drops in heat transfer coefficient during condensation.

10-58C The modified latent heat of vaporization h_{fg}^* is the amount of heat released as a unit mass of vapor condenses at a specified temperature, plus the amount of heat released as the condensate is cooled further to some average temperature between T_{sat} and T_s . It is defined as $h_{fg}^* = h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s)$ where c_{pl} is the specific heat of the liquid at the average film temperature.

10-59C During film condensation on a vertical plate, heat flux at the top will be higher since the thickness of the film at the top, and thus its thermal resistance, is lower.

10-60C The condensation heat transfer coefficient for the tubes will be the highest for the case of horizontal side by side (case b) since (1) for long tubes, the horizontal position gives the highest heat transfer coefficients, and (2) for tubes in a vertical tier, the average thickness of the liquid film at the lower tubes is much larger as a result of condensate falling on top of them from the tubes directly above, and thus the average heat transfer coefficient at the lower tubes in such arrangements is smaller.

10-61C In condensate flow, the wetted perimeter is defined as the length of the surface-condensate interface at a cross-section of condensate flow. It differs from the ordinary perimeter in that the latter refers to the entire circumference of the condensate at some cross-section.

10-62 The hydraulic diameter D_h for all 4 cases are expressed in terms of the boundary layer thickness δ as follows:

$$(a) \text{ Vertical plate: } D_h = \frac{4A_c}{p} = \frac{4w\delta}{w} = 4\delta$$

$$(b) \text{ Tilted plate: } D_h = \frac{4A_c}{p} = \frac{4w\delta}{w} = 4\delta$$

$$(c) \text{ Vertical cylinder: } D_h = \frac{4A_c}{p} = \frac{4\pi D\delta}{\pi D} = 4\delta$$

$$(d) \text{ Horizontal cylinder: } D_h = \frac{4A_c}{p} = \frac{4(2L\delta)}{2L} = 4\delta$$

$$(e) \text{ Sphere: } D_h = \frac{4A_c}{p} = \frac{4\pi D\delta}{\pi D} = 4\delta$$

Therefore, the Reynolds number for all 5 cases can be expressed as

$$\text{Re} = \frac{4\dot{m}}{p\mu_l} = \frac{4A_c\rho_l V_l}{p\mu_l} = \frac{D_h\rho_l V_l}{\mu_l} = \frac{4\delta\rho_l V_l}{\mu_l}$$

10-63 The local heat transfer coefficients at the middle and at the bottom of a vertical plate undergoing film condensation are to be determined.

Assumptions 1 Steady operating condition exists. 2 The plate surface has uniform temperature. 3 The flow is laminar.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \text{ kJ/kg}$ (Table A-2) and $\rho_v = 0.5978 \text{ kg/m}^3$ (Table A-9). The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 90^\circ\text{C}$ are, from Table A-9,

$$\begin{aligned}\rho_l &= 965.3 \text{ kg/m}^3 & c_{pl} &= 4206 \text{ J/kg}\cdot\text{K} \\ \mu_l &= 0.315 \times 10^{-3} \text{ kg/m}\cdot\text{s} & k_l &= 0.675 \text{ W/m}\cdot\text{K} \\ \nu_l &= \mu_l / \rho_l = 0.326 \times 10^{-6} \text{ m}^2/\text{s}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 + 0.68(4206)(100 - 80) \\ &= 2314 \times 10^3 \text{ J/kg}\end{aligned}$$

The local heat transfer coefficient can be calculated using

$$\begin{aligned}h_x &= \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{4\mu_l(T_{\text{sat}} - T_s)x} \right]^{1/4} \\ &= \left[\frac{(9.81)(965.3)(965.3 - 0.5978)(2314 \times 10^3)(0.675)^3}{4(0.315 \times 10^{-3})(100 - 80)x} \right]^{1/4} \\ &= 4008 \left(\frac{1}{x} \right)^{1/4} \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

The local heat transfer coefficient at the middle of the plate ($x = 0.1 \text{ m}$) is

$$h_x = 4008 \left(\frac{1}{x} \right)^{1/4} \text{ W/m}^2 \cdot \text{K} = 4008 \left(\frac{1}{0.1} \right)^{1/4} \text{ W/m}^2 \cdot \text{K} = \mathbf{7130 \text{ W/m}^2 \cdot \text{K}}$$

The local heat transfer coefficient at the bottom of the plate ($x = 0.2 \text{ m}$) is

$$h_x = 4008 \left(\frac{1}{x} \right)^{1/4} \text{ W/m}^2 \cdot \text{K} = 4008 \left(\frac{1}{0.2} \right)^{1/4} \text{ W/m}^2 \cdot \text{K} = \mathbf{5990 \text{ W/m}^2 \cdot \text{K}}$$

Discussion The assumption that the flow is laminar is verified to be appropriate:

$$\text{Re} \cong \frac{4g\rho_l^2}{3\mu_l^2} \left(\frac{k_l}{h_{x=L}} \right)^3 = \frac{4(9.81)(965.3)^2}{3(0.315 \times 10^{-3})^2} \left(\frac{0.675}{5990} \right)^3 = 176 < 1800$$

10-64 The necessary surface temperature of the plate used to condensate saturated water vapor at a desired condensation rate is to be determined.

Assumptions 1 Steady operating condition exists. 2 The plate surface has uniform temperature. 3 The film temperature is 90°C.

Properties Based on the problem statement, we take film temperature to be $T_f = (T_{\text{sat}} + T_s)/2 = 90^\circ\text{C}$ and the surface temperature to be $T_s = 80^\circ\text{C}$. The properties of liquid water at the film temperature of $T_f = 90^\circ\text{C}$ are, from Table A-9,

$$\begin{aligned}\rho_l &= 965.3 \text{ kg/m}^3 & c_{pl} &= 4206 \text{ J/kg}\cdot\text{K} \\ \mu_l &= 0.315 \times 10^{-3} \text{ kg/m}\cdot\text{s} & k_l &= 0.675 \text{ W/m}\cdot\text{K} \\ \nu_l &= \mu_l / \rho_l = 0.326 \times 10^{-6} \text{ m}^2/\text{s}\end{aligned}$$

The properties of water at the saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ are $h_{fg} = 2257 \text{ kJ/kg}$ (Table A-2) and $\rho_v = 0.5978 \text{ kg/m}^3$ (Table A-9).

Analysis The calculation of the modified latent heat of vaporization requires the knowledge of the T_s . Hence, we assume $T_s = 80^\circ\text{C}$, and iterate the solution, if necessary, until good agreement with the calculated value of T_s is achieved:

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 + 0.68(4206)(100 - 80) \\ &= 2314 \times 10^3 \text{ J/kg}\end{aligned}$$

The Reynolds number is

$$\text{Re} = \frac{4\dot{m}}{p\mu_l} = \frac{4(0.016 \text{ kg/s})}{(0.5 \text{ m})(0.315 \times 10^{-3} \text{ kg/m}\cdot\text{s})} = 406.3$$

which is between 30 and 1800, and thus the flow is wavy-laminar. The heat transfer coefficient is

$$\begin{aligned}h &= h_{\text{vert, wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{(406.3)(0.675 \text{ W/m}\cdot\text{K})}{1.08(406.3)^{1.22} - 5.2} \left[\frac{9.81 \text{ m/s}^2}{(0.326 \times 10^{-6} \text{ m}^2/\text{s})^2} \right]^{1/3} \\ &= 7558 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Hence, the surface temperature can be calculated using

$$\begin{aligned}hA_s(T_{\text{sat}} - T_s) &= \dot{m}h_{fg}^* \quad \rightarrow \quad T_s = T_{\text{sat}} - \frac{\dot{m}h_{fg}^*}{hA_s} \\ T_s &= 100^\circ\text{C} - \frac{(0.016 \text{ kg/s})(2314 \times 10^3 \text{ J/kg})}{(7558 \text{ W/m}^2 \cdot \text{K})(0.5 \text{ m})^2} = \mathbf{80.4^\circ\text{C}}\end{aligned}$$

Discussion The assumed $T_s = 80^\circ\text{C}$ and $T_f = 90^\circ\text{C}$ are good, thus the solution does not require iteration.

10-65 Saturated ammonia at a saturation temperature of $T_{\text{sat}} = 30^\circ\text{C}$ condenses on vertical plates which are maintained at 10°C . The average heat transfer coefficient and the rate of condensation of ammonia are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the entire plate (this assumption will be verified). 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of ammonia at the saturation temperature of 30°C are $h_{fg} = 1144 \times 10^3 \text{ J/kg}$ and $\rho_v = 9.055 \text{ kg/m}^3$. The properties of liquid ammonia at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (30 + 10)/2 = 20^\circ\text{C}$ are (Table A-11),

$$\rho_l = 610.2 \text{ kg/m}^3$$

$$\mu_l = 1.519 \times 10^{-4} \text{ kg/m} \cdot \text{s}$$

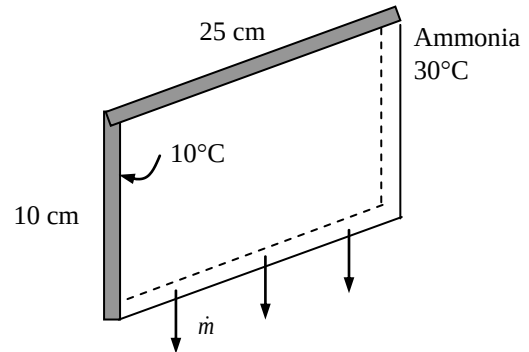
$$\nu_l = \mu_l / \rho_l = 2.489 \times 10^{-7} \text{ m}^2/\text{s}$$

$$c_{pl} = 4745 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_l = 0.4927 \text{ W/m} \cdot ^\circ\text{C}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 1144 \times 10^3 \text{ J/kg} + 0.68 \times 4745 \text{ J/kg} \cdot ^\circ\text{C}(30 - 10)^\circ\text{C} \\ &= 1209 \times 10^3 \text{ J/kg} \end{aligned}$$



Assuming wavy-laminar flow, the Reynolds number is determined from

$$\begin{aligned} \text{Re} &= \text{Re}_{\text{vertical, wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s) \left(\frac{g}{\nu_l^2} \right)^{1/3}}{\mu_l h_{fg}^*} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (0.1 \text{ m}) \times (0.4927 \text{ W/m} \cdot ^\circ\text{C}) \times (30 - 10)^\circ\text{C} \left(\frac{9.8 \text{ m/s}^2}{(2.489 \times 10^{-7} \text{ m}^2/\text{s})^2} \right)^{1/3}}{(1.519 \times 10^{-4} \text{ kg/m} \cdot \text{s})(1209 \times 10^3 \text{ J/kg})} \right]^{0.82} = 307.0 \end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined to be

$$\begin{aligned} h &= h_{\text{vertical, wavy}} = \frac{\text{Re } k_l}{1.08 \text{ Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{307 \times (0.4927 \text{ W/m} \cdot ^\circ\text{C})}{1.08(307)^{1.22} - 5.2} \left(\frac{9.8 \text{ m/s}^2}{(2.489 \times 10^{-7} \text{ m}^2/\text{s})^2} \right)^{1/3} = 7032 \text{ W/m}^2 \cdot ^\circ\text{C} \end{aligned}$$

The total heat transfer surface area of the plates is

$$A_s = W \times L = 30(0.25 \text{ m})(0.10 \text{ m}) = 0.75 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (7032 \text{ W/m}^2 \cdot ^\circ\text{C})(0.75 \text{ m}^2)(30 - 10)^\circ\text{C} = 105,480 \text{ W}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{105,480 \text{ J/s}}{1209 \times 10^3 \text{ J/kg}} = 0.0872 \text{ kg/s}$$

10-66 Saturated steam at atmospheric pressure thus at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ condenses on a vertical plate which is maintained at 90°C by circulating cooling water through the other side. The rate of heat transfer to the plate and the rate of condensation of steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the entire plate (this assumption will be verified). 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (100 + 90) / 2 = 95^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 961.5 \text{ kg/m}^3 \\ \mu_l &= 0.297 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.309 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4212 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.677 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4212 \text{ J/kg} \cdot ^\circ\text{C} (100 - 90)^\circ\text{C} = 2,286 \times 10^3 \text{ J/kg}\end{aligned}$$

Assuming wavy-laminar flow, the Reynolds number is determined from

$$\begin{aligned}\text{Re} &= \text{Re}_{\text{vertical, wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (2 \text{ m}) \times (0.677 \text{ W/m} \cdot ^\circ\text{C}) \times (100 - 90)^\circ\text{C}}{(0.297 \times 10^{-3} \text{ kg/m} \cdot \text{s}) (2,286 \times 10^3 \text{ J/kg})} \left(\frac{9.81 \text{ m/s}^2}{(0.309 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.82} = 798.0\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined to be

$$\begin{aligned}h &= h_{\text{vertical, wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{798 \times (0.677 \text{ W/m} \cdot ^\circ\text{C})}{1.08(798)^{1.22} - 5.2} \left(\frac{9.81 \text{ m/s}^2}{(0.309 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 6757 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the plate is

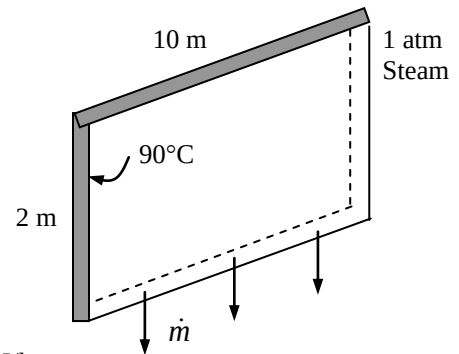
$$A_s = W \times L = (2 \text{ m})(10 \text{ m}) = 20 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (6757 \text{ W/m}^2 \cdot ^\circ\text{C})(20 \text{ m}^2)(100 - 90)^\circ\text{C} = 1,351,500 \text{ W} = \mathbf{1352 \text{ kW}}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{1,351,500 \text{ J/s}}{2,286 \times 10^3 \text{ J/kg}} = \mathbf{0.591 \text{ kg/s}}$$



10-67 Saturated steam at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ condenses on a plate which is tilted 60° from the vertical and maintained at 90°C by circulating cooling water through the other side. The rate of heat transfer to the plate and the rate of condensation of the steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the entire plate (this assumption will be verified). 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (100 + 90) / 2 = 95^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 961.5 \text{ kg/m}^3 \\ \mu_l &= 0.297 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.309 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4212 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.677 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4212 \text{ J/kg} \cdot ^\circ\text{C}(100 - 90)^\circ\text{C} \\ &= 2,286 \times 10^3 \text{ J/kg}\end{aligned}$$

Assuming wavy-laminar flow, the Reynolds number is determined from the vertical plate relation by replacing g by $g \cos \theta$ where $\theta = 60^\circ$ to be

$$\begin{aligned}\text{Re} &= \text{Re}_{\text{tilted,wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g \cos 60}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (2 \text{ m}) \times (0.677 \text{ W/m} \cdot ^\circ\text{C}) \times (100 - 90)^\circ\text{C}}{(0.297 \times 10^{-3} \text{ kg/m} \cdot \text{s})(2286 \times 10^3 \text{ J/kg})} \left(\frac{(9.81 \text{ m/s}^2) \cos 60}{(0.309 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.82} = 660.5\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{tilted,wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g \cos \theta}{\nu_l^2} \right)^{1/3} \\ &= \frac{660.5 \times (0.677 \text{ W/m} \cdot ^\circ\text{C})}{1.08(660.5)^{1.22} - 5.2} \left(\frac{(9.81 \text{ m/s}^2) \cos 60}{(0.309 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 5593 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the plate is $A_s = W \times L = (2 \text{ m})(10 \text{ m}) = 20 \text{ m}^2$.

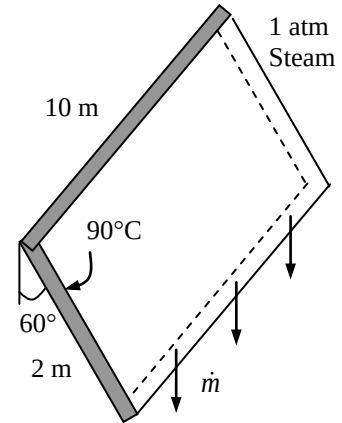
Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (5593 \text{ W/m}^2 \cdot ^\circ\text{C})(20 \text{ m}^2)(100 - 90)^\circ\text{C} = 1,118,600 \text{ W} = \mathbf{1119 \text{ kW}}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{1,118,600 \text{ J/s}}{2286 \times 10^3 \text{ J/kg}} = \mathbf{0.489 \text{ kg/s}}$$

Discussion Using the heat transfer coefficient determined in the previous problem for the vertical plate, we could also determine the heat transfer coefficient from $h_{\text{inclined}} = h_{\text{vert}} (\cos \theta)^{1/4}$. It would give $5682 \text{ W/m}^2 \cdot ^\circ\text{C}$, which is 1.6% different than the value determined above.



10-68 C&S Saturated ammonia vapor condenses on a vertical ASTM A240 410S stainless steel plate at 190 kPa. The highest rate of condensation that can be produced by the plate, without cooling the plate below the minimum suitable temperature set by the ASME Code for Process Piping, is to be determined.

Assumptions 1 Steady state conditions. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the plate. 4 The vapor density is much smaller than the liquid density.

Properties The saturation temperature of ammonia at 190 kPa is -20°C (Table A-11), and the properties of saturated ammonia are from Table A-11:

At the saturation temperature, $T_{\text{sat}} = -20^{\circ}\text{C}$,
 $h_{fg} = 1329 \text{ kJ/kg}$, $\rho_v = 1.603 \text{ kg/m}^3$.

At the film temperature, $T_f = (-20 - 30)^{\circ}\text{C}/2 = -25^{\circ}\text{C}$,
 $c_{pl} = 4489 \text{ J/kg}\cdot\text{K}$, $k_l = 0.5968 \text{ W/m}\cdot\text{K}$, $\rho_l = 671.5 \text{ kg/m}^3$, $\mu_l = 2.492 \times 10^{-4} \text{ kg/m}\cdot\text{s}$
 $\nu_l = \mu_l / \rho_l = 3.711 \times 10^{-7} \text{ m}^2/\text{s}$

Analysis The modified latent heat of vaporization is

$$h_{fg}^* = h_{fg} + 0.68 c_{pl}(T_{\text{sat}} - T_s) = 1329000 + 0.68(4489)(-20 + 30) = 1359500 \text{ J/kg}$$

For wavy-laminar flow, the Reynolds number is

$$\text{Re} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s) \left(\frac{g}{\nu_l^2} \right)^{1/3}}{\mu_l h_{fg}^*} \right]^{0.820}$$

$$\text{Re} = \left[4.81 + \frac{3.70(2)(0.5968)(-20 + 30) \left(\frac{9.81}{(3.711 \times 10^{-7})^2} \right)^{1/3}}{(2.492 \times 10^{-4})(1359500)} \right]^{0.820} = 1151$$

The Reynolds number is $30 < \text{Re} < 1800$, so the assumption for wavy-laminar flow is verified. The condensation heat transfer coefficient is then

$$h = h_{\text{vert,wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} = \frac{(1151)(0.5968)}{1.08(1151)^{1.22} - 5.2} \left(\frac{9.81}{(3.711 \times 10^{-7})^2} \right)^{1/3} = 4863 \text{ W/m}^2 \cdot \text{K}$$

The rate of heat transfer on the plate during the condensation process is

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (4863 \text{ W/m}^2 \cdot \text{K})(2 \text{ m} \times 1.5 \text{ m})(-20 + 30)\text{K} = 1.4589 \times 10^5 \text{ W}$$

The highest rate of condensation that can be produced by the plate, without cooling it below -30°C is

$$\dot{m}_{\text{condensate}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{1.4589 \times 10^5 \text{ J/s}}{1359500 \text{ J/kg}} = 0.1073 \text{ kg/s}$$

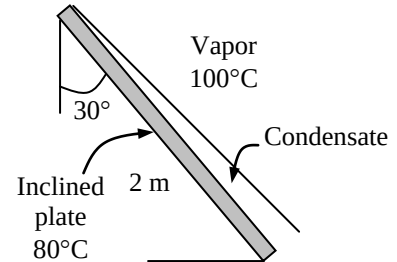
Discussion The highest condensation rate that can be achieved from the vertical ASTM A240 410S stainless steel plate is 0.107 kg/s. If the condensation rate is above 0.107 kg/s, it would increase the heat transfer rate on the plate. This would require the plate surface temperature to decrease below the minimum suitable temperature of -30°C set by the ASME Code for Process Piping.

10-69 Saturated steam at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ condenses on a plate which is tilted 30° from the vertical and maintained at 80°C by circulating cooling water through the other side. The rate of heat transfer to the plate and the rate of condensation of the steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the entire plate (this assumption will be verified). 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (100 + 80) / 2 = 90^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 965.3 \text{ kg/m}^3 \\ \mu_l &= 0.315 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.326 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4206 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.675 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4206 \text{ J/kg} \cdot ^\circ\text{C} (100 - 80)^\circ\text{C} = 2,314 \times 10^3 \text{ J/kg}\end{aligned}$$

Assuming wavy-laminar flow, the Reynolds number is determined from the vertical plate relation by replacing g by $g \cos \theta$ where $\theta = 30^\circ$ to be

$$\begin{aligned}\text{Re} &= \text{Re}_{\text{tilted, wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g \cos \theta}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (2 \text{ m}) \times (0.675 \text{ W/m} \cdot ^\circ\text{C}) \times (100 - 80)^\circ\text{C}}{(0.315 \times 10^{-3} \text{ kg/m} \cdot \text{s}) (2314 \times 10^3 \text{ J/kg})} \left(\frac{(9.81 \text{ m/s}^2) \cos 30}{(0.326 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.82} = 1237\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{tilted, wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g \cos \theta}{\nu_l^2} \right)^{1/3} \\ &= \frac{1237 \times (0.675 \text{ W/m} \cdot ^\circ\text{C})}{1.08(1237)^{1.22} - 5.2} \left(\frac{(9.81 \text{ m/s}^2) \cos 30}{(0.326 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 5623 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the plate is:

$$A = w \times L = (2 \text{ m})(2 \text{ m}) = 4 \text{ m}^2.$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA(T_{\text{sat}} - T_s) = (5623 \text{ W/m}^2 \cdot ^\circ\text{C})(4 \text{ m}^2)(100 - 80)^\circ\text{C} = 449,900 \text{ W}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{449,900 \text{ J/s}}{2314 \times 10^3 \text{ J/kg}} = 0.194 \text{ kg/s}$$

Discussion We could also determine the heat transfer coefficient from $h_{\text{inclined}} = h_{\text{vert}} (\cos \theta)^{1/4}$.



10-70 Prob. 10-69 is reconsidered. The effects of plate temperature and the angle of the plate from the vertical on the average heat transfer coefficient and the rate at which the condensate drips off are to be investigated.

Analysis The problem is solved using EES, and the solution is given below.

"GIVEN"

$T_{\text{sat}}=100$ [C]

$L=2$ [m]

$\theta=30$ [degrees]

$T_s=80$ [C]

"PROPERTIES"

Fluid\$='steam_IAPWS'

$T_f=1/2*(T_{\text{sat}}+T_s)$

$P_{\text{sat}}=\text{pressure}(\text{Fluid}\$, T=T_{\text{sat}}, x=1)$

$\rho_l=\text{density}(\text{Fluid}\$, T=T_f, x=0)$

$\mu_l=\text{Viscosity}(\text{Fluid}\$, T=T_f, x=0)$

$\nu_l=\mu_l/\rho_l$

$c_l=\text{CP}(\text{Fluid}\$, T=T_f, x=0)*\text{Convert}(\text{kJ/kg}\cdot\text{C}, \text{J/kg}\cdot\text{C})$

$k_l=\text{Conductivity}(\text{Fluid}\$, T=T_f, P=P_{\text{sat}}+1)$

$h_f=\text{enthalpy}(\text{Fluid}\$, T=T_{\text{sat}}, x=0)$

$h_g=\text{enthalpy}(\text{Fluid}\$, T=T_{\text{sat}}, x=1)$

$h_{fg}=(h_g-h_f)*\text{Convert}(\text{kJ/kg}, \text{J/kg})$

$g=9.81$ [m/s²]

"ANALYSIS"

"(a)"

$h_{fg_star}=h_{fg}+0.68*c_l*(T_{\text{sat}}-T_s)$

$Re=(4.81+(3.7*L*k_l*(T_{\text{sat}}-T_s))/(\mu_l*h_{fg_star})*((g*\cos(\theta))/\nu_l^2)^{1/3})^{0.820}$

$h=(Re*k_l)/(1.08*Re^{1.22-5.2}*((g*\cos(\theta))/\nu_l^2)^{1/3})$

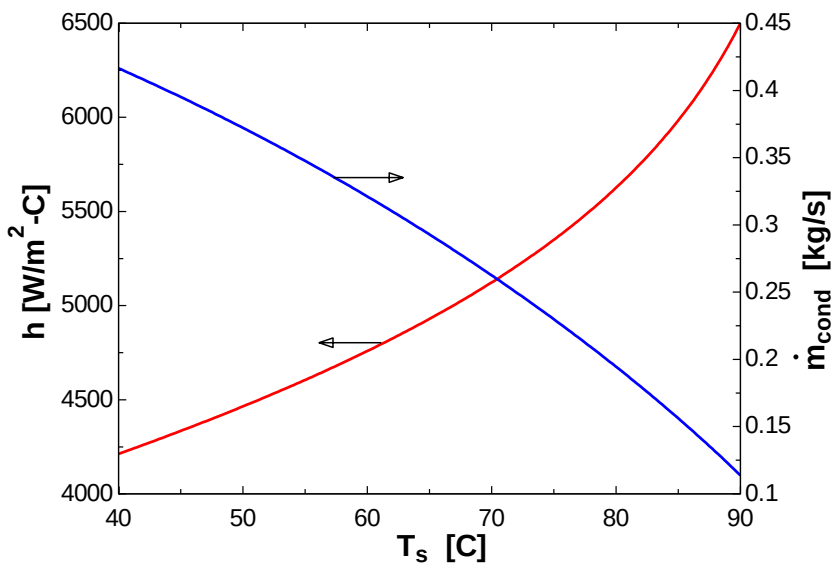
$\dot{Q}=h*A*(T_{\text{sat}}-T_s)$

$A=L^2$

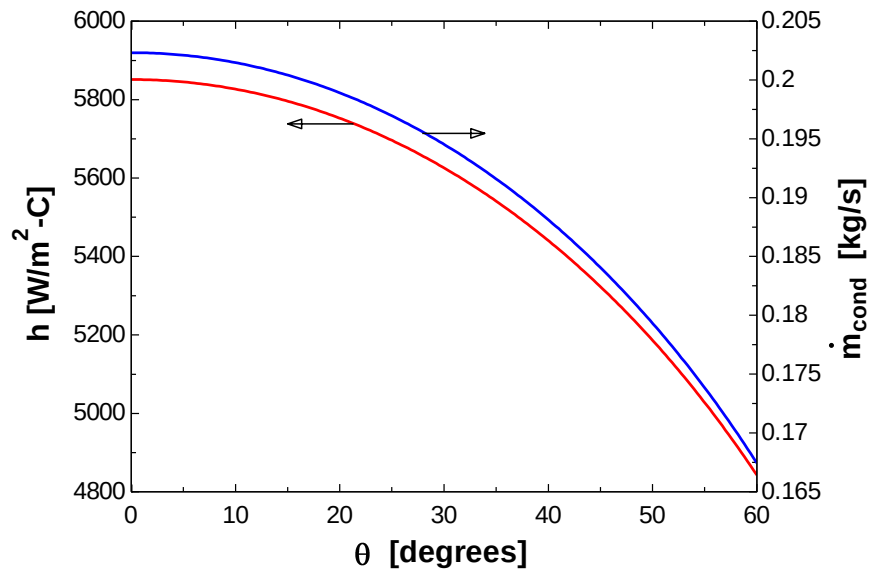
"(b)"

$\dot{m}_{\text{dot_cond}}=\dot{Q}/h_{fg_star}$

T_s [C]	h [W/m ² ·C]	$\dot{m}_{\text{cond}}$ [kg/s]
40	4212	0.4165
42.5	4272	0.406
45	4334	0.3951
47.5	4398	0.3839
50	4464	0.3722
52.5	4533	0.3601
55	4605	0.3476
57.5	4680	0.3346
60	4759	0.3212
62.5	4841	0.3073
65	4929	0.2929
67.5	5023	0.2779
70	5123	0.2625
72.5	5231	0.2465
75	5350	0.2298
77.5	5480	0.2125
80	5626	0.1945
82.5	5791	0.1758
85	5984	0.1561
87.5	6213	0.1355
90	6500	0.1138



θ [degrees]	h [W/m ² .C]	$\dot{m}_{\text{cond}}$ [kg/s]
0	5851	0.2023
3	5849	0.2022
6	5842	0.202
9	5831	0.2016
12	5816	0.2011
15	5796	0.2004
18	5771	0.1996
21	5742	0.1986
24	5708	0.1974
27	5670	0.196
30	5626	0.1945
33	5577	0.1928
36	5522	0.1909
39	5462	0.1889
42	5396	0.1866
45	5323	0.1841
48	5243	0.1813
51	5156	0.1783
54	5061	0.175
57	4957	0.1714
60	4842	0.1674



10-71 Saturated steam condenses outside of vertical tube. The rate of heat transfer to the coolant, the rate of condensation and the thickness of the condensate layer at the bottom are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal. 3 The tube can be treated as a vertical plate. 4 The condensate flow is wavy-laminar over the entire tube (this assumption will be verified). 5 Nusselt's analysis can be used to determine the thickness of the condensate film layer. 6 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 30°C are $h_{fg} = 2431 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.03 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (30 + 20)/2 = 25^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 997.0 \text{ kg/m}^3 \\ \mu_l &= 0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.894 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4180 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.607 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$

Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2431 \times 10^3 \text{ J/kg} + 0.68 \times 4180 \text{ J/kg} \cdot ^\circ\text{C}(30 - 20)^\circ\text{C} = 2459 \times 10^3 \text{ J/kg}\end{aligned}$$

Assuming wavy-laminar flow, the Reynolds number is determined from

$$\begin{aligned}\text{Re} = \text{Re}_{\text{vertical, wavy}} &= \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (2 \text{ m}) \times (0.607 \text{ W/m} \cdot ^\circ\text{C}) \times (30 - 20)^\circ\text{C}}{(0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s})(2459 \times 10^3 \text{ J/kg})} \left(\frac{9.81 \text{ m/s}^2}{(0.894 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.820} = 157.3\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined to be

$$\begin{aligned}h &= h_{\text{vertical, wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{157.3 \times (0.607 \text{ W/m} \cdot ^\circ\text{C})}{1.08(157.3)^{1.22} - 5.2} \left(\frac{9.81 \text{ m/s}^2}{(0.894 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 4302 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the tube is $A_s = \pi DL = \pi(0.04 \text{ m})(2 \text{ m}) = 0.2513 \text{ m}^2$. Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (4302 \text{ W/m}^2 \cdot ^\circ\text{C})(0.2513 \text{ m}^2)(30 - 20)^\circ\text{C} = \mathbf{10,811 \text{ W}}$$

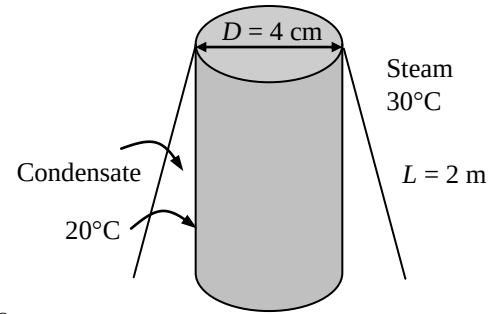
(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{10,811 \text{ J/s}}{2459 \times 10^3 \text{ J/kg}} = \mathbf{4.40 \times 10^{-3} \text{ kg/s}}$$

(c) Combining equations $\delta_L = k_l / h_l$ and $h = (4/3)h_L$, the thickness of the liquid film at the bottom of the tube is determined to be

$$\delta_L = \frac{4k_l}{3h} = \frac{4(0.607 \text{ W/m} \cdot ^\circ\text{C})}{3(4302 \text{ W/m}^2 \cdot ^\circ\text{C})} = 0.188 \times 10^{-3} \text{ m} = \mathbf{0.2 \text{ mm}}$$

Discussion The assumption of wavy laminar flow is verified since Reynolds number is between 30 and 1800. The assumption that the tube diameter is large relative to the thickness of the liquid film at the bottom of the tube is verified since the thickness of the liquid film is 0.2 mm, which is much smaller than the diameter of the tube (4 cm).



10-72 The rate of condensation and the heat transfer rate for a vertical pipe, with specified surface temperature, are to be determined.

Assumptions 1 Steady operating condition exists. 2 The surface has uniform temperature. 3 The pipe can be treated as a vertical plate. 4 The condensate flow is wavy-laminar over the entire tube (this assumption will be verified). 5 Nusselt's analysis can be used to determine the thickness of the condensate film layer. 6 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \text{ kJ/kg}$ (Table A-2) and $\rho_v = 0.5978 \text{ kg/m}^3$ (Table A-9). The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 90^\circ\text{C}$ are, from Table A-9,

$$\begin{aligned}\rho_l &= 965.3 \text{ kg/m}^3 & c_{pl} &= 4206 \text{ J/kg}\cdot\text{K} \\ \mu_l &= 0.315 \times 10^{-3} \text{ kg/m}\cdot\text{s} & k_l &= 0.675 \text{ W/m}\cdot\text{K} \\ \nu_l &= \mu_l / \rho_l = 0.326 \times 10^{-6} \text{ m}^2/\text{s}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 + 0.68(4206)(100 - 80) \\ &= 2314 \times 10^3 \text{ J/kg}\end{aligned}$$

Assuming wavy-laminar flow, the Reynolds number is determined from

$$\begin{aligned}\text{Re}_{\text{vert, wavy}} &= \left[4.81 + \frac{3.70Lk_l(T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70(1)(0.675)(100 - 80)}{(0.315 \times 10^{-3})(2314 \times 10^3)} \left(\frac{9.81}{(0.326 \times 10^{-6})^2} \right)^{1/3} \right]^{0.820} = 729.7\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined to be

$$\begin{aligned}h &= h_{\text{vert, wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} = \frac{(729.7)(0.675 \text{ W/m}\cdot\text{K})}{1.08(729.7)^{1.22} - 5.2} \left[\frac{9.81 \text{ m/s}^2}{(0.326 \times 10^{-6} \text{ m}^2/\text{s})^2} \right]^{1/3} \\ &= 6633 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Then the rate of heat transfer during this condensation process becomes

$$\begin{aligned}\dot{Q} &= \pi D L h (T_{\text{sat}} - T_s) \\ &= \pi(0.1 \text{ m})(1 \text{ m})(6633 \text{ W/m}^2 \cdot \text{K})(100 - 80) \text{ K} \\ &= 4.168 \times 10^4 \text{ W}\end{aligned}$$

The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{4.168 \times 10^4 \text{ W}}{2314 \times 10^3 \text{ J/kg}} = 0.018 \text{ kg/s}$$

Discussion Combining equations $\delta_L = k_l / h_L$ and $h = (4/3)h_L$, the thickness of the liquid film at the bottom of the tube is determined to be

$$\delta_L = \frac{4k_l}{3h} = \frac{4(0.675 \text{ W/m}\cdot\text{K})}{3(6633 \text{ W/m}^2 \cdot \text{K})} = 0.136 \text{ mm} \ll 100 \text{ mm}$$

Since $\delta_L \ll D$, the pipe can be treated as a vertical plate.

10-73 Saturated steam at a saturation temperature of $T_{\text{sat}} = 55^\circ\text{C}$ condenses on the outer surface of a vertical tube which is maintained at 45°C . The required tube length to condense steam at a rate of 10 kg/h is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal. 3 The vertical tube can be treated as a vertical plate. 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 55°C are $h_{fg} = 2371 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.1045 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (55 + 45) / 2 = 50^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 988.1 \text{ kg/m}^3 \\ \mu_l &= 0.547 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.554 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4181 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.644 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2371 \times 10^3 \text{ J/kg} + 0.68 \times 4181 \text{ J/kg} \cdot ^\circ\text{C}(55 - 45)^\circ\text{C} = 2399 \times 10^3 \text{ J/kg}\end{aligned}$$

The Reynolds number is determined from its definition to be

$$\text{Re} = \frac{4\dot{m}}{\pi \mu_l} = \frac{4(10/3600 \text{ kg/s})}{\pi(0.03 \text{ m})(0.547 \times 10^{-3} \text{ kg/m} \cdot \text{s})} = 215.5$$

which is between 30 and 1800. Therefore the condensate flow is wavy laminar, and the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{vertical, wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{215.5 \times (0.644 \text{ W/m} \cdot ^\circ\text{C})}{1.08(215.5)^{1.22} - 5.2} \left(\frac{9.81 \text{ m/s}^2}{(0.554 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 5644 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The rate of heat transfer during this condensation process is

$$\dot{Q} = \dot{m} h_{fg}^* = (10/3600 \text{ kg/s})(2399 \times 10^3 \text{ J/kg}) = 6,664 \text{ W}$$

Heat transfer can also be expressed as

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = h(\pi DL)(T_{\text{sat}} - T_s)$$

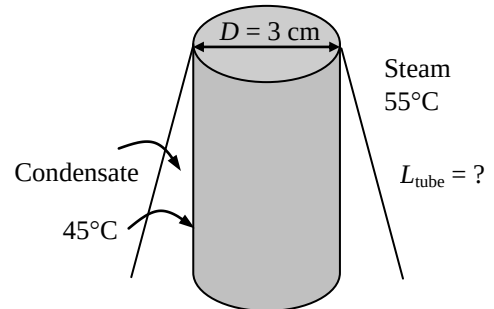
Then the required length of the tube becomes

$$L = \frac{\dot{Q}}{h(\pi D)(T_{\text{sat}} - T_s)} = \frac{6664 \text{ W}}{(5644 \text{ W/m}^2 \cdot ^\circ\text{C})\pi(0.03 \text{ m})(55 - 45)^\circ\text{C}} = \mathbf{1.21 \text{ m}}$$

Discussion Combining equations $\delta_L = k_l / h_l$ and $h = (4/3)h_L$, the thickness of the liquid film at the bottom of the tube is determined to be

$$\delta_L = \frac{4k_l}{3h} = \frac{4(0.644 \text{ W/m} \cdot ^\circ\text{C})}{3(5644 \text{ W/m}^2 \cdot ^\circ\text{C})} = 0.147 \times 10^{-3} \text{ m} = 0.15 \text{ mm}$$

The assumption that the tube diameter is large relative to the thickness of the liquid film at the bottom of the tube is verified since the thickness of the liquid film is 0.15 mm, which is much smaller than the diameter of the tube (3 cm). Also, the assumption of wavy laminar flow is verified since Reynolds number is between 30 and 1800.

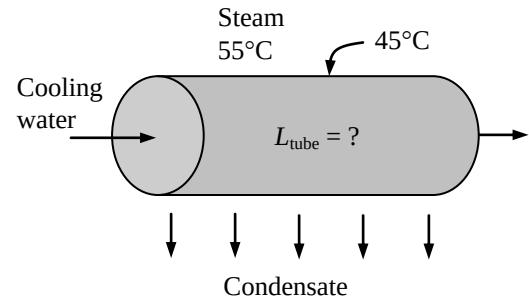


10-74 Saturated steam at a saturation temperature of $T_{\text{sat}} = 55^\circ\text{C}$ condenses on the outer surface of a horizontal tube which is maintained at 45°C . The required tube length to condense steam at a rate of 10 kg/h is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal.

Properties The properties of water at the saturation temperature of 55°C are $h_{fg} = 2371 \times 10^3\text{ J/kg}$ and $\rho_v = 0.1045\text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (55 + 45)/2 = 50^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 988.1\text{ kg/m}^3 \\ \mu_l &= 0.547 \times 10^{-3}\text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.554 \times 10^{-6}\text{ m}^2/\text{s} \\ c_{pl} &= 4181\text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.644\text{ W/m}\cdot^\circ\text{C}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2371 \times 10^3\text{ J/kg} + 0.68 \times 4181\text{ J/kg}\cdot^\circ\text{C}(55 - 45)^\circ\text{C} = 2399 \times 10^3\text{ J/kg}\end{aligned}$$

Noting that the tube is horizontal, the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81\text{ m/s}^2)(988.1\text{ kg/m}^3)(988.1 - 0.10\text{ kg/m}^3)(2399 \times 10^3\text{ J/kg})(0.644\text{ W/m}\cdot^\circ\text{C})^3}{(0.547 \times 10^{-3}\text{ kg/m}\cdot\text{s})(55 - 45)^\circ\text{C}(0.03\text{ m})} \right]^{1/4} \\ &= 10,135\text{ W/m}^2\cdot^\circ\text{C}\end{aligned}$$

The rate of heat transfer during this condensation process is

$$\dot{Q} = \dot{m}h_{fg}^* = (10/3600\text{ kg/s})(2399 \times 10^3\text{ J/kg}) = 6,664\text{ W}$$

Heat transfer can also be expressed as

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = h(\pi DL)(T_{\text{sat}} - T_s)$$

Then the required length of the tube becomes

$$L = \frac{\dot{Q}}{h(\pi D)(T_{\text{sat}} - T_s)} = \frac{6664\text{ W}}{(10,135\text{ W/m}^2\cdot^\circ\text{C})\pi(0.03\text{ m})(55 - 45)^\circ\text{C}} = \mathbf{0.70\text{ m}}$$

10-75 Saturated vapor condenses on the outer surface of a 1.5-m-long vertical tube at 60°C that is maintained with a surface temperature of 40°C. The rate of heat transfer to the tube and the required tube diameter to condense 12 kg/h of steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Isothermal tube surface. 3 The vertical tube can be treated as a vertical plate (this assumption will be verified). 4 The condensate flow is wavy-laminar over the entire plate (this assumption will be verified). 5 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 60°C are $h_{fg} = 2359 \times 10^3$ J/kg and $\rho_v = 0.1304$ kg/m³ (Table A-9). The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (60 + 40)/2 = 50^\circ\text{C}$ are (Table A-9)

$$\rho_l = 988.1 \text{ kg/m}^3$$

$$\mu_l = 0.547 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

$$\nu_l = \mu_l / \rho_l = 0.554 \times 10^{-6} \text{ m}^2/\text{s}$$

$$c_{pl} = 4181 \text{ J/kg} \cdot \text{K}$$

$$k_l = 0.644 \text{ W/m} \cdot \text{K}$$

Analysis The modified latent heat of vaporization is

$$h_{fg}^* = h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) = 2359 \times 10^3 \text{ J/kg} + 0.68(4181 \text{ J/kg} \cdot \text{K})(60 - 40) \text{ K} = 2.4159 \times 10^6 \text{ J/kg}$$

Assuming wavy-laminar flow, the Reynolds number is determined from

$$\begin{aligned} \text{Re} &= \text{Re}_{\text{vertical, wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{(3.70)(1.5 \text{ m})(0.644 \text{ W/m} \cdot \text{K})(60 - 40) \text{ K}}{(0.547 \times 10^{-3} \text{ kg/m} \cdot \text{s})(2.4159 \times 10^6 \text{ J/kg})} \left(\frac{9.81 \text{ m/s}^2}{(0.554 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.820} = 454.73 \end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined to be

$$\begin{aligned} h &= h_{\text{vertical, wavy}} = \frac{\text{Re } k_l}{1.08 \text{ Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{(454.73)(0.644 \text{ W/m} \cdot \text{K})}{1.08(454.73)^{1.22} - 5.2} \left[\frac{9.81 \text{ m/s}^2}{(0.554 \times 10^{-6} \text{ m}^2/\text{s})^2} \right]^{1/3} = 4937.5 \text{ W/m}^2 \cdot \text{K} \end{aligned}$$

The rate of heat transfer to the tube during this condensation process is

$$\dot{Q} = \dot{m} h_{fg}^* = (12/3600 \text{ kg/s})(2.4159 \times 10^6 \text{ J/kg}) = \mathbf{8053 \text{ W}}$$

Heat transfer can also be expressed as

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = h(\pi D L)(T_{\text{sat}} - T_s)$$

Then the required diameter of the tube becomes

$$D = \frac{\dot{Q}}{h(\pi L)(T_{\text{sat}} - T_s)} = \frac{8053 \text{ W}}{(4937.5 \text{ W/m}^2 \cdot \text{K})\pi(1.5 \text{ m})(60 - 40) \text{ K}} = \mathbf{0.0173 \text{ m}}$$

To verify the assumption that vertical tube can be treated as a vertical plate ($D \gg \delta$), calculate δ from

$$\delta = (4k_l)/3h = 4(0.664 \text{ W/m} \cdot \text{K})/(3)(4937.5 \text{ W/m}^2 \cdot \text{K}) = 1.79 \times 10^{-4} \text{ m} \ll D = 0.0173 \text{ m}$$

Thus our assumption of $D \gg \delta$ is verified.

Discussion With diameter known, the Reynolds number can also be verified to be wavy-laminar flow

$$\text{Re} = \frac{4\dot{m}}{\pi D \mu_l} = \frac{4(12/3600 \text{ kg/s})}{\pi(0.0173 \text{ m})(0.547 \times 10^{-3} \text{ kg/m} \cdot \text{s})} = 448$$

Note that the Re value obtained based on equation 10-27 will not exactly match the above calculated value of Re since equation 10-27 is based on experimental data and several approximations. However, the two values are very close (within 2%).

10-76 Saturated vapor condenses on the outer surface of a 1.5-m-long horizontal tube at 60°C that is maintained with a surface temperature of 40°C. The rate of heat transfer to the tube and the required tube diameter to condense 12 kg/h of steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Isothermal tube surface.

Properties The properties of water at the saturation temperature of 60°C are $h_{fg} = 2359 \times 10^3$ J/kg and $\rho_v = 0.1304$ kg/m³ (Table A-9). The properties of liquid water at the film temperature of $T_f = (T_{sat} + T_s)/2 = (60 + 40)/2 = 50^\circ\text{C}$ are (Table A-9)

$$\begin{aligned}\rho_l &= 988.1 \text{ kg/m}^3 \\ \mu_l &= 0.547 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.554 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4181 \text{ J/kg} \cdot \text{K} \\ k_l &= 0.644 \text{ W/m} \cdot \text{K}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{sat} - T_s) \\ &= 2359 \times 10^3 \text{ J/kg} + 0.68(4181 \text{ J/kg} \cdot \text{K})(60 - 40) \text{ K} = 2.4159 \times 10^6 \text{ J/kg}\end{aligned}$$

The rate of heat transfer to the tube during this condensation process is

$$\dot{Q} = \dot{m}h_{fg}^* = (12/3600 \text{ kg/s})(2.4159 \times 10^6 \text{ J/kg}) = \mathbf{8053 \text{ W}}$$

Heat transfer can also be expressed as

$$\dot{Q} = hA_s(T_{sat} - T_s) = h(\pi DL)(T_{sat} - T_s) \rightarrow h = \frac{\dot{Q}}{(\pi DL)(T_{sat} - T_s)}$$

Noting that the tube is horizontal, the condensation heat transfer coefficient is determined from

$$h = h_{horiz} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{sat} - T_s)D} \right]^{1/4} = \frac{\dot{Q}}{(\pi DL)(T_{sat} - T_s)}$$

Solving for the required tube diameter,

$$\begin{aligned}D &= \left\{ 0.729 \frac{(\pi L)(T_{sat} - T_s)}{\dot{Q}} \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{sat} - T_s)} \right]^{1/4} \right\}^{-4/3} \\ &= \left\{ 0.729 \frac{\pi(1.5 \text{ m})(60 - 40) \text{ K}}{8053 \text{ J/s}} \right. \\ &\quad \times \left. \left[\frac{(9.81 \text{ m/s}^2)(988.1 \text{ kg/m}^3)(988.1 - 0.1304) \text{ kg/m}^3(2.4159 \times 10^6 \text{ J/kg})(0.644 \text{ W/m} \cdot \text{K})^3}{(0.547 \times 10^{-3} \text{ kg/m} \cdot \text{s})(60 - 40) \text{ K}} \right]^{1/4} \right\}^{-4/3} \\ &= \mathbf{0.00694 \text{ m}}\end{aligned}$$

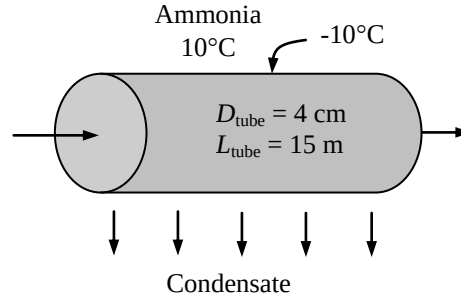
Discussion When placed vertically, the required tube diameter is about 2.5 times larger than that of a horizontal tube. Due to the higher heat transfer coefficient for a horizontal tube, in comparison with a vertical tube, the horizontal tube requires a smaller diameter for the same length to achieve the same rate of condensation.

10-77 Saturated ammonia vapor at a saturation temperature of $T_{\text{sat}} = 10^\circ\text{C}$ condenses on the outer surface of a horizontal tube which is maintained at -10°C . The rate of heat transfer from the ammonia and the rate of condensation of ammonia are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal.

Properties The properties of ammonia at the saturation temperature of 10°C are $h_{fg} = 1226 \times 10^3 \text{ J/kg}$ and $\rho_v = 4.870 \text{ kg/m}^3$ (Table A-11). The properties of liquid ammonia at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (10 + (-10))/2 = 0^\circ\text{C}$ are (Table A-11),

$$\begin{aligned}\rho_l &= 638.6 \text{ kg/m}^3 \\ \mu_l &= 1.896 \times 10^{-4} \text{ kg/m} \cdot \text{s} \\ \nu_l &= 0.2969 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4617 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.5390 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 1226 \times 10^3 \text{ J/kg} + 0.68 \times 4617 \text{ J/kg} \cdot ^\circ\text{C} [10 - (-10)]^\circ\text{C} = 1288 \times 10^3 \text{ J/kg}\end{aligned}$$

Noting that the tube is horizontal, the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(638.6 \text{ kg/m}^3)(638.6 - 4.870 \text{ kg/m}^3)(1288 \times 10^3 \text{ J/kg})(0.5390 \text{ W/m} \cdot ^\circ\text{C})^3}{(1.896 \times 10^{-4} \text{ kg/m} \cdot \text{s})[10 - (-10)]^\circ\text{C}(0.02 \text{ m})} \right]^{1/4} \\ &= 7390 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the tube is

$$A_s = \pi DL = \pi(0.04 \text{ m})(15 \text{ m}) = 1.885 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (7390 \text{ W/m}^2 \cdot ^\circ\text{C})(1.885 \text{ m}^2)[10 - (-10)]^\circ\text{C} = \mathbf{278,600 \text{ W}}$$

(b) The rate of condensation of ammonia is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{278,600 \text{ J/s}}{1288 \times 10^3 \text{ J/kg}} = \mathbf{0.216 \text{ kg/s}}$$

10-78 C&S Saturated R-134a vapor condenses on a horizontal ASTM A268 TP443 stainless steel tube at 133 kPa. Determine whether the tube would comply with the ASME Code for Process Piping if the condensation rate of R-134a on the tube surface is 3 g/s.

Assumptions 1 Steady state conditions. 2 The tube is isothermal.

Properties The saturation temperature of R-134a at 133 kPa is -20°C (Table A-10), and the properties of saturated R-134a are from Table A-10:

At the saturation temperature, $T_{\text{sat}} = -20^{\circ}\text{C}$,
 $h_{fg} = 213.0 \text{ kJ/kg}$, $\rho_v = 6.787 \text{ kg/m}^3$.

At the film temperature, $T_f = (-20 - 30)^{\circ}\text{C}/2 = -25^{\circ}\text{C}$,
 $c_{pl} = 1283 \text{ J/kg}\cdot\text{K}$, $k_l = 0.1047 \text{ W/m}\cdot\text{K}$, $\rho_l = 1374 \text{ kg/m}^3$, $\mu_l = 3.882 \times 10^{-4} \text{ kg/m}\cdot\text{s}$

Analysis The modified latent heat of vaporization is

$$h_{fg}^* = h_{fg} + 0.68 c_{pl}(T_{\text{sat}} - T_s) = 213000 + 0.68(1283)(-20 + 30) = 221724 \text{ J/kg}$$

For condensation on a horizontal tube, the heat transfer coefficient is

$$h = 0.729 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4}$$

$$h = 0.729 \left[\frac{(9.81)(1374)(1374 - 6.787)(221724)(0.1047)^3}{(3.882 \times 10^{-4})(-20 + 30)(0.015)} \right]^{1/4} = 2184 \text{ W/m}^2 \cdot \text{K}$$

The rate of heat transfer on the tube surface during the condensation process is

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (2184 \text{ W/m}^2 \cdot \text{K}) \pi (0.015 \text{ m})(0.50 \text{ m})(-20 + 30) \text{ K} = 514.6 \text{ W}$$

The highest rate of condensation on the tube surface, without cooling it below -30°C , is

$$\dot{m}_{\text{condensate}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{514.6 \text{ J/s}}{221724 \text{ J/kg}} = 0.00232 \text{ kg/s} = 2.32 \text{ g/s} < 3 \text{ g/s}$$

Discussion If the condensation rate of R-134a on the ASTM A268 TP443 tube surface is 3 g/s, the tube surface temperature would be below the minimum suitable temperature of -30°C . This would violate the ASME Code for Process Piping. Since the highest rate of condensation on the tube surface, without cooling it below -30°C , is 2.32 g/s.

10-79  A spherical tank containing cold fluid is causing condensation of moist air on the outer surface. The rate of moisture condensation is to be determined whether or not risk of electrical hazard exists.

Assumptions 1 Steady operating conditions exist. 2 Isothermal tank surface. 3 Film condensation occurs on the tank surface.

Properties The properties of water at the saturation temperature of 25°C are $h_{fg} = 2442 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.0231 \text{ kg/m}^3$ (Table A-9). The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (25 + 5)/2 = 15^\circ\text{C}$ are (Table A-9)

$$\begin{aligned}\rho_l &= 999.1 \text{ kg/m}^3 \\ \mu_l &= 1.138 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 1.139 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4185 \text{ J/kg} \cdot \text{K} \\ k_l &= 0.589 \text{ W/m} \cdot \text{K}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2442 \times 10^3 \text{ J/kg} + 0.68(4185 \text{ J/kg} \cdot \text{K})(25 - 5) \text{ K} = 2.4989 \times 10^6 \text{ J/kg}\end{aligned}$$

The film condensation heat transfer coefficient for a sphere is determined from

$$\begin{aligned}h &= h_{\text{horiz}} = 0.815 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.815 \left[\frac{(9.81 \text{ m/s}^2)(999.1 \text{ kg/m}^3)(999.1 - 0.0231) \text{ kg/m}^3 (2.4989 \times 10^6 \text{ J/kg})(0.589 \text{ W/m} \cdot \text{K})^3}{(1.138 \times 10^{-3} \text{ kg/m} \cdot \text{s})(25 - 5) \text{ K}(3 \text{ m})} \right]^{1/4} \\ &= 2384.1 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Thus, the rate of film condensation is

$$\begin{aligned}\dot{m} &= \frac{\dot{Q}}{h_{fg}^*} = \frac{h(\pi D^2)(T_{\text{sat}} - T_s)}{h_{fg}^*} \\ &= \frac{(2384.1 \text{ W/m}^2 \cdot \text{K})\pi(3 \text{ m})^2(25 - 5) \text{ K}}{2.4989 \times 10^6 \text{ J/kg}} = \mathbf{0.5395 \text{ kg/s}} > 0.5 \text{ kg/s}\end{aligned}$$

Discussion The rate of condensation from the tank surface is greater than the capability of the system in removing the condensate. Thus, there is a risk of excess condensate coming in contact with the high voltage device and cause electrical hazard. To prevent electrical hazard, the preventive system should be capable of removing more than 0.54 kg/s of condensate.

10-80 There is film condensation on the outer surfaces of N horizontal tubes arranged in a vertical tier. The value of N for which the average heat transfer coefficient for the entire tier be equal to half of the value for a single horizontal tube is to be determined.

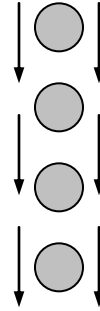
Assumptions Steady operating conditions exist.

Analysis The relation between the heat transfer coefficients for the two cases is given to be

$$h_{\text{horizontal, N tubes}} = \frac{h_{\text{horizontal, 1 tube}}}{N^{1/4}}$$

Therefore,

$$\frac{h_{\text{horizontal, N tubes}}}{h_{\text{horizontal, 1 tube}}} = \frac{1}{2} = \frac{1}{N^{1/4}} \longrightarrow N = \mathbf{16}$$



10-81 Saturated steam at a saturation temperature of $T_{\text{sat}} = 50^\circ\text{C}$ condenses on the outer surfaces of a tube bank with 33 tubes in each column maintained at 20°C . The average heat transfer coefficient and the rate of condensation of steam on the tubes are to be determined.

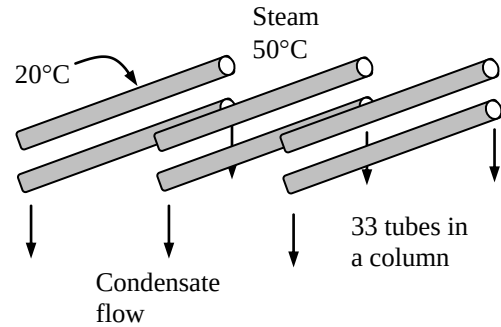
Assumptions 1 Steady operating conditions exist. 2 The tubes are isothermal.

Properties The properties of water at the saturation temperature of 50°C are $h_{fg} = 2383 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.0831 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (50 + 20) / 2 = 35^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 994.0 \text{ kg/m}^3 \\ \mu_l &= 0.720 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.724 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4178 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.623 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$

Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2383 \times 10^3 \text{ J/kg} + 0.68 \times 4178 \text{ J/kg} \cdot ^\circ\text{C}(50 - 20)^\circ\text{C} \\ &= 2468 \times 10^3 \text{ J/kg}\end{aligned}$$



The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(994 \text{ kg/m}^3)(994 - 0.08 \text{ kg/m}^3)(2468 \times 10^3 \text{ J/kg})(0.623 \text{ W/m} \cdot ^\circ\text{C})^3}{(0.720 \times 10^{-3} \text{ kg/m} \cdot \text{s})(50 - 20)^\circ\text{C}(0.015 \text{ m})} \right]^{1/4} \\ &= 8425 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

Then the average heat transfer coefficient for a 33-tube high vertical tier becomes

$$h_{\text{horiz, N tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{33^{1/4}} (8425 \text{ W/m}^2 \cdot ^\circ\text{C}) = 3515 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The surface area for all 33 tubes per unit length is

$$A_s = N_{\text{total}} \pi D L = 33 \pi (0.015 \text{ m})(1 \text{ m}) = 1.555 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (3515 \text{ W/m}^2 \cdot ^\circ\text{C})(1.555 \text{ m}^2)(50 - 20)^\circ\text{C} = 164,000 \text{ W}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{164,000 \text{ J/s}}{2468 \times 10^3 \text{ J/kg}} = 0.0664 \text{ kg/s}$$

10-82 Saturated steam at a pressure of 4.25 kPa and thus at a saturation temperature of $T_{\text{sat}} = 30^\circ\text{C}$ (Table A-9) condenses on the outer surfaces of 144 horizontal tubes arranged in a 12×12 square array maintained at 20°C by circulating cooling water. The rate of heat transfer to the cooling water and the rate of condensation of steam on the tubes are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tubes are isothermal.

Properties The properties of water at the saturation temperature of 30°C are $h_{fg} = 2431 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.03 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (30 + 20) / 2 = 25^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 997.0 \text{ kg/m}^3 \\ \mu_l &= 0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.894 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4180 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.607 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$

Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2431 \times 10^3 \text{ J/kg} + 0.68 \times 4180 \text{ J/kg} \cdot ^\circ\text{C}(30 - 20)^\circ\text{C} = 2,459 \times 10^3 \text{ J/kg}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(997 \text{ kg/m}^3)(997 - 0.03 \text{ kg/m}^3)(2459 \times 10^3 \text{ J/kg})(0.607 \text{ W/m} \cdot ^\circ\text{C})^3}{(0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s})(30 - 20)^\circ\text{C}(0.03 \text{ m})} \right]^{1/4} \\ &= 8674 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

Then the average heat transfer coefficient for a 10-pipe high vertical tier becomes

$$h_{\text{horiz, N tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{12^{1/4}} (8674 \text{ W/m}^2 \cdot ^\circ\text{C}) = 4661 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The surface area for all 144 tubes is

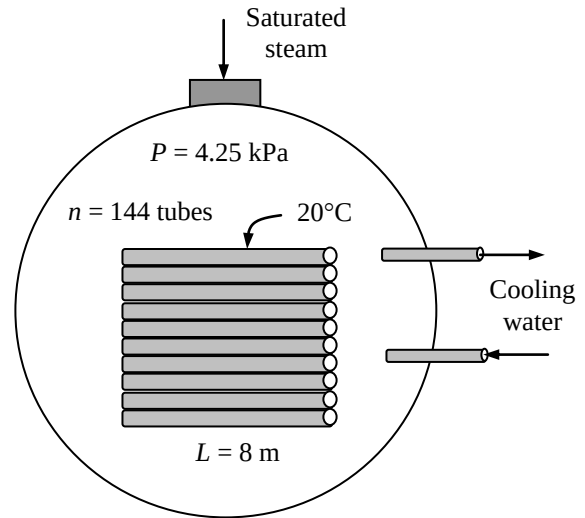
$$A_s = N_{\text{total}} \pi D L = 144 \pi (0.03 \text{ m})(8 \text{ m}) = 108.6 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (4661 \text{ W/m}^2 \cdot ^\circ\text{C})(108.6 \text{ m}^2)(30 - 20)^\circ\text{C} = 5,060,000 \text{ W} = \mathbf{5060 \text{ kW}}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{5,060,000 \text{ J/s}}{2459 \times 10^3 \text{ J/kg}} = \mathbf{2.06 \text{ kg/s}}$$





10-83 Prob. 10-82 is reconsidered. The effect of the condenser pressure on the rate of heat transfer and the rate of condensation of the steam is to be investigated.

Analysis The problem is solved using EES, and the solution is given below.

"GIVEN"

P_sat=4.25 [kPa]
 n_tube=144
 N=12
 L=8 [m]
 D=0.03 [m]
 T_s=20 [C]

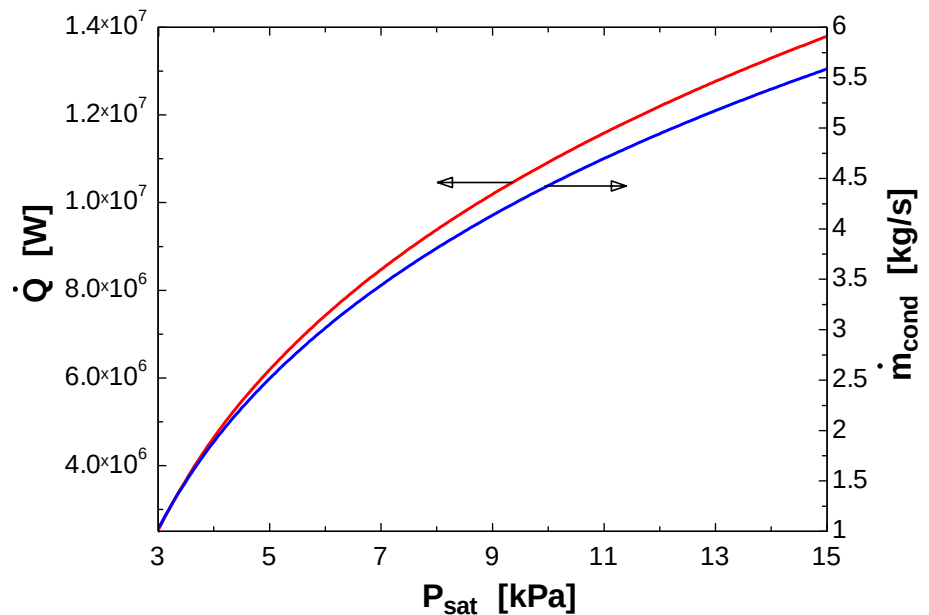
"PROPERTIES"

Fluid\$='steam_IAPWS'
 T_sat=temperature(Fluid\$, P=P_sat, x=1)
 T_f=1/2*(T_sat+T_s)
 h_f=enthalpy(Fluid\$, T=T_sat, x=0)
 h_g=enthalpy(Fluid\$, T=T_sat, x=1)
 h_fg=(h_g-h_f)*Convert(kJ/kg, J/kg)
 rho_v=density(Fluid\$, T=T_sat, x=1)
 rho_l=density(Fluid\$, T=T_f, x=0)
 mu_l=Viscosity(Fluid\$, T=T_f, x=0)
 nu_l=mu_l/rho_l
 c_l=CP(Fluid\$, T=T_f, x=0)*Convert(kJ/kg-C, J/kg-C)
 k_l=Conductivity(Fluid\$, T=T_f, P=P_sat+1)
 g=9.81 [m/s^2]

"ANALYSIS"

h_fg_star=h_fg+0.68*c_l*(T_sat-T_s)
 $h_{1tube}=0.729*((g*\rho_l*(\rho_l-\rho_v)*h_{fg_star}*k_l^3)/(\mu_l*(T_{sat}-T_s)*D))^{0.25}$
 $h=1/N^{0.25}*h_{1tube}$
 $\dot{Q}_{dot}=h*A*(T_{sat}-T_s)$
 $A=n_{tube}*pi*D*L$
 $\dot{m}_{dot_cond}=\dot{Q}_{dot}/h_{fg_star}$

P _{sat} [kPa]	$\dot{Q}$ [W]	$\dot{m}_{cond}$ [kg/s]
3	2523819	1.028
4	4642913	1.889
5	6185636	2.515
6	7426047	3.017
7	8473057	3.441
8	9383654	3.809
9	10192213	4.136
10	10921191	4.43
11	11586164	4.698
12	12198509	4.945
13	12766595	5.174
14	13296988	5.387
15	13794833	5.587



10-84 Saturated steam at a saturation temperature of $T_{\text{sat}} = 35^\circ\text{C}$ condenses on the outer surfaces of horizontal pipes which are maintained at 15°C by circulating cooling water. The rate of heat transfer to the cooling water and the rate of condensation per unit length of a single horizontal pipe are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The pipe is isothermal. 3 There is no interference between the pipes (no drip of the condensate from one tube to another).

Properties The properties of water at the saturation temperature of 35°C are $h_{fg} = 2419 \text{ kJ/kg}$ and $\rho_v = 0.0397 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (35 + 15)/2 = 25^\circ\text{C}$ are (Table A-9),

$$\rho_l = 997.0 \text{ kg/m}^3$$

$$\mu_l = 0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

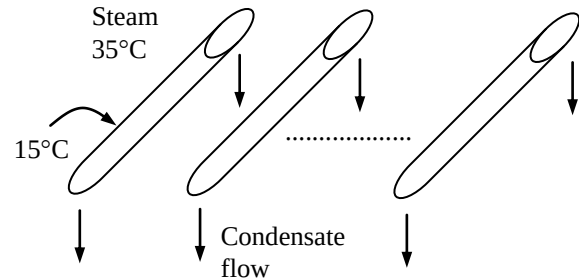
$$\nu_l = \mu_l / \rho_l = 0.894 \times 10^{-6} \text{ m}^2/\text{s}$$

$$c_{pl} = 4180 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_l = 0.607 \text{ W/m} \cdot ^\circ\text{C}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2419 \text{ kJ/kg} + 0.68 \times (4.180 \text{ kJ/kg} \cdot ^\circ\text{C})(35 - 15)^\circ\text{C} \\ &= 2476 \text{ kJ/kg} \end{aligned}$$



Noting that we have condensation on a horizontal tube, the heat transfer coefficient is determined from

$$\begin{aligned} h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(997.0 \text{ kg/m}^3)(997.0 - 0.0397 \text{ kg/m}^3)(2476 \times 10^3 \text{ J/kg})(0.607 \text{ W/m} \cdot ^\circ\text{C})^3}{(0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s})(35 - 15)^\circ\text{C}(0.025 \text{ m})} \right]^{1/4} \\ &= 8088 \text{ W/m}^2 \cdot ^\circ\text{C} \end{aligned}$$

The heat transfer surface area of the tube per unit length is

$$A_s = \pi DL = \pi(0.02 \text{ m})(1 \text{ m}) = 0.06283 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (8088 \text{ W/m}^2 \cdot ^\circ\text{C})(0.06283 \text{ m}^2)(35 - 15)^\circ\text{C} = \mathbf{10,160 \text{ W}}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{10.16 \text{ kJ/s}}{2476 \text{ kJ/kg}} = 0.004105 \text{ kg/s} = \mathbf{14.8 \text{ kg/h}}$$

10-85 Saturated steam at a saturation temperature of $T_{\text{sat}} = 35^\circ\text{C}$ condenses on the outer surfaces of 32 horizontal pipes which are maintained at 15°C by circulating cooling water and arranged in a rectangular array of 4 pipes high and 5 pipes wide. The rate of heat transfer to the cooling water and the rate of condensation per unit length of the pipes are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The pipes are isothermal.

Properties The properties of water at the saturation temperature of 35°C are $h_{fg} = 2419 \text{ kJ/kg}$ and $\rho_v = 0.0397 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (35 + 15) / 2 = 25^\circ\text{C}$ are (Table A-9),

$$\rho_l = 997.0 \text{ kg/m}^3$$

$$\mu_l = 0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

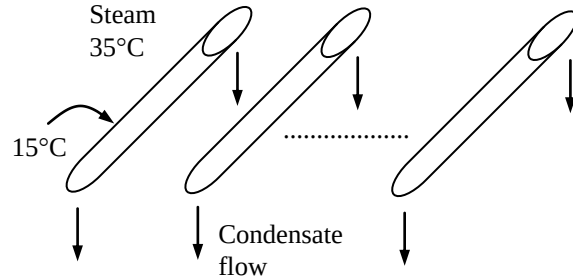
$$\nu_l = \mu_l / \rho_l = 0.894 \times 10^{-6} \text{ m}^2/\text{s}$$

$$c_{pl} = 4180 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_l = 0.607 \text{ W/m} \cdot ^\circ\text{C}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2419 \text{ kJ/kg} + 0.68 \times (4.180 \text{ kJ/kg} \cdot ^\circ\text{C})(35 - 15)^\circ\text{C} \\ &= 2476 \text{ kJ/kg} \end{aligned}$$



Noting that we have condensation on a horizontal tube, the heat transfer coefficient is determined from

$$\begin{aligned} h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(997.0 \text{ kg/m}^3)(997.0 - 0.0397 \text{ kg/m}^3)(2476 \times 10^3 \text{ J/kg})(0.607 \text{ W/m} \cdot ^\circ\text{C})^3}{(0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s})(35 - 15)^\circ\text{C}(0.02 \text{ m})} \right]^{1/4} \\ &= 8088 \text{ W/m}^2 \cdot ^\circ\text{C} \end{aligned}$$

Then the average heat transfer coefficient for a 4-pipe high vertical tier becomes

$$h_{\text{horiz, N tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{4^{1/4}} (8088 \text{ W/m}^2 \cdot ^\circ\text{C}) = 5719 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The surface area for all 32 pipes per unit length of the pipes is

$$A_s = N_{\text{total}} \pi D L = 32 \pi (0.02 \text{ m})(1 \text{ m}) = 2.011 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (5719 \text{ W/m}^2 \cdot ^\circ\text{C})(2.011 \text{ m}^2)(35 - 15)^\circ\text{C} = 229,975 \text{ W}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{229.975 \text{ kJ/s}}{2476 \text{ kJ/kg}} = 0.0929 \text{ kg/s} = 334 \text{ kg/h}$$

10-86 Saturated steam at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ condenses on the outer surfaces of a tube bank maintained at 80°C . The rate of condensation of steam on the tubes are to be determined.

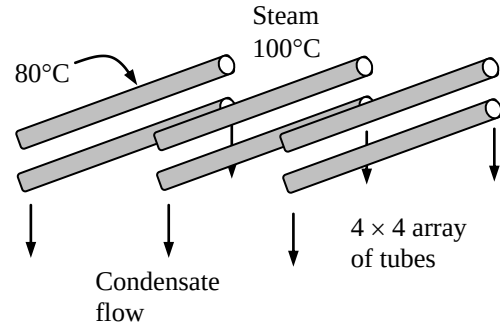
Assumptions 1 Steady operating conditions exist. 2 The tubes are isothermal.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.5978 \text{ kg/m}^3$ (Table A-9). The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (100 + 80)/2 = 90^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 965.3 \text{ kg/m}^3 \\ \mu_l &= 0.315 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ c_{pl} &= 4206 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.675 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$

Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4206 \text{ J/kg} \cdot ^\circ\text{C}(100 - 80)^\circ\text{C} \\ &= 2314 \times 10^3 \text{ J/kg}\end{aligned}$$



The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(965.3 \text{ kg/m}^3)(965.3 - 0.5978 \text{ kg/m}^3)(2314 \times 10^3 \text{ J/kg})(0.675 \text{ W/m} \cdot ^\circ\text{C})^3}{(0.315 \times 10^{-3} \text{ kg/m} \cdot \text{s})(100 - 80)^\circ\text{C}(0.05 \text{ m})} \right]^{1/4} \\ &= 8736 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

Then the average heat transfer coefficient for a 4-pipe high vertical tier becomes

$$h_{\text{horiz, N tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{4^{1/4}} (8736 \text{ W/m}^2 \cdot ^\circ\text{C}) = 6177 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The surface area for all 16 tubes is

$$A_s = N_{\text{total}} \pi D L = 16 \pi (0.05 \text{ m})(2 \text{ m}) = 5.027 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (6177 \text{ W/m}^2 \cdot ^\circ\text{C})(5.027 \text{ m}^2)(100 - 80)^\circ\text{C} = 621,000 \text{ W}$$

The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{621,000 \text{ J/s}}{2314 \times 10^3 \text{ J/kg}} = 0.2684 \text{ kg/s} = \mathbf{966 \text{ kg/h}}$$

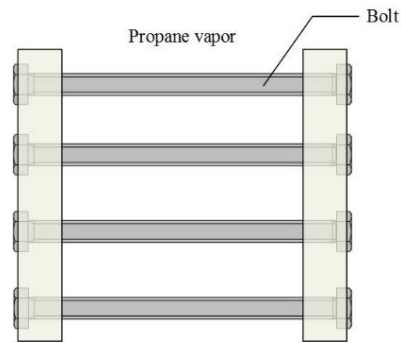
10-87 C&S Saturated propane vapor condenses on four ASTM A437 B4B stainless steel bolts that are arranged in a vertical tier. The condensation occurs at 344 kPa. The highest rate of condensation of propane on the bolts, without cooling the them below the minimum suitable temperature set by the ASME Code for Process Piping, is to be determined.

Assumptions 1 Steady state conditions. 2 The bolts are isothermal.

Properties The saturation temperature of saturated propane at 344 kPa is -10°C (Table A-12), and the properties of saturated propane are from Table A-12:

At the saturation temperature, $T_{\text{sat}} = -10^{\circ}\text{C}$,
 $h_{fg} = 387.8 \text{ kJ/kg}$, $\rho_v = 7.635 \text{ kg/m}^3$.

At the film temperature, $T_f = (-10 - 30)^{\circ}\text{C}/2 = -20^{\circ}\text{C}$,
 $c_{pl} = 2368 \text{ J/kg}\cdot\text{K}$, $k_l = 0.1163 \text{ W/m}\cdot\text{K}$, $\rho_l = 554.7 \text{ kg/m}^3$, $\mu_l = 1.551 \times 10^{-4} \text{ kg/m}\cdot\text{s}$



Analysis The modified latent heat of vaporization is

$$h_{fg}^* = h_{fg} + 0.68 c_{pl}(T_{\text{sat}} - T_s) = 387800 + 0.68(2368)(-10 + 30) = 420005 \text{ J/kg}$$

For average condensation heat transfer coefficient for the horizontal bolts arranged in a vertical tier of four is

$$h_N = \frac{0.729}{N^{1/4}} \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4}$$

$$h_N = \frac{0.729}{4^{1/4}} \left[\frac{(9.81)(554.7)(554.7 - 7.635)(420005)(0.1163)^3}{(1.551 \times 10^{-4})(-10 + 30)(0.013)} \right]^{1/4} = 1362 \text{ W/m}^2 \cdot \text{K}$$

The rate of heat transfer on the bolts during the condensation process is

$$\dot{Q} = h_N A_s (T_{\text{sat}} - T_s) = (1362 \text{ W/m}^2 \cdot \text{K}) \pi (4 \times 0.013 \text{ m} \times 0.15 \text{ m}) (-10 + 30) \text{ K} = 667.5 \text{ W}$$

where $A_s = N \pi D L$

The highest rate of condensation on the bolts, without cooling them below -30°C , is

$$\dot{m}_{\text{condensate}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{667.5 \text{ J/s}}{420005 \text{ J/kg}} = 0.00159 \text{ kg/s} = 1.59 \text{ g/s}$$

Discussion The highest condensation rate of propane that can occur on the ASTM A437 B4B bolts is 1.59 g/s. If the condensation rate is above that value, it would increase the heat transfer rate on the bolts. This means that the surface temperature of the bolts would be at below the minimum suitable temperature of -30°C set by the ASME Code for Process Piping.

10-88 Saturated water vapor at a pressure of 12.4 kPa condenses on a rectangular array of 100 horizontal tubes at 30°C. The condensation rate per unit length of is to be determined.

Assumptions 1 Steady operating condition exists. 2 The tube surfaces are isothermal.

Properties The properties of water at the saturation temperature of 50°C corresponding to 12.4 kPa are $h_{fg} = 2383$ kJ/kg and $\rho_v = 0.0831$ kg/m³ (Table A-9). The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 40^\circ\text{C}$ are, from Table A-9,

$$\begin{aligned}\rho_l &= 992.1 \text{ kg/m}^3 & c_{pl} &= 4179 \text{ J/kg}\cdot\text{K} \\ \mu_l &= 0.653 \times 10^{-3} \text{ kg/m}\cdot\text{s} & k_l &= 0.631 \text{ W/m}\cdot\text{K}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2383 \times 10^3 + 0.68(4179)(50 - 30) \\ &= 2440 \times 10^3 \text{ J/kg}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h_{\text{horiz, 1 tube}} &= 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81)(992.1)(992.1 - 0.0831)(2440 \times 10^3)(0.631)^3}{(0.653 \times 10^{-3})(50 - 30)(0.008)} \right]^{1/4} \\ &= 11,250 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Then the average heat transfer coefficient for a 5-tube high vertical tier becomes

$$h = h_{\text{horiz, } N \text{ tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{5^{1/4}} (11,250 \text{ W/m}^2 \cdot \text{K}) = 7523 \text{ W/m}^2 \cdot \text{K}$$

The rate of heat transfer per unit length during this condensation process becomes

$$\begin{aligned}\dot{Q}/L &= N_{\text{total}} \pi D h (T_{\text{sat}} - T_s) \\ &= (100) \pi (0.008 \text{ m}) (7523 \text{ W/m}^2 \cdot \text{K}) (50 - 30) \text{ K} \\ &= 3.781 \times 10^5 \text{ W/m}\end{aligned}$$

The rate of condensation per unit length is

$$\frac{\dot{m}_{\text{condensation}}}{L} = \frac{\dot{Q}/L}{h_{fg}^*} = \frac{3.781 \times 10^5 \text{ W/m}}{2440 \times 10^3 \text{ J/kg}} = \mathbf{0.155 \text{ kg/s} \cdot \text{m}}$$

Discussion Therefore, water vapor condenses at a rate of 155 g/s per meter length of the tubes.

10-89 Saturated water vapor at a pressure of 12.4 kPa condenses on an array of 100 horizontal tubes at 30°C. The condensation rates for (a) a rectangular array of 5 tubes high and 20 tubes wide and (b) a square array of 10 tubes high and 10 tubes wide are to be determined.

Assumptions 1 Steady operating condition exists. 2 The tube surfaces are isothermal.

Properties The properties of water at the saturation temperature of 50°C corresponding to 12.4 kPa are $h_{fg} = 2383$ kJ/kg and $\rho_v = 0.0831$ kg/m³ (Table A-9). The properties of liquid water at the film temperature of $T_f = (T_{sat} + T_s)/2 = 40^\circ\text{C}$ are, from Table A-9,

$$\begin{aligned}\rho_l &= 992.1 \text{ kg/m}^3 & c_{pl} &= 4179 \text{ J/kg}\cdot\text{K} \\ \mu_l &= 0.653 \times 10^{-3} \text{ kg/m}\cdot\text{s} & k_l &= 0.631 \text{ W/m}\cdot\text{K}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{sat} - T_s) \\ &= 2383 \times 10^3 + 0.68(4179)(50 - 30) \\ &= 2440 \times 10^3 \text{ J/kg}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h_{\text{horiz, 1 tube}} &= 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{sat} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81)(992.1)(992.1 - 0.0831)(2440 \times 10^3)(0.631)^3}{(0.653 \times 10^{-3})(50 - 30)(0.008)} \right]^{1/4} \\ &= 11,250 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

(a) For a rectangular array, the average heat transfer coefficient for a 5-tube high vertical tier becomes

$$h = h_{\text{horiz, } N \text{ tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{5^{1/4}} (11,250 \text{ W/m}^2 \cdot \text{K}) = 7523 \text{ W/m}^2 \cdot \text{K}$$

The rate of heat transfer during this condensation process becomes

$$\dot{Q} = N_{\text{total}} \pi D L h (T_{sat} - T_s) = (100) \pi (0.008 \text{ m})(1 \text{ m})(7523 \text{ W/m}^2 \cdot \text{K})(50 - 30) \text{ K} = 3.781 \times 10^5 \text{ W}$$

The rate of condensation is

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{3.781 \times 10^5 \text{ W}}{2440 \times 10^3 \text{ J/kg}} = \mathbf{0.155 \text{ kg/s}} \quad (\text{rectangular array})$$

(b) For a square array, the average heat transfer coefficient for a 10-tube high vertical tier becomes

$$h = h_{\text{horiz, } N \text{ tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{10^{1/4}} (11,250 \text{ W/m}^2 \cdot \text{K}) = 6326 \text{ W/m}^2 \cdot \text{K}$$

The rate of heat transfer during this condensation process becomes

$$\dot{Q} = N_{\text{total}} \pi D L h (T_{sat} - T_s) = (100) \pi (0.008 \text{ m})(1 \text{ m})(6326 \text{ W/m}^2 \cdot \text{K})(50 - 30) \text{ K} = 3.180 \times 10^5 \text{ W}$$

The rate of condensation is

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{3.180 \times 10^5 \text{ W}}{2440 \times 10^3 \text{ J/kg}} = \mathbf{0.130 \text{ kg/s}} \quad (\text{square array})$$

Discussion The condensation rate of the rectangular array tube bank is about 20% higher than that of the square array tube bank:

$$\frac{\dot{m}_{\text{condensation}} (\text{rectangular})}{\dot{m}_{\text{condensation}} (\text{square})} = \frac{0.155 \text{ kg/s}}{0.130 \text{ kg/s}} = 1.19$$

10-90 Saturated refrigerant-134a vapor is condensed as it is flowing through a tube. With a given vapor flow rate at the entrance, the flow rate of the vapor at the exit is to be determined.

Assumptions 1 Steady operating condition exists. 2 The tube surfaces are isothermal.

Properties The properties of refrigerant-134a at the saturation temperature of 35°C corresponding to 888 kPa are $h_{fg} = 168.2$ kJ/kg, $\rho_v = 43.41$ kg/m³, and $\mu_v = 1.327 \times 10^{-5}$ kg/m·s (Table A-10). The properties of liquid refrigerant-134a at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 25^\circ\text{C}$ are, from Table A-10,

$$\begin{aligned}\rho_l &= 1207 \text{ kg/m}^3 & c_{pl} &= 1427 \text{ J/kg}\cdot\text{K} \\ \mu_l &= 2.012 \times 10^{-4} \text{ kg/m}\cdot\text{s} & k_l &= 0.0833 \text{ W/m}\cdot\text{K}\end{aligned}$$

Analysis The modified latent heat of vaporization for film condensation inside horizontal tube is

$$\begin{aligned}h_{fg}^* &= h_{fg} + \frac{3}{8}c_{pl}(T_{\text{sat}} - T_s) \\ &= 168.2 \times 10^3 + \frac{3}{8}(1427)(35 - 15) \\ &= 178.9 \times 10^3 \text{ J/kg}\end{aligned}$$

The Reynolds number associated with film condensation inside a horizontal tube is

$$\text{Re}_{\text{vapor}} = \left(\frac{\rho_v V_v D}{\mu_v} \right)_{\text{inlet}} = \frac{4\dot{m}_{v,\text{inlet}}}{\pi D \mu_v} = \frac{4(0.003 \text{ kg/s})}{\pi(0.012 \text{ m})(1.327 \times 10^{-5} \text{ kg/m}\cdot\text{s})} = 24,000 < 35,000$$

Hence, the heat transfer coefficient for film condensation inside a horizontal tube can be determined using

$$\begin{aligned}h &= h_{\text{internal}} = 0.555 \left[\frac{g\rho_l(\rho_l - \rho_v)k_l^3 h_{fg}^*}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.555 \left[\frac{(9.81)(1207)(1207 - 43.41)(0.0833)^3(178.9 \times 10^3)}{(2.012 \times 10^{-4})(35 - 15)(0.012)} \right]^{1/4} \\ &= 1293 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

The rate of heat transfer during this condensation process becomes

$$\begin{aligned}\dot{Q} &= \pi D L h (T_{\text{sat}} - T_s) \\ &= \pi(0.012 \text{ m})(0.25 \text{ m})(1293 \text{ W/m}^2 \cdot \text{K})(35 - 15) \text{ K} \\ &= 243.7 \text{ W}\end{aligned}$$

Then, the rate of condensation can be calculated as

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{243.7 \text{ W}}{178.9 \times 10^3 \text{ J/kg}} = 0.00136 \text{ kg/s}$$

Applying the conservation of mass, the flow rate of vapor leaving the tube can be determined as

$$\begin{aligned}\dot{m}_{v,\text{inlet}} &= \dot{m}_{v,\text{outlet}} + \dot{m}_{\text{condensation}} \quad \rightarrow \quad \dot{m}_{v,\text{outlet}} = \dot{m}_{v,\text{inlet}} - \dot{m}_{\text{condensation}} \\ \dot{m}_{v,\text{outlet}} &= 0.003 \text{ kg/s} - 0.00136 \text{ kg/s} = \mathbf{0.00164 \text{ kg/s}}\end{aligned}$$

Discussion About 45% of the refrigerant-134a vapor that entered the tube is condensed inside it.

10-91 C&S Saturated ammonia vapor condenses inside a horizontal ASTM A268 TP443 stainless steel tube at 190 kPa. Determine whether the tube would comply with the minimum suitable temperature of the ASME Code for Process Piping, if the ammonia vapor exits the tube at a flow rate of 0.5 g/s.

Assumptions 1 Steady state conditions. 2 The tube is isothermal.

Properties The saturation temperature of ammonia at 190 kPa is -20°C (Table A-11), and the properties of saturated ammonia are from Table A-11:

At the saturation temperature, $T_{\text{sat}} = -20^{\circ}\text{C}$,
 $h_{fg} = 1329 \text{ kJ/kg}$, $\rho_v = 1.603 \text{ kg/m}^3$, $\mu_v = 8.669 \times 10^{-6} \text{ kg/m}\cdot\text{s}$

At the film temperature, $T_f = (-20 - 30)^{\circ}\text{C}/2 = -25^{\circ}\text{C}$,
 $c_{pl} = 4489 \text{ J/kg}\cdot\text{K}$, $k_l = 0.5968 \text{ W/m}\cdot\text{K}$, $\rho_l = 671.5 \text{ kg/m}^3$, $\mu_l = 2.492 \times 10^{-4} \text{ kg/m}\cdot\text{s}$

Analysis The modified latent heat of vaporization for film condensation inside a horizontal tube is

$$h_{fg}^* = h_{fg} + \frac{3}{8} c_{pl} (T_{\text{sat}} - T_s) = 1329000 + \frac{3}{8} (4489) (-20 + 30) = 1345834 \text{ J/kg}$$

The vapor Reynolds number associated with the film condensation inside the tube is

$$\text{Re}_{\text{vapor}} = \left(\frac{\rho_v V_v D}{\mu_v} \right) = \frac{4 \dot{m}_{v,\text{in}}}{\pi D \mu_v} = \frac{4(0.005 \text{ kg/s})}{\pi (0.025 \text{ m}) (8.669 \times 10^{-6} \text{ kg/m}\cdot\text{s})} = 29375 < 35000$$

Thus, the heat transfer coefficient for film condensation inside the horizontal tube can be determine as

$$h = 0.555 \left[\frac{g \rho_l (\rho_l - \rho_v) k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} h_{fg}^* \right]^{1/4}$$

$$h = 0.555 \left[\frac{(9.81)(671.5)(671.5 - 1.603)(0.5968)^3}{(2.492 \times 10^{-4})(-20 + 30)(0.025)} (1345834) \right]^{1/4} = 6621.7 \text{ W/m}^2 \cdot \text{K}$$

The rate of heat transfer during the condensation process is

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (6622 \text{ W/m}^2 \cdot \text{K}) \pi (0.025 \text{ m}) (1 \text{ m}) (-20 + 30) \text{ K} = 5201 \text{ W}$$

The highest rate of condensation inside the tube that can occur without cooling the tube wall below -30°C is

$$\dot{m}_{\text{condensate}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{5201 \text{ J/s}}{1345834 \text{ J/kg}} = 0.00386 \text{ kg/s} = 3.86 \text{ g/s}$$

So, the minimum flow rate of ammonia vapor at the tube exit is

$$\dot{m}_{v,\text{out}} = \dot{m}_{v,\text{in}} - \dot{m}_{\text{condensate}} = 5 \text{ g/s} - 3.86 \text{ g/s} = 1.14 \text{ g/s} > 0.5 \text{ g/s}$$

Discussion If the ammonia vapor flow rate at the tube exit is 0.5 g/s, this means that the rate of condensation of ammonia inside the tube is 4.5 g/s. This rate of condensation is greater than 3.86 g/s, which occurs at a tube wall temperature of -30°C . For a condensation rate of ammonia greater than 3.86 g/s to occur inside the horizontal tube, the tube wall temperature would need to be below -30°C , thus violating the ASME Code for Process Piping. For the ASTM A268 TP443 tube to comply with the ASME code, the flow rate of ammonia vapor at the tube exit needs to be 1.14 g/s or higher.

10-92 Saturated ammonia vapor is condensed as it flows through a tube. With a given vapor flow rate at the exit, the flow rate of the vapor at the inlet is to be determined.

Assumptions 1 Steady operating condition exists. 2 The tube surfaces are isothermal. 3 The Reynolds number of the vapor at the inlet is less than 35,000 (this assumption will be verified).

Properties The properties of ammonia at the saturation temperature of 25°C corresponding to 1003 kPa are $h_{fg} = 1166 \text{ kJ/kg}$, $\rho_v = 7.809 \text{ kg/m}^3$, and $\mu_v = 1.037 \times 10^{-5} \text{ kg/m}\cdot\text{s}$ (Table A-11). The properties of liquid ammonia at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 15^\circ\text{C}$ are, from Table A-11,

$$\rho_l = 617.5 \text{ kg/m}^3$$

$$c_{pl} = 4709 \text{ J/kg}\cdot\text{K}$$

$$\mu_l = 1.606 \times 10^{-4} \text{ kg/m}\cdot\text{s}$$

$$k_l = 0.5042 \text{ W/m}\cdot\text{K}$$

Analysis The modified latent heat of vaporization for film condensation inside horizontal tube is

$$\begin{aligned} h_{fg}^* &= h_{fg} + \frac{3}{8} c_{pl} (T_{\text{sat}} - T_s) \\ &= 1166 \times 10^3 + \frac{3}{8} (4709)(25 - 5) \\ &= 1201 \times 10^3 \text{ J/kg} \end{aligned}$$

Assuming $\text{Re}_{\text{vapor}} < 35,000$ and the heat transfer coefficient for film condensation inside a horizontal tube can be determined using

$$\begin{aligned} h &= h_{\text{internal}} = 0.555 \left[\frac{g \rho_l (\rho_l - \rho_v) k_l^3 h_{fg}^*}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4} \\ &= 0.555 \left[\frac{(9.81)(617.5)(617.5 - 7.809)(0.5042)^3 (1201 \times 10^3)}{(1.606 \times 10^{-4})(25 - 5)(0.025)} \right]^{1/4} \\ &= 5091 \text{ W/m}^2 \cdot \text{K} \end{aligned}$$

The rate of heat transfer during this condensation process becomes

$$\begin{aligned} \dot{Q} &= \pi D L h (T_{\text{sat}} - T_s) \\ &= \pi (0.025 \text{ m})(0.5 \text{ m})(5091 \text{ W/m}^2 \cdot \text{K})(25 - 5) \text{ K} \\ &= 3998 \text{ W} \end{aligned}$$

Then, the rate of condensation can be calculated as

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{3998 \text{ W}}{1201 \times 10^3 \text{ J/kg}} = 0.00333 \text{ kg/s}$$

Applying the conservation of mass, the flow rate of vapor leaving the tube can be determined as

$$\begin{aligned} \dot{m}_{v, \text{inlet}} &= \dot{m}_{v, \text{outlet}} + \dot{m}_{\text{condensation}} \\ \dot{m}_{v, \text{inlet}} &= 0.002 \text{ kg/s} + 0.00333 \text{ kg/s} = \mathbf{0.00533 \text{ kg/s}} \end{aligned}$$

Discussion The Reynolds number associated with film condensation inside a horizontal tube is

$$\text{Re}_{\text{vapor}} = \left(\frac{\rho_v V_v D}{\mu_v} \right)_{\text{inlet}} = \frac{4 \dot{m}_{v, \text{inlet}}}{\pi D \mu_v} = \frac{4(0.00533 \text{ kg/s})}{\pi (0.025 \text{ m})(1.037 \times 10^{-5} \text{ kg/m}\cdot\text{s})} = 26,200 < 35,000$$

Thus, the $\text{Re}_{\text{vapor}} < 35,000$ assumption is appropriate for this problem.

10-93 A copper tube transporting cold coolant has a surface temperature of 5°C and condenses moist air at 25°C. The rate of condensation is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Isothermal tube surface.

Properties The required property of water at the saturation temperature $T_{\text{sat}} = 25^\circ\text{C}$ is $h_{fg} = 2442 \times 10^3 \text{ J/kg}$ (Table A-9). The required property of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (25 + 5)/2 = 15^\circ\text{C}$ is $c_{pl} = 4185 \text{ J/kg}\cdot\text{K}$ (Table A-9).

Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2442 \times 10^3 \text{ J/kg} + 0.68(4185 \text{ J/kg}\cdot\text{K})(25 - 5) \text{ K} = 2.4989 \times 10^6 \text{ J/kg} \end{aligned}$$

The dropwise condensation heat transfer coefficient, for $22^\circ\text{C} < T_{\text{sat}} < 100^\circ\text{C}$, is determined to be

$$\begin{aligned} h &= h_{\text{dropwise}} = 51,104 + 2044T_{\text{sat}} \\ &= 51,104 + 2044(25^\circ\text{C}) = 102,204 \text{ W/m}^2 \cdot \text{K} \end{aligned}$$

The rate of heat transfer to the tube during this condensation process is

$$\begin{aligned} \dot{Q} &= hA_s(T_{\text{sat}} - T_s) = h(\pi DL)(T_{\text{sat}} - T_s) \\ &= (102,204 \text{ W/m}^2 \cdot \text{K})\pi(0.025 \text{ m})(10 \text{ m})(25 - 5) \text{ K} \\ &= 1.6054 \times 10^6 \text{ W} \end{aligned}$$

Thus, the rate of condensation is

$$\dot{m} = \frac{\dot{Q}}{h_{fg}^*} = \frac{1.6054 \times 10^6 \text{ J/s}}{2.4989 \times 10^6 \text{ J/kg}} = \mathbf{0.6424 \text{ kg/s}}$$

Discussion The heat transfer rate during the dropwise condensation can be increased by increasing the surface area of the tube, and thereby increasing the rate of condensation as well.

10-94  A copper tube that is used for transporting cold fluid is causing condensation of moist air on its outer surface. The rate of moisture condensation is to be determined so that a system for removing the condensate can be sized to alleviate the risk of electrical hazard.

Assumptions 1 Steady operating conditions exist. 2 Isothermal tube surface.

Properties The required property of water at the saturation temperature $T_{\text{sat}} = 27^\circ\text{C}$ is $h_{fg} = 2438 \times 10^3 \text{ J/kg}$ (Table A-9). The required property of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (27 + 3)/2 = 15^\circ\text{C}$ is $c_{pl} = 4185 \text{ J/kg}\cdot\text{K}$ (Table A-9).



Analysis Since dropwise condensation can occur, and it will have higher rate of heat transfer and therefore higher condensation rate than film condensation. The system must be able to handle the dropwise condensation rate.

The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2438 \times 10^3 \text{ J/kg} + 0.68(4185 \text{ J/kg}\cdot\text{K})(27 - 3) \text{ K} = 2.5063 \times 10^6 \text{ J/kg} \end{aligned}$$

The dropwise condensation heat transfer coefficient, for $22^\circ\text{C} < T_{\text{sat}} < 100^\circ\text{C}$, is determined to be

$$\begin{aligned} h &= h_{\text{dropwise}} = 51,104 + 2044T_{\text{sat}} \\ &= 51,104 + 2044(27^\circ\text{C}) = 106,292 \text{ W/m}^2 \cdot \text{K} \end{aligned}$$

The rate of heat transfer during the condensation process is

$$\begin{aligned} \dot{Q} &= hA_s(T_{\text{sat}} - T_s) \\ &= h(\pi DL)(T_{\text{sat}} - T_s) \\ &= (106,292 \text{ W/m}^2 \cdot \text{K})\pi(0.025 \text{ m})(10 \text{ m})(27 - 3) \text{ K} \\ &= 2.0036 \times 10^6 \text{ W} \end{aligned}$$

Thus, the rate of dropwise condensation is

$$\dot{m} = \frac{\dot{Q}}{h_{fg}^*} = \frac{2.0036 \times 10^6 \text{ J/s}}{2.5063 \times 10^6 \text{ J/kg}} = \mathbf{0.799 \text{ kg/s}}$$

Discussion In order to alleviate the risk of electrical hazard, the system must be able to remove the condensate at a rate of 0.8 kg/s. Note that dropwise condensation rate are higher than film condensation rate (by 10 times or more), thus a system capable of removing the condensate at a rate of 0.8 kg/s can also handle the condensate from film condensation.

10-95 Steam condenses at 60°C on a copper tube that is maintained with a surface temperature of 40°C. The condensation rates during film condensation and dropwise condensation are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Isothermal tube surface.

Properties The properties of water at the saturation temperature of 60°C are $h_{fg} = 2359 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.1304 \text{ kg/m}^3$ (Table A-9). The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (60 + 40)/2 = 50^\circ\text{C}$ are (Table A-9)

$$\begin{aligned}\rho_l &= 988.1 \text{ kg/m}^3 \\ \mu_l &= 0.547 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.554 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4181 \text{ J/kg} \cdot \text{K} \\ k_l &= 0.644 \text{ W/m} \cdot \text{K}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2359 \times 10^3 \text{ J/kg} + 0.68(4181 \text{ J/kg} \cdot \text{K})(60 - 40) \text{ K} = 2.4159 \times 10^6 \text{ J/kg}\end{aligned}$$

(a) The film condensation heat transfer coefficient is determined from

$$\begin{aligned}h_{\text{film}} &= h_{\text{horiz}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(988.1 \text{ kg/m}^3)(988.1 - 0.1304) \text{ kg/m}^3 (2.4159 \times 10^6 \text{ J/kg})(0.644 \text{ W/m} \cdot \text{K})^3}{(0.547 \times 10^{-3} \text{ kg/m} \cdot \text{s})(60 - 40) \text{ K}(0.025 \text{ m})} \right]^{1/4} \\ &= 8937.7 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Thus, the rate of film condensation is

$$\begin{aligned}\dot{m}_{\text{film}} &= \frac{\dot{Q}}{h_{fg}^*} = \frac{h_{\text{film}}(\pi DL)(T_{\text{sat}} - T_s)}{h_{fg}^*} \\ &= \frac{(8937.7 \text{ W/m}^2 \cdot \text{K})\pi(0.025 \text{ m})(15 \text{ m})(60 - 40) \text{ K}}{2.4159 \times 10^6 \text{ J/kg}} = \mathbf{0.08717 \text{ kg/s}}\end{aligned}$$

(b) The dropwise condensation heat transfer coefficient, for $22^\circ\text{C} < T_{\text{sat}} < 100^\circ\text{C}$, is determined from

$$\begin{aligned}h_{\text{dropwise}} &= 51,104 + 2044T_{\text{sat}} \\ &= 51,104 + 2044(60^\circ\text{C}) = 173,744 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Thus, the rate of dropwise condensation is

$$\begin{aligned}\dot{m}_{\text{dropwise}} &= \frac{\dot{Q}}{h_{fg}^*} = \frac{h_{\text{dropwise}}(\pi DL)(T_{\text{sat}} - T_s)}{h_{fg}^*} \\ &= \frac{(173,744 \text{ W/m}^2 \cdot \text{K})\pi(0.025 \text{ m})(15 \text{ m})(60 - 40) \text{ K}}{2.4159 \times 10^6 \text{ J/kg}} = \mathbf{1.695 \text{ kg/s}}\end{aligned}$$

Discussion The dropwise condensation rate is 19.4 times greater than that of film condensation. This is because the convection heat transfer coefficient for dropwise condensation is greater than that of film condensation at the same factor, $h_{\text{dropwise}}/h_{\text{film}} = 19.4$.

With dropwise condensation, there is no liquid film to impede heat transfer. Therefore, a much higher heat transfer coefficient can be achieved in dropwise condensation than in film condensation.

Special Topic: Non-Boiling Two-Phase Flow Heat Transfer

10-96 The flow quality of a non-boiling two-phase flow in a tube with $\dot{m}_l / \dot{m}_g = 300$ is to be determined.

Assumptions **1** Steady operating condition exists. **2** Two-phase flow is non-boiling and it does not involve phase change. **3** Fluid properties are constant.

Analysis The flow quality is given as

$$x = \frac{\dot{m}_g}{\dot{m}_l + \dot{m}_g}$$

Hence, the equation can be rearranged as

$$x = \frac{\dot{m}_g}{\dot{m}_l + \dot{m}_g} = \frac{\dot{m}_g / \dot{m}_g}{(\dot{m}_l + \dot{m}_g) / \dot{m}_g} = \frac{1}{\dot{m}_l / \dot{m}_g + 1}$$

Thus, the flow quality is

$$x = \frac{1}{300 + 1} = \mathbf{0.00332}$$

Discussion The flow quality is a dimensionless parameter.

10-97 The flow quality and the mass flow rates of the gas and the liquid for a non-boiling two-phase flow, where $V_{sl} = 3V_{sg}$, are to be determined.

Assumptions **1** Steady operating condition exists. **2** Two-phase flow is non-boiling and it does not involve phase change. **3** Fluid properties are constant.

Properties The densities of the gas and liquid are given to be $\rho_g = 8.5 \text{ kg/m}^3$ and $\rho_l = 855 \text{ kg/m}^3$, respectively.

Analysis The mass flow rate of gas can be calculated using

$$\begin{aligned} \dot{m}_g &= \rho_g V_{sg} A_c = \rho_g V_{sg} \pi \frac{D^2}{4} \\ &= (8.5 \text{ kg/m}^3)(0.8 \text{ m/s})\pi \frac{(0.102 \text{ m})^2}{4} = \mathbf{0.0556 \text{ kg/s}} \end{aligned}$$

Then, the mass flow rate of liquid is (with $V_{sl} = 3V_{sg}$)

$$\begin{aligned} \dot{m}_l &= \rho_l V_{sl} A_c = 3\rho_l V_{sg} \pi \frac{D^2}{4} \\ &= 3(855 \text{ kg/m}^3)(0.8 \text{ m/s})\pi \frac{(0.102 \text{ m})^2}{4} = \mathbf{16.8 \text{ kg/s}} \end{aligned}$$

Thus, the flow quality is

$$x = \frac{\dot{m}_g}{\dot{m}_l + \dot{m}_g} = \frac{0.0556}{16.8 + 0.0556} = \mathbf{0.00330}$$

Discussion The total mass flow rate of gas and liquid for this two-phase flow is simply

$$\dot{m}_{\text{tot}} = \dot{m}_l + \dot{m}_g = 16.8 \text{ kg/s} + 0.0556 \text{ kg/s} = 16.86 \text{ kg/s}$$

10-98 The mass flow rate of air and the superficial velocities of air and engine oil for a non-boiling two-phase flow in a tube are to be determined.

Assumptions 1 Steady operating condition exists. 2 Two-phase flow is non-boiling and it does not involve phase change. 3 Fluid properties are constant.

Properties The densities of air and engine oil at the bulk mean temperature $T_b = 140^\circ\text{C}$ are $\rho_g = 0.8542 \text{ kg/m}^3$ (Table A-15) and $\rho_l = 816.8 \text{ kg/m}^3$ (Table A-13), respectively.

Analysis The flow quality is given as

$$x = \frac{\dot{m}_g}{\dot{m}_l + \dot{m}_g} = \frac{\dot{m}_g / \dot{m}_g}{(\dot{m}_l + \dot{m}_g) / \dot{m}_g} = \frac{1}{\dot{m}_l / \dot{m}_g + 1}$$

or

$$\frac{\dot{m}_l}{\dot{m}_g} + 1 = \frac{1}{x} \quad \rightarrow \quad \frac{\dot{m}_l}{\dot{m}_g} = \frac{1-x}{x}$$

With known liquid (engine oil) mass flow rate and flow quality, the gas (air) mass flow rate is determined using

$$\dot{m}_g = \frac{x}{1-x} \dot{m}_l = \frac{2.1 \times 10^{-3}}{1 - 2.1 \times 10^{-3}} (0.9 \text{ kg/s}) = \mathbf{0.00189 \text{ kg/s}}$$

From the gas and liquid mass flow rates, the superficial gas and liquid velocities can be calculated:

$$V_{sg} = \frac{\dot{m}_g}{\rho_g A} = \frac{4\dot{m}_g}{\rho_g \pi D^2} = \frac{4(0.00189 \text{ kg/s})}{(0.8542 \text{ kg/m}^3) \pi (0.025 \text{ m})^2} = \mathbf{4.51 \text{ m/s}}$$

$$V_{sl} = \frac{\dot{m}_l}{\rho_l A} = \frac{4\dot{m}_l}{\rho_l \pi D^2} = \frac{4(0.9 \text{ kg/s})}{(816.8 \text{ kg/m}^3) \pi (0.025 \text{ m})^2} = \mathbf{2.25 \text{ m/s}}$$

Discussion The superficial velocity of air is twice that of the engine oil.

10-99 Starting with the two-phase non-boiling heat transfer correlation, the expression that is appropriate for the case when only water is flowing in the tube is to be determined.

Assumptions 1 Steady operating condition exists. 2 Two-phase flow is non-boiling and it does not involve phase change. 3 Fluid properties are constant.

Analysis The two-phase non-boiling heat transfer correlation is given as

$$h_{tp} = h_l F_p \left[1 + 0.55 \left(\frac{x}{1-x} \right)^{0.1} \left(\frac{1-F_p}{F_p} \right)^{0.4} \left(\frac{Pr_g}{Pr_l} \right)^{0.25} \left(\frac{\mu_l}{\mu_g} \right)^{0.25} (I^*)^{0.25} \right]$$

where

$$F_p = (1-\alpha) + \alpha \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{\rho_g (V_g - V_l)^2}{gD(\rho_l - \rho_g)}} \right) \right]^2 \quad \text{and} \quad x = \frac{\dot{m}_g}{\dot{m}_l + \dot{m}_g}$$

For the situation when the air flow is shut off and only water is flowing in the pipe, we have $\dot{m}_g = 0$ and $\alpha = 0$. Hence, we get

$$F_p = (1-0) + 0 = 1 \quad \text{and} \quad x = 0$$

Thus, the two-phase non-boiling heat transfer correlation becomes

$$h_{tp, \alpha=0} = h_l (1) \left[1 + 0.55 \left(\frac{0}{1-0} \right)^{0.1} \left(\frac{1-1}{1} \right)^{0.4} \left(\frac{Pr_g}{Pr_l} \right)^{0.25} \left(\frac{\mu_l}{\mu_g} \right)^{0.25} (I^*)^{0.25} \right] = h_l (1) [1+0] = h_l$$

The liquid phase heat transfer coefficient is calculated using:

$$h_l = 0.027 \text{Re}_l^{4/5} \text{Pr}_l^{1/3} \left(\frac{k_l}{D} \right) \left(\frac{\mu_l}{\mu_s} \right)^{0.14}$$

where the *in situ* liquid Reynolds number is

$$\text{Re}_l = \frac{4\dot{m}_l}{\pi \sqrt{1-\alpha} \mu_l D} = \frac{4\dot{m}_l}{\pi \sqrt{1-0} \mu_l D} = \frac{4\dot{m}_l}{\pi \mu_l D}$$

Therefore, we have

$$h_{tp, \alpha=0} = h_l = 0.027 \text{Re}_l^{4/5} \text{Pr}_l^{1/3} \left(\frac{k_l}{D} \right) \left(\frac{\mu_l}{\mu_s} \right)^{0.14}$$

Discussion When only water is flowing in the tube, the two-phase non-boiling heat transfer correlation is reduced to a familiar equation for internal forced convection.

10-100 Air-water slug flows through a 25.4-mm diameter horizontal tube in microgravity condition. Using the non-boiling two-phase heat transfer correlation, the two-phase heat transfer coefficient (h_{tp}) is to be determined

Assumptions 1 Steady operating condition exists. 2 Two-phase flow is non-boiling and it does not involve phase change. 3 Fluid properties are constant.

Properties The properties of water (liquid) are given to be $\mu_l = 85.5 \times 10^{-5} \text{ kg/m} \cdot \text{s}$, $\mu_s = 73.9 \times 10^{-5} \text{ kg/m} \cdot \text{s}$, $\rho_l = 997 \text{ kg/m}^3$, $k_l = 0.613 \text{ W/m} \cdot \text{K}$, and $\text{Pr}_l = 5.0$. The properties of air (gas) are given to be $\mu_g = 18.5 \times 10^{-6} \text{ kg/m} \cdot \text{s}$, $\rho_g = 1.16 \text{ kg/m}^3$, and $\text{Pr}_g = 0.71$.

Analysis From the superficial gas and liquid velocities, and void fraction, the gas and liquid velocities can be calculated as

$$V_g = \frac{V_{sg}}{\alpha} = \frac{0.3 \text{ m/s}}{0.27} = 1.11 \text{ m/s}$$

$$V_l = \frac{V_{sl}}{1 - \alpha} = \frac{0.544 \text{ m/s}}{1 - 0.27} = 0.745 \text{ m/s}$$

The gas and liquid mass flow rates are calculated as

$$\dot{m}_g = \rho_g V_{sg} A_c = \rho_g V_{sg} \pi \frac{D^2}{4} = (1.16 \text{ kg/m}^3)(0.3 \text{ m/s})\pi \frac{(0.0254 \text{ m})^2}{4} = 1.76 \times 10^{-4} \text{ kg/s}$$

$$\dot{m}_l = \rho_l V_{sl} A_c = \rho_l V_{sl} \pi \frac{D^2}{4} = (997 \text{ kg/m}^3)(0.544 \text{ m/s})\pi \frac{(0.0254 \text{ m})^2}{4} = 0.275 \text{ kg/s}$$

Using the gas and liquid mass flow rates, the quality is determined to be

$$x = \frac{\dot{m}_g}{\dot{m}_l + \dot{m}_g} = \frac{1.76 \times 10^{-4}}{0.275 + 1.76 \times 10^{-4}} = 6.40 \times 10^{-4}$$

The flow pattern factor (F_p) can be calculated using

$$F_p = (1 - \alpha) + \alpha \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{\rho_g (V_g - V_l)^2}{gD(\rho_l - \rho_g)}} \right) \right]^2$$

$$= (1 - 0.27) + (0.27) \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{(1.16 \text{ kg/m}^3)(1.11 \text{ m/s} - 0.745 \text{ m/s})^2}{(9.81 \text{ m/s}^2)(0.0254 \text{ m})(997 \text{ kg/m}^3 - 1.16 \text{ kg/m}^3)}} \right) \right]^2 = 0.730$$

The liquid phase heat transfer coefficient is calculated using:

$$h_l = 0.027 \text{Re}_l^{4/5} \text{Pr}_l^{1/3} \left(\frac{k_l}{D} \right) \left(\frac{\mu_l}{\mu_s} \right)^{0.14}$$

$$= 0.027(18870)^{4/5} (5.0)^{1/3} \left(\frac{0.613 \text{ W/m} \cdot \text{K}}{0.0254 \text{ m}} \right) \left(\frac{85.5 \times 10^{-5} \text{ kg/m} \cdot \text{s}}{73.9 \times 10^{-5} \text{ kg/m} \cdot \text{s}} \right)^{0.14} = 2995 \text{ W/m}^2 \cdot \text{K}$$

where the *in situ* liquid Reynolds number is

$$\text{Re}_l = \frac{4\dot{m}_l}{\pi \sqrt{1 - \alpha} \mu_l D} = \frac{4(0.275 \text{ kg/s})}{\pi \sqrt{1 - 0.27} (85.5 \times 10^{-5} \text{ kg/m} \cdot \text{s})(0.0254 \text{ m})} = 18870$$

The inclination factor (I^*) has a value of one for horizontal tube ($\theta = 0$). Thus, using the general two-phase heat transfer correlation, the value for h_{tp} is estimated to be

$$\frac{h_{tp}}{h_l} = F_p \left[1 + 0.55 \left(\frac{x}{1 - x} \right)^{0.1} \left(\frac{1 - F_p}{F_p} \right)^{0.4} \left(\frac{\text{Pr}_g}{\text{Pr}_l} \right)^{0.25} \left(\frac{\mu_l}{\mu_g} \right)^{0.25} (I^*)^{0.25} \right]$$

$$= (0.730) \left[1 + 0.55 \left(\frac{6.40 \times 10^{-4}}{1 - 6.40 \times 10^{-4}} \right)^{0.1} \left(\frac{1 - 0.730}{0.730} \right)^{0.4} \left(\frac{0.71}{5.0} \right)^{0.25} \left(\frac{85.5 \times 10^{-5} \text{ kg/m} \cdot \text{s}}{18.5 \times 10^{-6} \text{ kg/m} \cdot \text{s}} \right)^{0.25} \right] = 0.9369$$

$$\text{or } h_{tp} = 0.9369 h_l = 0.9369(2995 \text{ W/m}^2 \cdot \text{K}) = \mathbf{2810 \text{ W/m}^2 \cdot \text{K}}$$

Discussion The non-boiling two-phase heat transfer coefficient (h_{tp}) is about 7% lower than the liquid phase heat transfer coefficient (h_l).

10-101 A two-phase flow of air and silicone (Dow Corning 200® Fluid, 5 cs) is being transported in an 11.7-mm diameter horizontal tube, where condensation occurs on the tube outer surface. The overall convection heat transfer coefficient is to be determined.

Assumptions 1 Steady operating condition exists. 2 Two-phase flow inside tube is non-boiling and does not involve phase change. 3 Fluid properties are constant. 4 The thermal resistance of the tube wall is negligible. 5 Isothermal tube surface. 6 Film condensation occurs on the tube outer surface.

Properties Inside the tube: The properties of liquid silicone (liquid) are given to be $\mu_l = 45.7 \times 10^{-4} \text{ kg/m} \cdot \text{s}$, $\rho_l = 913 \text{ kg/m}^3$, $k_l = 0.117 \text{ W/m} \cdot \text{K}$, $\sigma = 19.7 \times 10^{-3} \text{ N/m}$ and $\text{Pr}_l = 64$. The properties of air (gas) are given to be $\mu_g = 18.4 \times 10^{-6} \text{ kg/m} \cdot \text{s}$, $\rho_g = 1.19 \text{ kg/m}^3$, $\text{Pr}_g = 0.71$.

Outside the tube: The properties of water at the saturation temperature of 40°C are $h_{fg} = 2407 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.0512 \text{ kg/m}^3$ (Table A-9). The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (40 + 20)/2 = 30^\circ\text{C}$ are (Table A-9),

$$\rho_{l,f} = 996.0 \text{ kg/m}^3$$

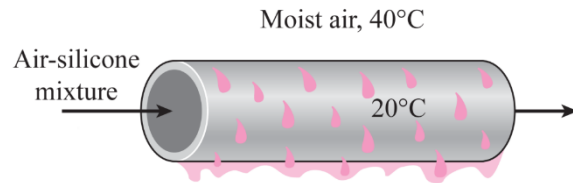
$$\mu_{l,f} = 0.798 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

$$\nu_{l,f} = \mu_l / \rho_l = 0.801 \times 10^{-6} \text{ m}^2/\text{s}$$

$$c_{pl,f} = 4178 \text{ J/kg} \cdot \text{K}$$

$$k_{l,f} = 0.615 \text{ W/m} \cdot \text{K}$$

$$\text{Pr}_f = 5.42$$



Analysis Inside the tube: The flow pattern factor (F_P) can be calculated using

$$\begin{aligned} F_P &= (1 - \alpha) + \alpha \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{\rho_g (V_g - V_l)^2}{gD(\rho_l - \rho_g)}} \right) \right]^2 \\ &= (1 - 0.011) + (0.011) \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{(1.19 \text{ kg/m}^3)(13.5 \text{ m/s} - 9.34 \text{ m/s})^2}{(9.81 \text{ m/s}^2)(0.0117 \text{ m})(913 \text{ kg/m}^3 - 1.19 \text{ kg/m}^3)}} \right) \right]^2 \\ &= 0.9898 \end{aligned}$$

The inclination factor (I^*) for horizontal tube ($\theta = 0^\circ$) is calculated to be

$$I^* = 1 + \frac{(\rho_l - \rho_g)gD^2}{\sigma} |\sin \theta| = 1$$

The liquid phase heat transfer coefficient is calculated using the Seder and Tate (1936) equation:

$$\begin{aligned} h_l &= 0.027 \text{Re}_l^{0.8} \text{Pr}_l^{1/3} \left(\frac{\mu_l}{\mu_s} \right)^{0.14} \left(\frac{k_l}{D} \right) \\ &= 0.027(21718)^{0.8} (64)^{1/3} \left(\frac{45.7}{39.8} \right)^{0.14} \left(\frac{0.117 \text{ W/m} \cdot \text{K}}{0.0117 \text{ m}} \right) \\ &= 3246 \text{ W/m}^2 \cdot \text{K} \end{aligned}$$

where the *in situ* liquid Reynolds number is

$$\text{Re}_l = \frac{4\dot{m}_l}{\pi \sqrt{1 - \alpha} \mu_l D} = \frac{4(0.907 \text{ kg/s})}{\pi \sqrt{1 - 0.011} (45.7 \times 10^{-4} \text{ kg/m} \cdot \text{s})(0.0117 \text{ m})} = 21718$$

Using the general two-phase heat transfer correlation, Eq. (10-38), the two-phase heat transfer coefficient (h_{tp}) is estimated to be

$$\begin{aligned}
 h_{tp} &= h_l F_P \left[1 + 0.55 \left(\frac{x}{1-x} \right)^{0.1} \left(\frac{1-F_P}{F_P} \right)^{0.4} \left(\frac{Pr_g}{Pr_l} \right)^{0.25} \left(\frac{\mu_l}{\mu_g} \right)^{0.25} (I^*)^{0.25} \right] \\
 &= (3246 \text{ W/m}^2 \cdot \text{K})(0.9898) \left[1 + 0.55 \left(\frac{2.08 \times 10^{-5}}{1 - 2.08 \times 10^{-5}} \right)^{0.1} \left(\frac{1 - 0.9898}{0.9898} \right)^{0.4} \left(\frac{0.71}{64} \right)^{0.25} \left(\frac{45.7 \times 10^{-4}}{18.4 \times 10^{-6}} \right)^{0.25} \right] \\
 h_i &= h_{tp} = 3337 \text{ W/m}^2 \cdot \text{K}
 \end{aligned}$$

Outside the tube: The modified latent heat of vaporization is

$$h_{fg}^* = h_{fg} + 0.68 c_{p,l,f} (T_{\text{sat}} - T_s) = 2407 \times 10^3 \text{ J/kg} + 0.68 \times (4182 \text{ J/kg} \cdot \text{K})(40 - 20) \text{ K} = 2464 \times 10^3 \text{ J/kg}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}
 h_o &= h_{\text{horizontal}} = 0.729 \left[\frac{g \rho_{l,f} (\rho_{l,f} - \rho_v) h_{fg}^* k_{l,f}^3}{\mu_{l,f} (T_{\text{sat}} - T_s) D} \right]^{1/4} \\
 &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(996 \text{ kg/m}^3)(996 - 0.05 \text{ kg/m}^3)(2464 \times 10^3 \text{ J/kg})(0.615 \text{ W/m} \cdot \text{K})^3}{(0.798 \times 10^{-3} \text{ kg/m} \cdot \text{s})(40 - 20) \text{ K}(0.0117 \text{ m})} \right]^{1/4} \\
 &= 9584 \text{ W/m}^2 \cdot \text{K}
 \end{aligned}$$

Noting that the thermal resistance of the tube is negligible, the overall heat transfer coefficient becomes

$$U = \frac{1}{1/h_i + 1/h_o} = \frac{1}{1/3337 + 1/9584} = 2475 \text{ W/m}^2 \cdot \text{K}$$

Discussion The condensation heat transfer coefficient is almost 3 times higher than the non-boiling heat transfer coefficient. This is expected, as heat transfer coefficient involving phase change is much higher than that without phase change.

10-102 Air-water mixture is flowing in a 5° inclined tube with diameter of 25.4 mm, and the mixture superficial gas and liquid velocities are 1 m/s and 2 m/s, respectively. The two-phase heat transfer coefficient (h_{tp}) is to be determined.

Assumptions 1 Steady operating condition exists. 2 Two-phase flow is non-boiling and it does not involve phase change. 3 Fluid properties are constant.

Properties The properties of water (liquid) at bulk mean temperature $T_b = (T_i + T_e)/2 = 45^\circ\text{C}$ are, from Table A-9, $\mu_l = 0.596 \times 10^{-3} \text{ kg/m}\cdot\text{s}$, $\rho_l = 990.1 \text{ kg/m}^3$, $k_l = 0.637 \text{ W/m}\cdot\text{K}$, and $\text{Pr}_l = 3.91$. The properties of air (gas) at bulk mean temperature $T_b = 45^\circ\text{C}$ are, from Table A-15, $\mu_g = 1.941 \times 10^{-5} \text{ kg/m}\cdot\text{s}$, $\rho_g = 1.109 \text{ kg/m}^3$, and $\text{Pr}_g = 0.7241$. Also, at $T_s = 80^\circ\text{C}$ we get $\mu_s = 0.355 \times 10^{-3} \text{ kg/m}\cdot\text{s}$ from Table A-9.

Analysis From the superficial gas and liquid velocities, and void fraction, the gas and liquid velocities can be calculated as

$$V_g = \frac{V_{sg}}{\alpha} = \frac{1 \text{ m/s}}{0.33} = 3.030 \text{ m/s} \quad V_l = \frac{V_{sl}}{1 - \alpha} = \frac{2 \text{ m/s}}{1 - 0.33} = 2.985 \text{ m/s}$$

The gas and liquid mass flow rates are calculated as

$$\dot{m}_g = \rho_g V_{sg} A_c = \rho_g V_{sg} \pi \frac{D^2}{4} = (1.109 \text{ kg/m}^3)(1 \text{ m/s})\pi \frac{(0.0254 \text{ m})^2}{4} = 5.619 \times 10^{-4} \text{ kg/s}$$

$$\dot{m}_l = \rho_l V_{sl} A_c = \rho_l V_{sl} \pi \frac{D^2}{4} = (990.1 \text{ kg/m}^3)(2 \text{ m/s})\pi \frac{(0.0254 \text{ m})^2}{4} = 1.003 \text{ kg/s}$$

Using the gas and liquid mass flow rates, the quality is determined to be

$$x = \frac{\dot{m}_g}{\dot{m}_l + \dot{m}_g} = \frac{5.619 \times 10^{-4}}{1.003 + 5.619 \times 10^{-4}} = 5.599 \times 10^{-4}$$

The flow pattern factor (F_p) can be calculated using

$$F_p = (1 - \alpha) + \alpha \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{\rho_g (V_g - V_l)^2}{gD(\rho_l - \rho_g)}} \right) \right]^2$$

$$= (1 - 0.33) + (0.33) \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{(1.109 \text{ kg/m}^3)(3.03 \text{ m/s} - 2.985 \text{ m/s})^2}{(9.81 \text{ m/s}^2)(0.0254 \text{ m})(990.1 \text{ kg/m}^3 - 1.109 \text{ kg/m}^3)}} \right) \right]^2 = 0.670$$

The inclination factor (I^*) for $\theta = 5^\circ$ is calculated to be

$$I^* = 1 + \frac{(\rho_l - \rho_g)gD^2}{\sigma} |\sin \theta| = 1 + \frac{(990.1 \text{ kg/m}^3 - 1.109 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(0.0254 \text{ m})^2}{0.068 \text{ N/m}} |\sin 5^\circ| = 9.023$$

The liquid phase heat transfer coefficient is calculated using:

$$h_l = 0.027 \text{Re}_l^{4/5} \text{Pr}_l^{1/3} \left(\frac{k_l}{D} \right) \left(\frac{\mu_l}{\mu_s} \right)^{0.14}$$

$$= 0.027(103100)^{4/5} (3.91)^{1/3} \left(\frac{0.637 \text{ W/m}\cdot\text{K}}{0.0254 \text{ m}} \right) \left(\frac{0.596 \times 10^{-3} \text{ kg/m}\cdot\text{s}}{0.355 \times 10^{-3} \text{ kg/m}\cdot\text{s}} \right)^{0.14} = 11754 \text{ W/m}^2 \cdot \text{K}$$

where the *in situ* liquid Reynolds number is

$$\text{Re}_l = \frac{4\dot{m}_l}{\pi \sqrt{1 - \alpha} \mu_l D} = \frac{4(1.003 \text{ kg/s})}{\pi \sqrt{1 - 0.33} (0.596 \times 10^{-3} \text{ kg/m}\cdot\text{s})(0.0254 \text{ m})} = 103100$$

Thus, using the general two-phase heat transfer correlation, the value for h_{tp} is estimated to be

$$\frac{h_{tp}}{h_l} = F_p \left[1 + 0.55 \left(\frac{x}{1 - x} \right)^{0.1} \left(\frac{1 - F_p}{F_p} \right)^{0.4} \left(\frac{\text{Pr}_g}{\text{Pr}_l} \right)^{0.25} \left(\frac{\mu_l}{\mu_g} \right)^{0.25} (I^*)^{0.25} \right]$$

$$= (0.670) \left[1 + 0.55 \left(\frac{5.599 \times 10^{-4}}{1 - 5.599 \times 10^{-4}} \right)^{0.1} \left(\frac{1 - 0.670}{0.670} \right)^{0.4} \left(\frac{0.7241}{3.91} \right)^{0.25} \left(\frac{59.6}{1.941} \right)^{0.25} (9.023)^{0.25} \right] = 1.021$$

or $h_{tp} = 1.021 h_l = 1.021(11754 \text{ W/m}^2 \cdot \text{K}) = \mathbf{12,000 \text{ W/m}^2 \cdot \text{K}}$

Discussion The inclination factor is $I^* = 1$ when the tube is at horizontal position, since $\sin(0^\circ) = 0$. The two-phase heat transfer coefficient for horizontal tube would be $h_{tp} = 10300 \text{ W/m}^2 \cdot \text{K}$, which is about 14% lower than that of 5° inclined tube.

10-103 Mixture of petroleum and natural gas is being transported in a pipeline that is located in a terrain that caused it to have an average inclination angle of 10° . The two-phase heat transfer coefficient is to be determined.

Assumptions 1 Steady operating condition exists. 2 Two-phase flow is non-boiling and it does not involve phase change. 3 Fluid properties are constant.

Properties The properties of petroleum (liquid) are given to be $\mu_l = 297.5 \times 10^{-4} \text{ kg/m} \cdot \text{s}$, $\mu_s = 238 \times 10^{-4} \text{ kg/m} \cdot \text{s}$, $\rho_l = 853 \text{ kg/m}^3$, $k_l = 0.163 \text{ W/m} \cdot \text{K}$, $\sigma = 0.020 \text{ N/m}$, and $Pr_l = 405$. The properties of natural gas are given to be $\mu_g = 9.225 \times 10^{-6} \text{ kg/m} \cdot \text{s}$, $\rho_g = 9.0 \text{ kg/m}^3$, and $Pr_g = 0.80$.

Analysis From the gas and liquid mass flow rates, the superficial gas and liquid velocities can be calculated:

$$V_{sg} = \frac{\dot{m}_g}{\rho_g A} = \frac{4\dot{m}_g}{\rho_g \pi D^2} = \frac{4(0.055 \text{ kg/s})}{(9.0 \text{ kg/m}^3)\pi(0.102 \text{ m})^2} = 0.748 \text{ m/s}$$

$$V_{sl} = \frac{\dot{m}_l}{\rho_l A} = \frac{4\dot{m}_l}{\rho_l \pi D^2} = \frac{4(16 \text{ kg/s})}{(853 \text{ kg/m}^3)\pi(0.102 \text{ m})^2} = 2.296 \text{ m/s}$$

Using the superficial velocities and void fraction, the gas and liquid velocities are found to be

$$V_g = \frac{V_{sg}}{\alpha} = \frac{0.748 \text{ m/s}}{0.22} = 3.400 \text{ m/s} \quad V_l = \frac{V_{sl}}{1-\alpha} = \frac{2.296 \text{ m/s}}{1-0.22} = 2.944 \text{ m/s}$$

Using the gas and liquid mass flow rates, the quality is determined to be

$$x = \frac{\dot{m}_g}{\dot{m}_l + \dot{m}_g} = \frac{0.055}{16 + 0.055} = 3.426 \times 10^{-3}$$

The flow pattern factor (F_p) can be calculated using

$$F_p = (1-\alpha) + \alpha \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{\rho_g (V_g - V_l)^2}{gD(\rho_l - \rho_g)}} \right) \right]^2$$

$$= (1-0.22) + (0.22) \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{(9.0 \text{ kg/m}^3)(3.400 \text{ m/s} - 2.944 \text{ m/s})^2}{(9.81 \text{ m/s}^2)(0.102 \text{ m})(853 \text{ kg/m}^3 - 9.0 \text{ kg/m}^3)}} \right) \right]^2 = 0.7802$$

The liquid phase heat transfer coefficient is calculated using:

$$h_l = 0.027 \text{Re}_l^{4/5} \text{Pr}_l^{1/3} \left(\frac{k_l}{D} \right) \left(\frac{\mu_l}{\mu_s} \right)^{0.14}$$

$$= 0.027(7601)^{4/5} (405)^{1/3} \left(\frac{0.163 \text{ W/m} \cdot \text{K}}{0.102 \text{ m}} \right) \left(\frac{297.5 \times 10^{-4} \text{ kg/m} \cdot \text{s}}{238 \times 10^{-4} \text{ kg/m} \cdot \text{s}} \right)^{0.14} = 419.1 \text{ W/m}^2 \cdot \text{K}$$

where the *in situ* liquid Reynolds number is

$$\text{Re}_l = \frac{4\dot{m}_l}{\pi \sqrt{1-\alpha} \mu_l D} = \frac{4(16 \text{ kg/s})}{\pi \sqrt{1-0.22} (297.5 \times 10^{-4} \text{ kg/m} \cdot \text{s})(0.102 \text{ m})} = 7601$$

The inclination factor (I^*) for $\theta = 10^\circ$ is calculated to be

$$I^* = 1 + \frac{(\rho_l - \rho_g)gD^2}{\sigma} |\sin \theta| = 1 + \frac{(853 \text{ kg/m}^3 - 9.0 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(0.102 \text{ m})^2}{0.020 \text{ N/m}} |\sin 10^\circ| = 748.9$$

Thus, using the general two-phase heat transfer correlation, the value for h_{tp} is estimated to be

$$\frac{h_{tp}}{h_l} = F_p \left[1 + 0.55 \left(\frac{x}{1-x} \right)^{0.1} \left(\frac{1-F_p}{F_p} \right)^{0.4} \left(\frac{Pr_g}{Pr_l} \right)^{0.25} \left(\frac{\mu_l}{\mu_g} \right)^{0.25} (I^*)^{0.25} \right]$$

$$= (0.7802) \left[1 + 0.55 \left(\frac{3.426 \times 10^{-3}}{1-3.426 \times 10^{-3}} \right)^{0.1} \left(\frac{1-0.7802}{0.7802} \right)^{0.4} \left(\frac{0.80}{405} \right)^{0.25} \left(\frac{29750}{9.225} \right)^{0.25} (748.9)^{0.25} \right] = 1.999$$

or $h_{tp} = 1.999h_l = 1.999(419.1 \text{ W/m}^2 \cdot \text{K}) = \mathbf{838 \text{ W/m}^2 \cdot \text{K}}$

Discussion Since $V_g > V_b$, there will be slippage between the gas and liquid phases. When $V_g \neq V_b$, slippage between the gas and liquid phases exists. When $V_g = V_b$, slippage between the gas and liquid phases is negligible, and the flow is called homogeneous two-phase flow.

10-104 Air-water mixture is flowing in a tube with diameter of 25.4 mm, and the mixture superficial gas and liquid velocities are 1 m/s and 2 m/s, respectively. The two-phase heat transfer coefficient (h_{tp}) for (a) horizontal tube ($\theta = 0^\circ$) and (b) vertical tube ($\theta = 90^\circ$), are to be determined and compared.

Assumptions 1 Steady operating condition exists. 2 Two-phase flow is non-boiling and it does not involve phase change. 3 Fluid properties are constant.

Properties The properties of water (liquid) at bulk mean temperature $T_b = (T_i + T_e)/2 = 45^\circ\text{C}$ are, from Table A-9, $\mu_l = 0.596 \times 10^{-3} \text{ kg/m}\cdot\text{s}$, $\rho_l = 990.1 \text{ kg/m}^3$, $k_l = 0.637 \text{ W/m}\cdot\text{K}$, and $\text{Pr}_l = 3.91$. The properties of air (gas) at bulk mean temperature $T_b = 45^\circ\text{C}$ are, from Table A-15, $\mu_g = 1.941 \times 10^{-5} \text{ kg/m}\cdot\text{s}$, $\rho_g = 1.109 \text{ kg/m}^3$, and $\text{Pr}_g = 0.7241$. Also, at $T_s = 80^\circ\text{C}$ we get $\mu_s = 0.355 \times 10^{-3} \text{ kg/m}\cdot\text{s}$ from Table A-9.

Analysis From the superficial gas and liquid velocities, and void fraction, the gas and liquid velocities can be calculated as

$$V_g = \frac{V_{sg}}{\alpha} = \frac{1 \text{ m/s}}{0.33} = 3.030 \text{ m/s}$$

$$V_l = \frac{V_{sl}}{1 - \alpha} = \frac{2 \text{ m/s}}{1 - 0.33} = 2.985 \text{ m/s}$$

The gas and liquid mass flow rates are calculated as

$$\dot{m}_g = \rho_g V_{sg} A_c = \rho_g V_{sg} \pi \frac{D^2}{4} = (1.109 \text{ kg/m}^3)(1 \text{ m/s})\pi \frac{(0.0254 \text{ m})^2}{4} = 5.619 \times 10^{-4} \text{ kg/s}$$

$$\dot{m}_l = \rho_l V_{sl} A_c = \rho_l V_{sl} \pi \frac{D^2}{4} = (990.1 \text{ kg/m}^3)(2 \text{ m/s})\pi \frac{(0.0254 \text{ m})^2}{4} = 1.003 \text{ kg/s}$$

Using the gas and liquid mass flow rates, the quality is determined to be

$$x = \frac{\dot{m}_g}{\dot{m}_l + \dot{m}_g} = \frac{5.619 \times 10^{-4}}{1.003 + 5.619 \times 10^{-4}} = 5.599 \times 10^{-4}$$

The flow pattern factor (F_p) can be calculated using

$$\begin{aligned} F_p &= (1 - \alpha) + \alpha \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{\rho_g (V_g - V_l)^2}{gD(\rho_l - \rho_g)}} \right) \right]^2 \\ &= (1 - 0.33) + (0.33) \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{(1.109 \text{ kg/m}^3)(3.03 \text{ m/s} - 2.985 \text{ m/s})^2}{(9.81 \text{ m/s}^2)(0.0254 \text{ m})(990.1 \text{ kg/m}^3 - 1.109 \text{ kg/m}^3)}} \right) \right]^2 \\ &= 0.670 \end{aligned}$$

The liquid phase heat transfer coefficient is calculated using:

$$\begin{aligned} h_l &= 0.027 \text{Re}_l^{4/5} \text{Pr}_l^{1/3} \left(\frac{k_l}{D} \right) \left(\frac{\mu_l}{\mu_s} \right)^{0.14} \\ &= 0.027(103100)^{4/5} (3.91)^{1/3} \left(\frac{0.637 \text{ W/m}\cdot\text{K}}{0.0254 \text{ m}} \right) \left(\frac{0.596 \times 10^{-3} \text{ kg/m}\cdot\text{s}}{0.355 \times 10^{-3} \text{ kg/m}\cdot\text{s}} \right)^{0.14} \\ &= 11754 \text{ W/m}^2 \cdot \text{K} \end{aligned}$$

where the *in situ* liquid Reynolds number is

$$\text{Re}_l = \frac{4\dot{m}_l}{\pi \sqrt{1 - \alpha} \mu_l D} = \frac{4(1.003 \text{ kg/s})}{\pi \sqrt{1 - 0.33} (0.596 \times 10^{-3} \text{ kg/m}\cdot\text{s})(0.0254 \text{ m})} = 103100$$

(a) The inclination factor (I^*) for horizontal tube ($\theta = 0^\circ$) is $I^* = 1$. Thus, using the general two-phase heat transfer correlation, the value for h_{tp} is estimated to be

$$\begin{aligned}
 \frac{h_{tp, \text{horiz}}}{h_l} &= F_p \left[1 + 0.55 \left(\frac{x}{1-x} \right)^{0.1} \left(\frac{1-F_p}{F_p} \right)^{0.4} \left(\frac{Pr_g}{Pr_l} \right)^{0.25} \left(\frac{\mu_l}{\mu_g} \right)^{0.25} (I^*)^{0.25} \right] \\
 &= (0.670) \left[1 + 0.55 \left(\frac{5.599 \times 10^{-4}}{1 - 5.599 \times 10^{-4}} \right)^{0.1} \left(\frac{1-0.670}{0.670} \right)^{0.4} \left(\frac{0.7241}{3.91} \right)^{0.25} \left(\frac{59.6}{1.941} \right)^{0.25} \right] \\
 &= 0.8727
 \end{aligned}$$

or

$$h_{tp, \text{horiz}} = 0.8727 h_l = 0.8727 (11754 \text{ W/m}^2 \cdot \text{K}) = \mathbf{10,300 \text{ W/m}^2 \cdot \text{K}}$$

(b) The inclination factor (I^*) for vertical tube ($\theta = 90^\circ$) is calculated to be

$$\begin{aligned}
 I^* &= 1 + \frac{(\rho_l - \rho_g) g D^2}{\sigma} |\sin \theta| \\
 &= 1 + \frac{(990.1 \text{ kg/m}^3 - 1.109 \text{ kg/m}^3) (9.81 \text{ m/s}^2) (0.0254 \text{ m})^2}{0.068 \text{ N/m}} |\sin 90^\circ| \\
 &= 93.05
 \end{aligned}$$

Thus, using the general two-phase heat transfer correlation, the value for h_{tp} is estimated to be

$$\begin{aligned}
 \frac{h_{tp, \text{vert}}}{h_l} &= F_p \left[1 + 0.55 \left(\frac{x}{1-x} \right)^{0.1} \left(\frac{1-F_p}{F_p} \right)^{0.4} \left(\frac{Pr_g}{Pr_l} \right)^{0.25} \left(\frac{\mu_l}{\mu_g} \right)^{0.25} (I^*)^{0.25} \right] \\
 &= (0.670) \left[1 + 0.55 \left(\frac{5.599 \times 10^{-4}}{1 - 5.599 \times 10^{-4}} \right)^{0.1} \left(\frac{1-0.670}{0.670} \right)^{0.4} \left(\frac{0.7241}{3.91} \right)^{0.25} \left(\frac{59.6}{1.941} \right)^{0.25} (93.05)^{0.25} \right] \\
 &= 1.30
 \end{aligned}$$

or

$$h_{tp, \text{vert}} = 1.30 h_l = 1.30 (11754 \text{ W/m}^2 \cdot \text{K}) = \mathbf{15,300 \text{ W/m}^2 \cdot \text{K}}$$

Discussion The two-phase heat transfer coefficient of vertical pipe is about 49% higher than that of horizontal pipe:

$$\frac{h_{tp, \text{vert}}}{h_{tp, \text{horiz}}} = \frac{15,300}{10,300} = 1.49$$

10-105 Air-water mixture flows through a 0.0254 m stainless steel pipe at specified flow condition. Using the concept of Reynolds analogy, the two phase convective heat transfer coefficient is to be determined.

Assumptions 1 Steady state operating conditions exist. 2 Two phase flow is non-boiling in nature and does not undergo any phase change. 3 Fluid properties are constant.

Properties Use the following thermo physical properties for water and air:

$$\rho_l = 997.1 \text{ kg/m}^3, \mu_l = 8.9 \times 10^{-4} \text{ kg/m} \cdot \text{s}, \mu_s = 4.66 \times 10^{-4} \text{ kg/m} \cdot \text{s}, \text{Pr}_l = 6.26, k_l = 0.595 \text{ W/m} \cdot \text{K},$$

$$\sigma = 0.0719 \text{ N/m}, \rho_g = 2.35 \text{ kg/m}^3, \text{ and } \mu_g = 1.84 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

Analysis We first need to determine superficial Reynolds number and the actual velocities for each phase.

$$\text{Re}_{sl} = \frac{\rho_l V_{sl} D}{\mu_l} = \frac{997.1(\text{kg/m}^3) \times 0.3(\text{m/s}) \times 0.0254(\text{m})}{8.9 \times 10^{-4} (\text{kg/m} \cdot \text{s})} = 8537$$

$$\text{Re}_{sg} = \frac{\rho_g V_{sg} D}{\mu_g} = \frac{2.35(\text{kg/m}^3) \times 23(\text{m/s}) \times 0.0254(\text{m})}{1.84 \times 10^{-5} (\text{kg/m} \cdot \text{s})} = 74612$$

The actual phase velocities are calculated from the known values of superficial velocities and void fraction.

$$V_l = \frac{V_{sl}}{1 - \alpha} = \frac{0.3(\text{m/s})}{1 - 0.86} = 2.14 \text{ m/s} \quad \text{and} \quad V_g = \frac{V_{sg}}{\alpha} = \frac{23(\text{m/s})}{0.86} = 26.74 \text{ m/s}$$

The mass flow rate of each phase is,

$$\dot{m}_l = \rho_l V_{sl} \frac{\pi}{4} D^2 = 997.1(\text{kg/m}^3) \times 0.3(\text{m/s}) \times \frac{\pi}{4} \times 0.0254^2 (\text{m}^2) = 0.15 \text{ kg/s}$$

$$\dot{m}_g = \rho_g V_{sg} \frac{\pi}{4} D^2 = 2.35(\text{kg/m}^3) \times 23(\text{m/s}) \times \frac{\pi}{4} \times 0.0254^2 (\text{m}^2) = 0.027 \text{ kg/s}$$

Thus the total mass flow rate is, $\dot{m} = \dot{m}_l + \dot{m}_g = 0.15 + 0.027 = 0.177 \text{ kg/s}$

Since this is a vertical upward flow of air water we can use the Reynolds analogy given by Equation (10-40).

$$\frac{h_{tp}}{h_l} = F_p^m \left(\frac{\dot{m}_l}{\dot{m}} \right) \left(\frac{\rho_{tp}}{\rho_l} \right)^n \phi_l^p$$

The single phase heat transfer coefficient is calculated as,

$$h_l = 0.027 \text{Re}_l^{4/5} \text{Pr}_l^{1/3} (k_l / D) (\mu_l / \mu_s)^{0.14}$$

The in-situ Reynolds number required in single phase heat transfer equation is calculated as,

$$\text{Re}_l = \frac{4\dot{m}_l}{\mu_l D \pi \sqrt{1 - \alpha}} = \frac{4 \times 0.15(\text{kg/s})}{8.9 \times 10^{-4} (\text{kg/m} \cdot \text{s}) \times 0.0254(\text{m}) \times \pi \sqrt{1 - 0.86}} = 22579$$

$$h_l = 0.027 \times 22579^{4/5} \times 6.26^{1/3} \times \frac{0.595(\text{W/m} \cdot \text{K})}{0.0254(\text{m})} \times \left(\frac{8.9 \times 10^{-4}}{4.66 \times 10^{-4}} \right)^{0.14} = 3878 \text{ W/m}^2 \cdot \text{K}$$

$$F_p = (1 - \alpha) + \alpha \left(\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{\rho_g (V_g - V_l)^2}{gD(\rho_l - \rho_g)}} \right) \right)^2$$

$$F_p = (1 - 0.86) + 0.86 \left(\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{2.35(\text{kg/m}^3)(26.74 - 2.14)^2}{9.81 \times 0.0254(\text{m}) \times (997.1 - 2.35)(\text{kg/m}^3)}} \right) \right)^2$$

$$\therefore F_p = 0.621$$

The two phase density required in Reynolds analogy equation is calculated as,

$$\rho_{tp} = (1-\alpha)\rho_l + \alpha\rho_g = (1-0.86)\times 997.1(\text{kg/m}^3) + 0.86\times 2.35(\text{kg/m}^3) = 141.6\text{kg/m}^3$$

Single phase pressure drop for turbulent pipe flow (with superficial Reynolds number of $Re_{sl} = 8537$) is calculated by first calculating the single phase friction factor from either the Moody chart (Fig. A-20) or the Colebrook equation (Eq.8-76) as follows,

$$\frac{1}{\sqrt{f_l}} = -2\log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re_{sl}\sqrt{f_l}}\right) = -2\log\left(\frac{0.002/25.4}{3.7} + \frac{2.51}{8537\sqrt{f_l}}\right) \rightarrow f_l = 0.0323$$

Where from Table 8-3 for stainless steel pipe, the roughness is $\varepsilon = 0.002$.

Single phase pressure drop is then calculated as,

$$(dP/dL)_{f,l} = \frac{f_l \rho_l V_{sl}^2}{2D} = \frac{0.0323 \times 997.1(\text{kg/m}^3) \times 0.3^2(\text{m}^2)}{2 \times 0.0254(\text{m})} = 57.06 \text{ Pa/m}$$

Thus the two-phase friction multiplier is,

$$\phi_l = \sqrt{\frac{(dP/dL)_{f,tp}}{(dP/dL)_{f,l}}} = \sqrt{\frac{2700(\text{Pa/m})}{57.06(\text{Pa/m})}} = 6.87$$

Thus the two-phase heat transfer coefficient calculated using Reynolds analogy (Eq. 10-40) is,

$$\frac{h_{tp}}{h_l} = F_p^m \left(\frac{\dot{m}_l}{\dot{m}} \right) \left(\frac{\rho_{tp}}{\rho_l} \right)^n \phi_l^p = 0.621^{0.5} \left(\frac{0.15}{0.177} \right) \left(\frac{141.6(\text{kg/m}^3)}{997.1(\text{kg/m}^3)} \right)^{-0.5} 6.87^{0.2}$$

$$\frac{h_{tp}}{h_l} = 2.605$$

Thus the two phase heat transfer coefficient is: $h_{tp} = 10102 \text{ W/m}^2\cdot\text{K}$

Discussion The use of Reynolds analogy strongly depends on the correct estimation of the two-phase pressure drop. Most of the correlations available for two-phase pressure drop fail to correctly estimate the two-phase pressure drop and hence based on the calculated value of two-phase pressure drop, the Reynolds analogy may not predict the two- phase heat transfer coefficient correctly.

Review Problems

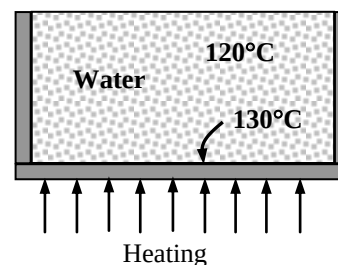
10-106 Water is boiled at $T_{\text{sat}} = 120^\circ\text{C}$ in a mechanically polished stainless steel pressure cooker whose inner surface temperature is maintained at $T_s = 130^\circ\text{C}$. The time it will take for the tank to empty is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the boiler are negligible.

Properties The properties of water at the saturation temperature of 120°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 943.4 \text{ kg/m}^3 & h_{fg} &= 2203 \times 10^3 \text{ J/kg} \\ \rho_v &= 1.121 \text{ kg/m}^3 & \mu_l &= 0.232 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \sigma &= 0.0550 \text{ N/m} & c_{pl} &= 4244 \text{ J/kg} \cdot ^\circ\text{C} \\ \text{Pr}_l &= 1.44\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.



Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 130 - 120 = 10^\circ\text{C}$ which is relatively low (less than 30°C). Therefore, nucleate boiling will occur. The heat flux in this case can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.232 \times 10^{-3})(2203 \times 10^3) \left[\frac{9.81(943.4 - 1.121)}{0.0550} \right]^{1/2} \left(\frac{4244(130 - 120)}{0.0130(2203 \times 10^3)1.44} \right)^3 \\ &= 228,400 \text{ W/m}^2\end{aligned}$$

The rate of heat transfer is

$$\dot{Q} = A \dot{q}_{\text{nucleate}} = \frac{1}{4} \pi (0.20 \text{ m})^2 (228,400 \text{ W/m}^2) = 7174 \text{ W}$$

The rate of evaporation is

$$\dot{m}_{\text{evap}} = \frac{\dot{Q}}{h_{fg}} = \frac{7174 \text{ W}}{2203 \times 10^3 \text{ kJ/kg}} = 0.003256 \text{ kg/s}$$

Noting that the tank is half-filled, the mass of the water and the time it will take for the tank to empty are

$$m = \frac{1}{2} \rho_l V = \frac{1}{2} (943.4 \text{ kg/m}^3) \left[\pi (0.20 \text{ m})^2 / 4 \times (0.30 \text{ m}) \right] = 4.446 \text{ kg}$$

$$t = \frac{m}{\dot{m}_{\text{evap}}} = \frac{4.446 \text{ kg}}{0.003256 \text{ kg/s}} = 1365 \text{ s} = \mathbf{22.8 \text{ min}}$$

10-107 Water is boiled at sea level (1 atm pressure) and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 100^\circ\text{C}$ in a mechanically polished AISI 304 stainless steel pan placed on top of a 3-kW electric burner. Only 60% of the heat (1.8 kW) generated is transferred to the water. The inner surface temperature of the pan and the temperature difference across the bottom of the pan are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later). 4 Heat transfer through the bottom of the pan is one-dimensional.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\rho_l = 957.9 \text{ kg/m}^3$$

$$\rho_v \approx 0.60 \text{ kg/m}^3$$

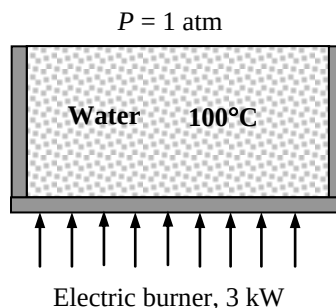
$$\sigma = 0.0589 \text{ N/m}$$

$$\text{Pr}_l = 1.75$$

$$h_{fg} = 2257 \times 10^3 \text{ J/kg}$$

$$\mu_l = 0.282 \times 10^{-3} \text{ kg} \cdot \text{m/s}$$

$$c_{pl} = 4217 \text{ J/kg} \cdot \text{K}$$



Also, $k_{\text{steel}} = 14.9 \text{ W/m} \cdot \text{K}$ (Table A-3), $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis The rate of heat transfer to the water and the heat flux are

$$\dot{Q} = 0.60 \times 3 \text{ kW} = 1.8 \text{ kW} = 1800 \text{ W}$$

$$A_s = \pi D^2 / 4 = \pi (0.30 \text{ m})^2 / 4 = 0.07069 \text{ m}^2$$

$$\dot{q} = \dot{Q} / A_s = (1800 \text{ W}) / (0.07069 \text{ m}^2) = 25.46 \text{ W/m}^2$$

Then temperature difference across the bottom of the pan is determined directly from the steady one-dimensional heat conduction relation to be

$$\dot{q} = k_{\text{steel}} \frac{\Delta T}{L} \rightarrow \Delta T = \frac{\dot{q} L}{k_{\text{steel}}} = \frac{(25,460 \text{ W/m}^2)(0.006 \text{ m})}{14.9 \text{ W/m} \cdot \text{K}} = \mathbf{10.3^\circ\text{C}}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given.

Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3$$

$$25,460 = (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3$$

It gives

$$T_s = \mathbf{105.7^\circ\text{C}}$$

which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

10-108 Water is boiled at 84.5 kPa pressure and thus at a saturation (or boiling) temperature of $T_{\text{sat}} = 95^\circ\text{C}$ in a mechanically polished AISI 304 stainless steel pan placed on top of a 3-kW electric burner. Only 60% of the heat (1.8 kW) generated is transferred to the water. The inner surface temperature of the pan and the temperature difference across the bottom of the pan are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible. 3 The boiling regime is nucleate boiling (this assumption will be checked later). 4 Heat transfer through the bottom of the pan is one-dimensional.

Properties The properties of water at the saturation temperature of 95°C are (Tables 10-1 and A-9)

$$\rho_l = 961.5 \text{ kg/m}^3$$

$$\rho_v \approx 0.50 \text{ kg/m}^3$$

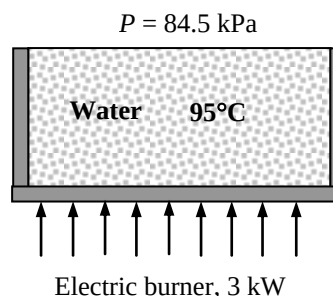
$$\sigma = 0.0599 \text{ N/m}$$

$$\text{Pr}_l = 1.85$$

$$h_{fg} = 2270 \times 10^3 \text{ J/kg}$$

$$\mu_l = 0.297 \times 10^{-3} \text{ kg} \cdot \text{m/s}$$

$$c_{pl} = 4212 \text{ J/kg} \cdot \text{K}$$



Also, $k_{\text{steel}} = 14.9 \text{ W/m} \cdot \text{K}$ (Table A-3), $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis The rate of heat transfer to the water and the heat flux are

$$\dot{Q} = 0.60 \times 3 \text{ kW} = 1.8 \text{ kW} = 1800 \text{ W}$$

$$A_s = \pi D^2 / 4 = \pi (0.30 \text{ m})^2 / 4 = 0.07069 \text{ m}^2$$

$$\dot{q} = \dot{Q} / A_s = (1800 \text{ W}) / (0.07069 \text{ m}^2) = 25,460 \text{ W/m}^2 = 25.46 \text{ kW/m}^2$$

Then temperature difference across the bottom of the pan is determined directly from the steady one-dimensional heat conduction relation to be

$$\dot{q} = k_{\text{steel}} \frac{\Delta T}{L} \rightarrow \Delta T = \frac{\dot{q} L}{k_{\text{steel}}} = \frac{(25,460 \text{ W/m}^2)(0.006 \text{ m})}{14.9 \text{ W/m} \cdot \text{K}} = \mathbf{10.3^\circ\text{C}}$$

The Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given.

Assuming nucleate boiling, the temperature of the inner surface of the pan is determined from Rohsenow relation to be

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3$$

$$25,460 = (0.297 \times 10^{-3})(2270 \times 10^3) \left[\frac{9.81(961.5 - 0.50)}{0.0599} \right]^{1/2} \left(\frac{4212(T_s - 95)}{0.0130(2270 \times 10^3)1.85} \right)^3$$

It gives

$$T_s = \mathbf{100.9^\circ\text{C}}$$

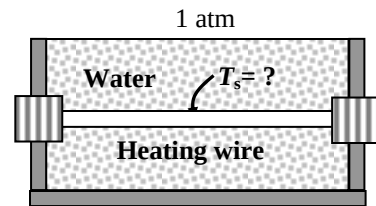
which is in the nucleate boiling range (5 to 30°C above surface temperature). Therefore, the nucleate boiling assumption is valid.

10-109 Water is boiled at 1 atm pressure and thus at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ by a nickel electric heater whose diameter is 2 mm. The highest temperature at which this heater can operate without burnout is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the water are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &\approx 0.60 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg} \cdot \text{m/s} \\ \sigma &= 0.0589 \text{ N/m} & c_{pl} &= 4217 \text{ J/kg} \cdot \text{K} \\ \text{Pr}_l &= 1.75\end{aligned}$$



Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a nickel surface (Table 10-3).

Analysis The maximum rate of heat transfer without the burnout is simply the critical heat flux. For a horizontal heating wire, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned}L^* &= L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.001) \left(\frac{9.81(957.9 - 0.60)}{0.0589} \right)^{1/2} = 0.399 < 1.2 \\ C_{cr} &= 0.12 L^{*-0.25} = 0.12(0.399)^{-0.25} = 0.151\end{aligned}$$

Then the maximum or critical heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.151(2257 \times 10^3) [0.0589 \times 9.8 \times (0.6)^2 (957.9 - 0.60)]^{1/4} \\ &= 1,280,000 \text{ W/m}^2\end{aligned}$$

Rohsenow relation which gives the nucleate boiling heat flux for a specified surface temperature can also be used to determine the surface temperature when the heat flux is given. Substituting the maximum heat flux into Rohsenow relation together with other properties gives

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ 1,280,000 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(T_s - 100)}{0.0060(2257 \times 10^3)1.75} \right)^3\end{aligned}$$

It gives the maximum temperature to be:

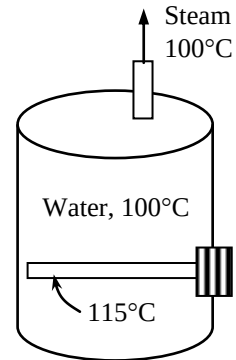
$$T_s = 109.6^\circ\text{C}$$

10-110 Water is boiled at $T_{\text{sat}} = 100^\circ\text{C}$ by a chemically etched stainless steel electric heater whose surface temperature is maintained at $T_s = 115^\circ\text{C}$. The rate of heat transfer to the water, the rate of evaporation of water, and the maximum rate of evaporation are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Tables 10-1 and A-9)

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 \\ \rho_v &= 0.60 \text{ kg/m}^3 \\ \sigma &= 0.0589 \text{ N/m} \\ \text{Pr}_l &= 1.75 \\ h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \mu_l &= 0.282 \times 10^{-3} \text{ kg} \cdot \text{m/s} \\ c_{pl} &= 4217 \text{ J/kg} \cdot ^\circ\text{C}\end{aligned}$$



Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a chemically etched stainless steel surface (Table 10-3). Note that we expressed the properties in units specified under Eq. 10-2 in connection with their definitions in order to avoid unit manipulations.

Analysis (a) The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 115 - 100 = 15^\circ\text{C}$ which is relatively low (less than 30°C). Therefore, nucleate boiling will occur. The heat flux in this case can be determined from Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left(\frac{c_{p,l}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right)^3 \\ &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.60)}{0.0589} \right]^{1/2} \left(\frac{4217(115 - 100)}{0.0130(2257 \times 10^3)1.75} \right)^3 \\ &= 474,900 \text{ W/m}^2\end{aligned}$$

The surface area of the bottom of the heater is $A_s = \pi DL = \pi(0.002 \text{ m})(0.8 \text{ m}) = 0.005027 \text{ m}^2$.

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A_s \dot{q}_{\text{nucleate}} = (0.005027 \text{ m}^2)(474,900 \text{ W/m}^2) = \mathbf{2387 \text{ W}}$$

The rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{2387 \text{ J/s}}{2257 \times 10^3 \text{ J/kg}} = \mathbf{1.058 \times 10^{-3} \text{ kg/s} = 3.81 \text{ kg/h}}$$

(b) For a horizontal heating wire, the coefficient C_{cr} is determined from Table 10-4 to be

$$\begin{aligned}L^* &= L \left(\frac{g(\rho_l - \rho_v)}{\sigma} \right)^{1/2} = (0.001) \left(\frac{9.81(957.9 - 0.60)}{0.0589} \right)^{1/2} = 0.399 < 1.2 \\ C_{cr} &= 0.12 L^{*-0.25} = 0.12(0.399)^{-0.25} = 0.151\end{aligned}$$

Then the maximum or critical heat flux is determined from

$$\begin{aligned}\dot{q}_{\text{max}} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} = 0.151(2257 \times 10^3)[0.0589 \times 9.8 \times (0.6)^2 (957.9 - 0.60)]^{1/4} \\ &= 1,280,000 \text{ W/m}^2 = \mathbf{1280 \text{ kW/m}^2}\end{aligned}$$

10-111 The initial boiling heat transfer coefficient and the total heat transfer coefficient, when a heated steel rod was submerged in a water bath, are to be determined.

Assumptions 1 Steady operating condition exists. 2 The steel rod has uniform initial surface temperature.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \text{ kJ/kg}$ (Table A-2) and $\rho_l = 957.9 \text{ kg/m}^3$ (Table A-9). The properties of vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 300^\circ\text{C}$ are, from Table A-16,

$$\begin{aligned}\rho_v &= 0.3831 \text{ kg/m}^3 & c_{pv} &= 1997 \text{ J/kg}\cdot\text{K} \\ \mu_v &= 2.045 \times 10^{-5} \text{ kg/m}\cdot\text{s} & k_v &= 0.04345 \text{ W/m}\cdot\text{K}\end{aligned}$$

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 400^\circ\text{C}$, which is much larger than 30°C for water from Fig. 10-6. Therefore, film boiling will occur. The film boiling heat flux in this case can be determined from

$$\begin{aligned}\dot{q}_{\text{film}} &= C_{\text{film}} \left[\frac{g k_v^3 \rho_l (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.62 \left[\frac{9.81 (0.04345)^3 (0.3831) (957.9 - 0.3831) [2257 \times 10^3 + 0.4 (1997) (400)]}{(2.045 \times 10^{-5}) (0.02) (400)} \right]^{1/4} (400) \\ &= 6.476 \times 10^4 \text{ W/m}^2\end{aligned}$$

Using the Newton's law of cooling, the boiling heat transfer coefficient is

$$\begin{aligned}\dot{q}_{\text{film}} &= h_{\text{film}} (T_s - T_{\text{sat}}) \quad \rightarrow \quad h_{\text{film}} = \frac{\dot{q}_{\text{film}}}{T_s - T_{\text{sat}}} \\ h_{\text{film}} &= \frac{6.476 \times 10^4 \text{ W/m}^2}{(500 - 100) \text{ K}} = \mathbf{162 \text{ W/m}^2 \cdot \text{K}}\end{aligned}$$

The radiation heat transfer coefficient can be determined using

$$\begin{aligned}\dot{q}_{\text{rad}} &= \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) = h_{\text{rad}} (T_s - T_{\text{sat}}) \quad \rightarrow \quad h_{\text{rad}} = \frac{\varepsilon \sigma (T_s^4 - T_{\text{sat}}^4)}{T_s - T_{\text{sat}}} \\ h_{\text{rad}} &= \frac{\varepsilon \sigma (T_s^4 - T_{\text{sat}}^4)}{T_s - T_{\text{sat}}} = \frac{(0.9) (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) (773^4 - 373^4) \text{ K}^4}{(500 - 100) \text{ K}} = 43.08 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Then, the total heat transfer coefficient can be determined using

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} \quad \rightarrow \quad h_{\text{total}} (T_s - T_{\text{sat}}) = h_{\text{film}} (T_s - T_{\text{sat}}) + \frac{3}{4} h_{\text{rad}} (T_s - T_{\text{sat}})$$

or

$$\begin{aligned}h_{\text{total}} &= h_{\text{film}} + \frac{3}{4} h_{\text{rad}} \\ &= 162 \text{ W/m}^2 \cdot \text{K} + \frac{3}{4} (43.08 \text{ W/m}^2 \cdot \text{K}) \\ &= \mathbf{194 \text{ W/m}^2 \cdot \text{K}}\end{aligned}$$

Discussion The boiling heat transfer coefficient (h_{film}) is 3.76 times the radiation heat transfer coefficient (h_{rad}).

10-112 The boiling heat transfer coefficient and the total heat transfer coefficient for water being boiled by a cylindrical metal rod are to be determined.

Assumptions 1 Steady operating condition exists. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \text{ kJ/kg}$ (Table A-2) and $\rho_l = 957.9 \text{ kg/m}^3$ (Table A-9). The properties of vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 300^\circ\text{C}$ are, from Table A-16,

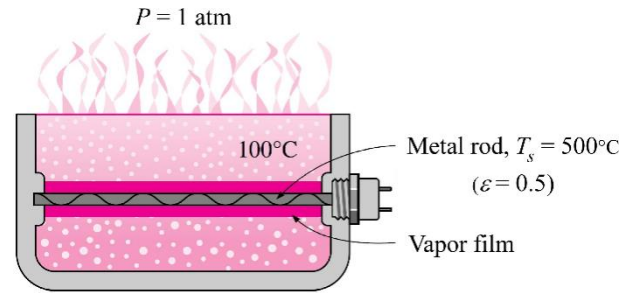
$$\rho_v = 0.3831 \text{ kg/m}^3$$

$$c_{pv} = 1997 \text{ J/kg}\cdot\text{K}$$

$$\mu_v = 2.045 \times 10^{-5} \text{ kg/m}\cdot\text{s}$$

$$k_v = 0.04345 \text{ W/m}\cdot\text{K}$$

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 400^\circ\text{C}$, which is much larger than 30°C for water from Fig 10-6. Therefore, film boiling will occur. The film boiling heat flux in this case can be determined from



$$\begin{aligned}\dot{q}_{\text{film}} &= C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.62 \left[\frac{9.81 (0.04345)^3 (0.3831) (957.9 - 0.3831) [2257 \times 10^3 + 0.4 (1997) (400)]}{(2.045 \times 10^{-5}) (0.002) (400)} \right]^{1/4} (400) \\ &= 1.152 \times 10^5 \text{ W/m}^2\end{aligned}$$

Using the Newton's law of cooling, the boiling heat transfer coefficient is

$$\begin{aligned}\dot{q}_{\text{film}} &= h_{\text{film}} (T_s - T_{\text{sat}}) \quad \rightarrow \quad h_{\text{film}} = \frac{\dot{q}_{\text{film}}}{T_s - T_{\text{sat}}} \\ h_{\text{film}} &= \frac{1.152 \times 10^5 \text{ W/m}^2}{(500 - 100) \text{ K}} = 288 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

The radiation heat transfer coefficient can be determined using

$$\begin{aligned}\dot{q}_{\text{rad}} &= \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) = h_{\text{rad}} (T_s - T_{\text{sat}}) \quad \rightarrow \quad h_{\text{rad}} = \frac{\varepsilon \sigma (T_s^4 - T_{\text{sat}}^4)}{T_s - T_{\text{sat}}} \\ h_{\text{rad}} &= \frac{\varepsilon \sigma (T_s^4 - T_{\text{sat}}^4)}{T_s - T_{\text{sat}}} = \frac{(0.5) (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) (773^4 - 373^4) \text{ K}^4}{(500 - 100) \text{ K}} = 23.93 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Then, the total heat transfer coefficient can be determined using

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} \quad \rightarrow \quad h_{\text{total}} (T_s - T_{\text{sat}}) = h_{\text{film}} (T_s - T_{\text{sat}}) + \frac{3}{4} h_{\text{rad}} (T_s - T_{\text{sat}})$$

or

$$\begin{aligned}h_{\text{total}} &= h_{\text{film}} + \frac{3}{4} h_{\text{rad}} \\ &= 288 \text{ W/m}^2 \cdot \text{K} + \frac{3}{4} (23.93 \text{ W/m}^2 \cdot \text{K}) \\ &= 306 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Discussion The boiling heat transfer coefficient (h_{film}) is about 12 times the radiation heat transfer coefficient (h_{rad}).

10-113 Water is boiled at $T_{\text{sat}} = 100^\circ\text{C}$ by a spherical platinum heating element immersed in water. The surface temperature is $T_s = 350^\circ\text{C}$. The boiling heat transfer coefficient is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are (Table A-9)

$$h_{fg} = 2257 \times 10^3 \text{ J/kg}$$

$$\rho_l = 957.9 \text{ kg/m}^3$$

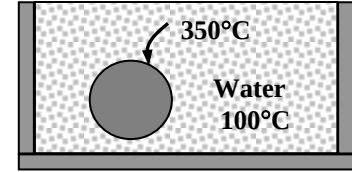
The properties of water vapor at $(350+100)/2 = 225^\circ\text{C}$ are (Table A-16)

$$\rho_v = 0.444 \text{ kg/m}^3$$

$$\mu_v = 1.749 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$c_{pv} = 1951 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_v = 0.03581 \text{ W/m} \cdot ^\circ\text{C}$$



Analysis The film boiling occurs since the temperature difference between the surface and the fluid. The heat flux in this case can be determined from

$$\begin{aligned} \dot{q}_{\text{film}} &= 0.67 \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.67 \left[\frac{(9.81)(0.03581)^3 (0.444)(957.9 - 0.444) [2257 \times 10^3 + 0.4(1951)(350 - 100)]}{(1.749 \times 10^{-5})(0.08)(350 - 100)} \right]^{1/4} (350 - 100) \\ &= 32,062 \text{ W/m}^2 \end{aligned}$$

The boiling heat transfer coefficient is

$$\dot{q}_{\text{film}} = h(T_s - T_{\text{sat}}) \longrightarrow h = \frac{\dot{q}_{\text{film}}}{T_s - T_{\text{sat}}} = \frac{32,062 \text{ W/m}^2}{(350 - 100)^\circ\text{C}} = \mathbf{128 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

10-114 The initial boiling heat transfer coefficient and the total heat transfer coefficient, when heated steel ball bearings are submerged in a water bath, are to be determined.

Assumptions 1 Steady operating condition exists. 2 The steel ball bearings have uniform initial surface temperature.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \text{ kJ/kg}$ (Table A-2) and $\rho_l = 957.9 \text{ kg/m}^3$ (Table A-9). The properties of vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = 400^\circ\text{C}$ are, from Table A-16,

$$\begin{aligned}\rho_v &= 0.3262 \text{ kg/m}^3 & c_{pv} &= 2066 \text{ J/kg}\cdot\text{K} \\ \mu_v &= 2.446 \times 10^{-5} \text{ kg/m}\cdot\text{s} & k_v &= 0.05467 \text{ W/m}\cdot\text{K}\end{aligned}$$

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 600^\circ\text{C}$, which is much larger than 30°C for water in Fig. 10-6. Therefore, film boiling will occur. The film boiling heat flux in this case can be determined from

$$\begin{aligned}\dot{q}_{\text{film}} &= C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.67 \left[\frac{9.81 (0.05467)^3 (0.3262) (957.9 - 0.3262) [2257 \times 10^3 + 0.4 (2066) (600)]}{(2.446 \times 10^{-5}) (0.02) (600)} \right]^{1/4} (600) \\ &= 1.052 \times 10^5 \text{ W/m}^2\end{aligned}$$

Using the Newton's law of cooling, the boiling heat transfer coefficient is

$$\begin{aligned}\dot{q}_{\text{film}} &= h_{\text{film}} (T_s - T_{\text{sat}}) \quad \rightarrow \quad h_{\text{film}} = \frac{\dot{q}_{\text{film}}}{T_s - T_{\text{sat}}} \\ h_{\text{film}} &= \frac{1.052 \times 10^5 \text{ W/m}^2}{(700 - 100) \text{ K}} = 175 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

The radiation heat transfer coefficient can be determined using

$$\begin{aligned}\dot{q}_{\text{rad}} &= \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) = h_{\text{rad}} (T_s - T_{\text{sat}}) \quad \rightarrow \quad h_{\text{rad}} = \frac{\varepsilon \sigma (T_s^4 - T_{\text{sat}}^4)}{T_s - T_{\text{sat}}} \\ h_{\text{rad}} &= \frac{\varepsilon \sigma (T_s^4 - T_{\text{sat}}^4)}{T_s - T_{\text{sat}}} = \frac{(0.75) (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) (973^4 - 373^4) \text{ K}^4}{(700 - 100) \text{ K}} = 62.15 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Then, the total heat transfer coefficient can be determined using

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} \quad \rightarrow \quad h_{\text{total}} (T_s - T_{\text{sat}}) = h_{\text{film}} (T_s - T_{\text{sat}}) + \frac{3}{4} h_{\text{rad}} (T_s - T_{\text{sat}})$$

or

$$\begin{aligned}h_{\text{total}} &= h_{\text{film}} + \frac{3}{4} h_{\text{rad}} \\ &= 175 \text{ W/m}^2 \cdot \text{K} + \frac{3}{4} (62.15 \text{ W/m}^2 \cdot \text{K}) \\ &= 222 \text{ W/m}^2 \cdot \text{K}\end{aligned}$$

Discussion The boiling heat transfer coefficient (h_{film}) is 2.82 times the radiation heat transfer coefficient (h_{rad}).

10-115 Steam at a saturation temperature of $T_{\text{sat}} = 35^\circ\text{C}$ condenses on a vertical plate which is maintained at 25°C . The rate of heat transfer to the plate and the rate of condensation of steam per m width of the plate are to be determined.

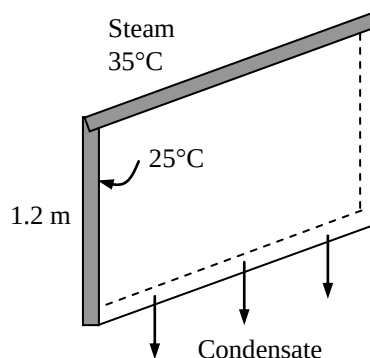
Assumptions 1 Steady operating conditions exist. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the entire plate (this assumption will be verified). 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 35°C are $h_{fg} = 2419 \text{ kJ/kg}$ and $\rho_v = 0.0397 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (35 + 25) / 2 = 30^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 996.0 \text{ kg/m}^3 \\ \mu_l &= 0.798 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ \nu_l &= \mu_l / \rho_l = 0.801 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4178 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.615 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2419 \text{ kJ/kg} + 0.68 \times (4178 \text{ J/kg} \cdot ^\circ\text{C})(35 - 25)^\circ\text{C} \\ &= 2447 \text{ kJ/kg}\end{aligned}$$



Assuming wavy-laminar flow, the Reynolds number is determined from

$$\begin{aligned}\text{Re} &= \text{Re}_{\text{vertical, wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70 \times (1.2 \text{ m}) \times (0.615 \text{ W/m} \cdot ^\circ\text{C}) \times (35 - 25)^\circ\text{C}}{(0.798 \times 10^{-3} \text{ kg/m} \cdot \text{s}) (2447 \times 10^3 \text{ J/kg})} \left(\frac{9.81 \text{ m/s}^2}{(0.801 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.820} = 122\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{vertical, wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{122 \times (0.615 \text{ W/m} \cdot ^\circ\text{C})}{1.08(122)^{1.22} - 5.2} \left(\frac{9.81 \text{ m/s}^2}{(0.801 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 4980 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

The heat transfer surface area of the plate is

$$A_s = W \times L = (1.2 \text{ m})(1 \text{ m}) = 1.2 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (4980 \text{ W/m}^2 \cdot ^\circ\text{C})(1.2 \text{ m}^2)(35 - 25)^\circ\text{C} = \mathbf{59,760 \text{ W}}$$

The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{59.76 \text{ kJ/s}}{2447 \text{ kJ/kg}} = 0.0244 \text{ kg/s} = \mathbf{87.9 \text{ kg/h}}$$

10-116 Saturated ammonia at a saturation temperature of $T_{\text{sat}} = 25^\circ\text{C}$ condenses on the outer surface of vertical tube which is maintained at 15°C by circulating cooling water. The rate of heat transfer to the coolant and the rate of condensation of ammonia are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal. 3 The tube can be treated as a vertical plate. 4 The condensate flow is turbulent over the entire tube (this assumption will be verified). 5 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of ammonia at the saturation temperature of 25°C are $h_{fg} = 1166 \times 10^3 \text{ J/kg}$ and $\rho_v = 7.809 \text{ kg/m}^3$. The properties of liquid ammonia at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (25 + 15)/2 = 20^\circ\text{C}$ are (Table A-11),

$$\rho_l = 610.2 \text{ kg/m}^3$$

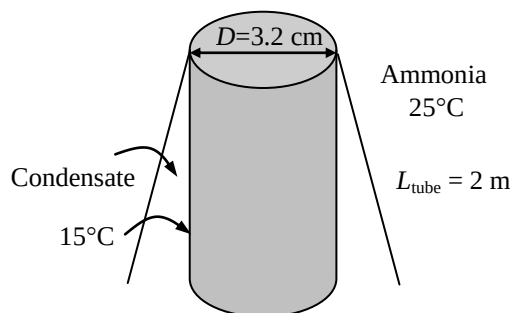
$$\mu_l = 1.519 \times 10^{-4} \text{ kg/m} \cdot \text{s}$$

$$\nu_l = \mu_l / \rho_l = 0.2489 \times 10^{-6} \text{ m}^2/\text{s}$$

$$c_{pl} = 4745 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_l = 0.4927 \text{ W/m} \cdot ^\circ\text{C}$$

$$\text{Pr}_l = 1.463$$



Analysis (a) The modified latent heat of vaporization is

$$h_{fg}^* = h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ = 1166 \times 10^3 \text{ J/kg} + 0.68 \times 4745 \text{ J/kg} \cdot ^\circ\text{C} (25 - 15)^\circ\text{C} = 1198 \times 10^3 \text{ J/kg}$$

Assuming turbulent flow, the Reynolds number is determined from

$$\text{Re} = \text{Re}_{\text{vertical, turb}} = \left[\frac{0.0690 L k_l \text{Pr}^{0.5} (T_{\text{sat}} - T_s) \left(\frac{g}{\nu_l^2} \right)^{1/3} - 151 \text{Pr}^{0.5} + 253}{\mu_l h_{fg}^*} \right]^{4/3} \\ = \left[\frac{0.0690 \times 2 \times 0.4927 (1.463)^{0.5} (25 - 15) \left(\frac{9.81}{(0.2489 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} - 151 (1.463)^{0.5} + 253}{(1.519 \times 10^{-4} \text{ kg/m} \cdot \text{s}) (1198 \times 10^3 \text{ J/kg})} \right]^{4/3} \\ = 2142$$

which is greater than 1800, and thus our assumption of turbulent flow is verified. Then the condensation heat transfer coefficient is determined from

$$h = h_{\text{vertical, turbulent}} = \frac{\text{Re} k_l}{8750 + 58 \text{Pr}^{-0.5} (\text{Re}^{0.75} - 253)} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ = \frac{2142 \times (0.4927 \text{ W/m} \cdot ^\circ\text{C})}{8750 + 58 \times 1.463^{-0.5} (2142^{0.75} - 253)} \left(\frac{9.81 \text{ m/s}^2}{(0.2489 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 4873 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The heat transfer surface area of the tube is $A_s = \pi DL = \pi(0.032 \text{ m})(2 \text{ m}) = 0.2011 \text{ m}^2$. Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (4873 \text{ W/m}^2 \cdot ^\circ\text{C})(0.2011 \text{ m}^2)(25 - 15)^\circ\text{C} = \mathbf{9800 \text{ W}}$$

(b) The rate of condensation of ammonia is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{9800 \text{ J/s}}{1198 \times 10^3 \text{ J/kg}} = \mathbf{8.180 \times 10^{-3} \text{ kg/s}}$$

Discussion Combining equations $\delta_L = k_l / h_l$ and $h = (4/3)h_L$, the thickness of the liquid film at the bottom of the tube is determined to be

$$\delta_L = \frac{4k_l}{3h} = \frac{4(0.4927 \text{ W/m} \cdot ^\circ\text{C})}{3(4873 \text{ W/m}^2 \cdot ^\circ\text{C})} = 0.135 \times 10^{-3} \text{ m} = 0.135 \text{ mm}$$

The assumption that the tube diameter is large relative to the thickness of the liquid film at the bottom of the tube is verified since the thickness of the liquid film is 0.135 mm, which is much smaller than the diameter of the tube (3.2 cm). Also, the assumption of turbulent flow is verified since Reynolds number is greater than 1800.

10-117 Saturated refrigerant-134a vapor condenses on the outside of a horizontal tube maintained at a specified temperature. The rate of condensation of the refrigerant is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal.

Properties The properties of refrigerant-134a at the saturation temperature of 35°C are $h_{fg} = 168.2 \times 10^3 \text{ J/kg}$ and $\rho_v = 43.41 \text{ kg/m}^3$. The properties of liquid R-134a at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (35 + 25) / 2 = 30^\circ\text{C}$ are (Table A-10),

$$\rho_l = 1188 \text{ kg/m}^3$$

$$\mu_l = 1.888 \times 10^{-4} \text{ kg/m}\cdot\text{s}$$

$$c_{pl} = 1448 \text{ J/kg}\cdot^\circ\text{C}$$

$$k_l = 0.0808 \text{ W/m}\cdot^\circ\text{C}$$

Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 168.2 \times 10^3 \text{ J/kg} + 0.68 \times 1448 \text{ J/kg}\cdot^\circ\text{C}(35 - 25)^\circ\text{C} = 178.0 \times 10^3 \text{ J/kg} \end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned} h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(1188 \text{ kg/m}^3)(1188 - 43.41 \text{ kg/m}^3)(178.0 \times 10^3 \text{ J/kg})(0.0808 \text{ W/m}\cdot^\circ\text{C})^3}{(1.888 \times 10^{-4} \text{ kg/m}\cdot\text{s})(35 - 25)^\circ\text{C}(0.012 \text{ m})} \right]^{1/4} \\ &= 1988 \text{ W/m}^2\cdot^\circ\text{C} \end{aligned}$$

The heat transfer surface area of the pipe is

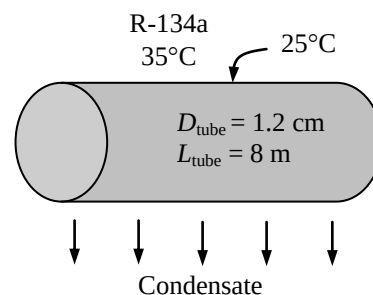
$$A_s = \pi DL = \pi(0.012 \text{ m})(8 \text{ m}) = 0.3016 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (1988 \text{ W/m}^2\cdot^\circ\text{C})(0.3016 \text{ m}^2)(35 - 25)^\circ\text{C} = 5996 \text{ W}$$

The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{5996 \text{ J/s}}{178.0 \times 10^3 \text{ J/kg}} = 0.03368 \text{ kg/s} = \mathbf{2.02 \text{ kg/min}}$$



10-118 Saturated refrigerant-134a vapor condenses on the outside of a horizontal tube maintained at a specified temperature. The rate of condensation of the refrigerant is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal.

Properties The properties of refrigerant-134a at the saturation temperature of 35°C are $h_{fg} = 168.2 \times 10^3 \text{ J/kg}$ and $\rho_v = 43.41 \text{ kg/m}^3$. The properties of liquid R-134a at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (35 + 25) / 2 = 30^\circ\text{C}$ are (Table A-10),

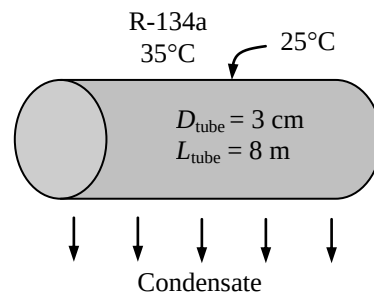
$$\rho_l = 1188 \text{ kg/m}^3$$

$$\mu_l = 1.888 \times 10^{-4} \text{ kg/m}\cdot\text{s}$$

$$\nu_l = \mu_l / \rho_l = 0.1590 \times 10^{-6} \text{ m}^2/\text{s}$$

$$c_{pl} = 1448 \text{ J/kg}\cdot^\circ\text{C}$$

$$k_l = 0.0808 \text{ W/m}\cdot^\circ\text{C}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 168.2 \times 10^3 \text{ J/kg} + 0.68 \times 1448 \text{ J/kg}\cdot^\circ\text{C}(35 - 25)^\circ\text{C} = 178.0 \times 10^3 \text{ J/kg} \end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned} h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(1188 \text{ kg/m}^3)(1188 - 43.41 \text{ kg/m}^3)(178.0 \times 10^3 \text{ J/kg})(0.0808 \text{ W/m}\cdot^\circ\text{C})^3}{(1.888 \times 10^{-4} \text{ kg/m}\cdot\text{s})(35 - 25)^\circ\text{C}(0.03 \text{ m})} \right]^{1/4} \\ &= 1581 \text{ W/m}^2\cdot^\circ\text{C} \end{aligned}$$

The heat transfer surface area of the pipe is

$$A_s = \pi DL = \pi(0.03 \text{ m})(8 \text{ m}) = 0.7540 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (1581 \text{ W/m}^2\cdot^\circ\text{C})(0.7540 \text{ m}^2)(35 - 25)^\circ\text{C} = 11,920 \text{ W}$$

The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{11,920 \text{ J/s}}{178.0 \times 10^3 \text{ J/kg}} = 0.06697 \text{ kg/s} = \mathbf{4.02 \text{ kg/min}}$$

10-119 Steam at a saturation temperature of $T_{\text{sat}} = 40^\circ\text{C}$ condenses on the outside of a thin horizontal tube. Heat is transferred to the cooling water that enters the tube at 10°C and exits at 30°C . The rate of condensation of steam, the average overall heat transfer coefficient, and the tube length are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube can be taken to be isothermal at the bulk mean fluid temperature in the evaluation of the condensation heat transfer coefficient. 3 Liquid flow through the tube is fully developed. 4 The thickness and the thermal resistance of the tube is negligible.

Properties The properties of water at the saturation temperature of 40°C are $h_{fg} = 2407 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.05 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (40 + 20) / 2 = 30^\circ\text{C}$ and at the bulk fluid temperature of $T_b = (T_{\text{in}} + T_{\text{out}}) / 2 = (10 + 30) / 2 = 20^\circ\text{C}$ are (Table A-9),

At 30°C :

$$\rho_l = 996.0 \text{ kg/m}^3$$

$$\mu_l = 0.798 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

$$\nu_l = \mu_l / \rho_l = 0.801 \times 10^{-6} \text{ m}^2/\text{s}$$

$$c_{pl} = 4178 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_l = 0.615 \text{ W/m} \cdot ^\circ\text{C}$$

$$\text{Pr} = 5.42$$

At 20°C :

$$\rho_l = 998.0 \text{ kg/m}^3$$

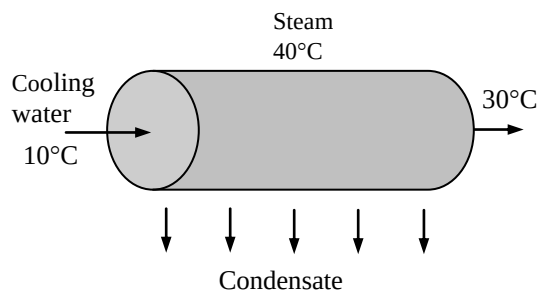
$$\mu_l = 1.002 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

$$\nu_l = \mu_l / \rho_l = 1.004 \times 10^{-6} \text{ m}^2/\text{s}$$

$$c_{pl} = 4182 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_l = 0.598 \text{ W/m} \cdot ^\circ\text{C}$$

$$\text{Pr} = 7.01$$



Analysis The mass flow rate of water and the rate of heat transfer to the water are

$$\dot{m}_{\text{water}} = \rho V A_c = (998 \text{ kg/m}^3)(2 \text{ m/s})[\pi(0.03 \text{ m})^2 / 4] = 1.411 \text{ kg/s}$$

$$\dot{Q} = \dot{m} c_p (T_{\text{out}} - T_{\text{in}}) = (1.411 \text{ kg/s})(4182 \text{ J/kg} \cdot ^\circ\text{C})(30 - 10)^\circ\text{C} = \mathbf{118,000 \text{ W}}$$

The modified latent heat of vaporization is

$$h_{fg}^* = h_{fg} + 0.68 c_{pl} (T_{\text{sat}} - T_s) = 2407 \times 10^3 \text{ J/kg} + 0.68 \times 4182 \text{ J/kg} \cdot ^\circ\text{C} (40 - 20)^\circ\text{C} = 2464 \times 10^3 \text{ J/kg}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$h_o = h_{\text{horizontal}} = 0.729 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4}$$

$$= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(996 \text{ kg/m}^3)(996 - 0.05 \text{ kg/m}^3)(2464 \times 10^3 \text{ J/kg})(0.615 \text{ W/m} \cdot ^\circ\text{C})^3}{(0.798 \times 10^{-3} \text{ kg/m} \cdot \text{s})(40 - 20)^\circ\text{C}(0.03 \text{ m})} \right]^{1/4} = 7572 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The average heat transfer coefficient for flow inside the tube is determined as follows:

$$\text{Re} = \frac{V_{\text{avg}} D}{\nu} = \frac{(2 \text{ m/s})(0.03 \text{ m})}{1.004 \times 10^{-6}} = 59,761$$

$$\text{Nu} = 0.023 \text{Re}^{0.8} \text{Pr}^{0.4} = 0.023(59,761)^{0.8} (7.01)^{0.4} = 332.0$$

$$h_i = \frac{k \text{Nu}}{D} = \frac{(0.598 \text{ W/m} \cdot ^\circ\text{C}) \times 332.0}{0.03 \text{ m}} = 6618 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Noting that the thermal resistance of the tube is negligible, the overall heat transfer coefficient becomes

$$U = \frac{1}{1/h_i + 1/h_o} = \frac{1}{1/6618 + 1/7572} = \mathbf{3531 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

The logarithmic mean temperature difference is:

$$\Delta T_{\text{lm}} = \frac{\Delta T_i - \Delta T_e}{\ln(\Delta T_i / \Delta T_e)} = \frac{30 - 10}{\ln(30/10)} = 18.20^\circ\text{C}$$

The tube length is determined from

$$\dot{Q} = U A_s \Delta T_{\text{lm}} \rightarrow L = \frac{\dot{Q}}{U(\pi D) \Delta T_{\text{lm}}} = \frac{118,000 \text{ W}}{(3531 \text{ W/m}^2 \cdot ^\circ\text{C}) \pi (0.03 \text{ m}) (18.20^\circ\text{C})} = \mathbf{19.5 \text{ m}}$$

Note that the flow is turbulent, and thus the entry length in this case is $10D = 0.3 \text{ m}$ is much shorter than the total tube length. This verifies our assumption of fully developed flow.

10-120 Saturated steam condenses on a suspended silver sphere which is initially at 25°C. The time needed for the temperature of the sphere to rise to 50°C and the amount of steam condenses are to be determined.

Assumptions 1 The temperature of the sphere changes uniformly and thus the lumped system analysis is applicable. **2** The average condensation heat transfer coefficient evaluated for the average temperature can be used for the entire process. **3** Constant properties at room temperature can be used for the silver ball.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of the silver ball at room temperature and the properties of liquid water at the average film temperature of $T_f = (T_{\text{sat}} + T_{s,\text{avg}})/2 = (100 + 37.5)/2 = 69^\circ\text{C} \approx 70^\circ\text{C}$ are (Tables A-3 and A-9),

Silver Ball:

$$\rho = 10,500 \text{ kg/m}^3$$

$$\alpha = 174 \times 10^{-6} \text{ m}^2/\text{s}$$

$$c_p = 235 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_l = 429 \text{ W/m} \cdot ^\circ\text{C}$$

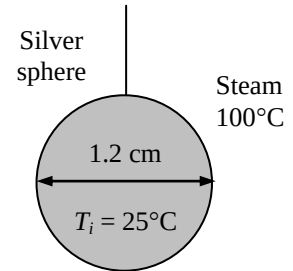
Liquid Water:

$$\rho_l = 977.5 \text{ kg/m}^3$$

$$\mu_l = 0.404 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

$$c_{pl} = 4190 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_l = 0.663 \text{ W/m} \cdot ^\circ\text{C}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4190 \text{ J/kg} \cdot ^\circ\text{C}(100 - 37.5)^\circ\text{C} = 2435 \times 10^3 \text{ J/kg} \end{aligned}$$

Noting that the tube is horizontal, the condensation heat transfer coefficient is determined from

$$\begin{aligned} h &= h_{\text{sph}} = 0.815 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.815 \left[\frac{(9.81 \text{ m/s}^2)(977.5 \text{ kg/m}^3)(977.5 - 0.60 \text{ kg/m}^3)(2435 \times 10^3 \text{ J/kg})(0.663 \text{ W/m} \cdot ^\circ\text{C})^3}{(0.404 \times 10^{-3} \text{ kg/m} \cdot \text{s})(100 - 37.5)^\circ\text{C}(0.012 \text{ m})} \right]^{1/4} \\ &= 9916 \text{ W/m}^2 \cdot ^\circ\text{C} \end{aligned}$$

The characteristic length and the Biot number for the lumped system analysis is (see Chap. 4)

$$\begin{aligned} L_c &= \frac{V}{A} = \frac{\pi D^3/6}{\pi D^2} = \frac{D}{6} = \frac{0.012 \text{ m}}{6} = 0.002 \text{ m} \\ Bi &= \frac{hL_c}{k} = \frac{(9916 \text{ W/m}^2 \cdot ^\circ\text{C})(0.002 \text{ m})}{(429 \text{ W/m} \cdot ^\circ\text{C})} = 0.0462 < 0.1 \end{aligned}$$

The lumped system analysis is applicable since $Bi < 0.1$. Then the time needed for the temperature of the sphere to rise from 25 to 50°C is determined to be

$$\begin{aligned} b &= \frac{hA_s}{\rho c_p V} = \frac{h}{\rho c_p L_c} = \frac{9916 \text{ W/m}^2 \cdot ^\circ\text{C}}{(10,500 \text{ kg/m}^3)(235 \text{ J/kg} \cdot ^\circ\text{C})(0.002 \text{ m})} = 2.009 \text{ s}^{-1} \\ \frac{T(t) - T_\infty}{T_i - T_\infty} &= e^{-bt} \longrightarrow \frac{50 - 100}{25 - 100} = e^{-2.009t} \longrightarrow t = \mathbf{0.202 \text{ s}} \end{aligned}$$

The total heat transfer to the ball and the amount of steam that condenses become

$$\begin{aligned} m_{\text{sphere}} &= \rho V = \rho \frac{\pi D^3}{6} = (10,500 \text{ kg/m}^3) \frac{\pi (0.012 \text{ m})^3}{6} = 0.009500 \text{ kg} \\ Q &= mc_p [T(t) - T_i]_{\text{sphere}} = (0.009500 \text{ kg})(235 \text{ J/kg} \cdot ^\circ\text{C})(50 - 25)^\circ\text{C} = 55.81 \text{ J} \\ \dot{m}_{\text{condensation}} &= \frac{\dot{Q}}{h_{fg}^*} = \frac{55.81 \text{ J/s}}{2435 \times 10^3 \text{ J/kg}} = \mathbf{2.29 \times 10^{-5} \text{ kg/s}} \end{aligned}$$

10-121 Steam at a saturation temperature of $T_{\text{sat}} = 100^\circ\text{C}$ condenses on a suspended silver sphere which is initially at 25°C . The time needed for the temperature of the sphere to rise to 50°C and the amount of steam condenses during this process are to be determined.

Assumptions 1 The temperature of the sphere changes uniformly and thus the lumped system analysis is applicable. 2 The average condensation heat transfer coefficient evaluated for the average temperature can be used for the entire process. 3 Constant properties at room temperature can be used for the silver ball.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of the silver ball at room temperature and the properties of liquid water at the average film temperature of $T_f = (T_{\text{sat}} + T_{s,\text{avg}})/2 = (100 + 37.5)/2 = 69^\circ\text{C} \approx 70^\circ\text{C}$ are (Tables A-3 and A-9),

Copper Ball:

Liquid Water :

$$\rho = 8933 \text{ kg/m}^3$$

$$\rho_l = 977.5 \text{ kg/m}^3$$

$$\alpha = 117 \times 10^{-6} \text{ m}^2/\text{s}$$

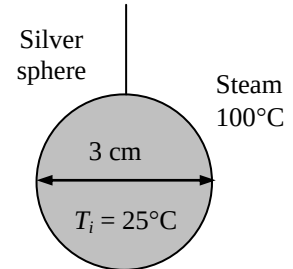
$$\mu_l = 0.404 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

$$c_p = 385 \text{ J/kg} \cdot ^\circ\text{C}$$

$$c_{pl} = 4190 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_l = 401 \text{ W/m} \cdot ^\circ\text{C}$$

$$k_l = 0.663 \text{ W/m} \cdot ^\circ\text{C}$$



Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times 4190 \text{ J/kg} \cdot ^\circ\text{C}(100 - 37.5)^\circ\text{C} = 2435 \times 10^3 \text{ J/kg} \end{aligned}$$

Noting that the tube is horizontal, the condensation heat transfer coefficient is determined from

$$\begin{aligned} h &= h_{\text{sph}} = 0.815 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.815 \left[\frac{(9.81 \text{ m/s}^2)(977.5 \text{ kg/m}^3)(977.5 - 0.60 \text{ kg/m}^3)(2435 \times 10^3 \text{ J/kg})(0.663 \text{ W/m} \cdot ^\circ\text{C})^3}{(0.404 \times 10^{-3} \text{ kg/m} \cdot \text{s})(100 - 37.5)^\circ\text{C}(0.03 \text{ m})} \right]^{1/4} \\ &= 7886 \text{ W/m}^2 \cdot ^\circ\text{C} \end{aligned}$$

The characteristic length and the Biot number for the lumped system analysis is (see Chap. 4)

$$\begin{aligned} L_c &= \frac{V}{A_s} = \frac{\pi D^3/6}{\pi D^2} = \frac{D}{6} = \frac{0.03 \text{ m}}{6} = 0.005 \text{ m} \\ Bi &= \frac{hL_c}{k} = \frac{(7886 \text{ W/m}^2 \cdot ^\circ\text{C})(0.005 \text{ m})}{(401 \text{ W/m} \cdot ^\circ\text{C})} = 0.098 < 0.1 \end{aligned}$$

The lumped system analysis is applicable since $Bi < 0.1$. Then the time needed for the temperature of the sphere to rise from 25 to 50°C is determined to be

$$\begin{aligned} b &= \frac{hA_s}{\rho c_p V} = \frac{h}{\rho c_p L_c} = \frac{7886 \text{ W/m}^2 \cdot ^\circ\text{C}}{(8933 \text{ kg/m}^3)(385 \text{ J/kg} \cdot ^\circ\text{C})(0.005 \text{ m})} = 0.4586 \text{ s}^{-1} \\ \frac{T(t) - T_\infty}{T_i - T_\infty} &= e^{-bt} \longrightarrow \frac{50 - 100}{25 - 100} = e^{-0.4586t} \longrightarrow t = \mathbf{0.884 \text{ s}} \end{aligned}$$

The total heat transfer to the ball and the amount of steam that condenses become

$$\begin{aligned} m_{\text{sphere}} &= \rho V = \rho \frac{\pi D^3}{6} = (8933 \text{ kg/m}^3) \frac{\pi (0.03 \text{ m})^3}{6} = 0.1263 \text{ kg} \\ Q &= mc_p [T(t) - T_i]_{\text{sphere}} = (0.1263 \text{ kg})(385 \text{ J/kg} \cdot ^\circ\text{C})(50 - 25)^\circ\text{C} = 1216 \text{ J} \\ \dot{m}_{\text{condensation}} &= \frac{\dot{Q}}{h_{fg}^*} = \frac{1216 \text{ J/s}}{2435 \times 10^3 \text{ J/kg}} = \mathbf{4.99 \times 10^{-4} \text{ kg/s}} \end{aligned}$$

10-122 There is film condensation on the outer surfaces of 8 horizontal tubes arranged in a horizontal or vertical tier. The ratio of the condensation rate for the cases of the tubes being arranged in a horizontal tier versus in a vertical tier is to be determined.

Assumptions Steady operating conditions exist.

Analysis The heat transfer coefficients for the two cases are related to the heat transfer coefficient on a single horizontal tube by

Horizontal tier:

$$h_{\text{horizontal tier of } N \text{ tubes}} = h_{\text{horizontal, 1 tube}}$$

Vertical tier:

$$h_{\text{vertical tier of } N \text{ tubes}} = \frac{h_{\text{horizontal, 1 tube}}}{N^{1/4}}$$

Therefore,

$$\begin{aligned} \text{Ratio} &= \frac{\dot{m}_{\text{horizontal tier of } N \text{ tubes}}}{\dot{m}_{\text{vertical tier of } N \text{ tubes}}} \\ &= \frac{h_{\text{horizontal tier of } N \text{ tubes}}}{h_{\text{vertical tier of } N \text{ tubes}}} \\ &= \frac{h_{\text{horizontal, 1 tube}}}{h_{\text{horizontal, 1 tube}} / N^{1/4}} \\ &= N^{1/4} \\ &= 8^{1/4} = \mathbf{1.68} \end{aligned}$$



Horizontal tier



Vertical tier

10-123 Ammonia is liquefied in a horizontal condenser at 37°C by a coolant at 20°C. The average value of overall heat transfer coefficient and the tube length are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tubes are isothermal. 3 The thermal resistance of the tube walls is negligible.

Properties The properties of ammonia at the saturation temperature of 310 K (37°C) are $h_{fg} = 1113 \times 10^3$ J/kg and $\rho_v = 11.09$ kg/m³ (Table A-11). We assume a tube outer surface temperature of 31°C. The properties of liquid ammonia at the film temperature of $T_f = (T_{sat} + T_s)/2 = (37 + 31)/2 = 34^\circ\text{C}$ are (Table A-11)

$$\rho_l = 589.0 \text{ kg/m}^3$$

$$\mu_l = 1.303 \times 10^{-4} \text{ kg/m} \cdot \text{s}$$

$$c_{pl} = 4867 \text{ J/kg} \cdot ^\circ\text{C}$$

$$k_l = 0.4602 \text{ W/m} \cdot ^\circ\text{C}$$

The thermal conductivity of copper is 401 W/m·°C (Table A-3).

Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{sat} - T_s) \\ &= 1113 \times 10^3 \text{ J/kg} + 0.68 \times 4867 \text{ J/kg} \cdot ^\circ\text{C}(37 - 31)^\circ\text{C} = 1133 \times 10^3 \text{ J/kg} \end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned} h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{sat} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(589.0 \text{ kg/m}^3)(589.0 - 11.09 \text{ kg/m}^3)(1133 \times 10^3 \text{ J/kg})(0.4602 \text{ W/m} \cdot ^\circ\text{C})^3}{(1.303 \times 10^{-4} \text{ kg/m} \cdot \text{s})(37 - 31)^\circ\text{C}(0.038 \text{ m})} \right]^{1/4} \\ &= 7693 \text{ W/m}^2 \cdot ^\circ\text{C} \end{aligned}$$

Noting that there are two 2-pipe high, two 3-pipe high, and one 4-pipe high vertical tiers in the tube-layout, the average heat transfer coefficient is to be determined as follows

$$\begin{aligned} h_1 &= \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{2^{1/4}} (7693 \text{ W/m}^2 \cdot ^\circ\text{C}) = 6469 \text{ W/m}^2 \cdot ^\circ\text{C} \\ h_2 &= \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{3^{1/4}} (7693 \text{ W/m}^2 \cdot ^\circ\text{C}) = 5845 \text{ W/m}^2 \cdot ^\circ\text{C} \\ h_3 &= \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{4^{1/4}} (7693 \text{ W/m}^2 \cdot ^\circ\text{C}) = 5440 \text{ W/m}^2 \cdot ^\circ\text{C} \\ h_o &= \frac{2 \times 2h_1 + 2 \times 3h_2 + 1 \times 4h_3}{2 \times 2 + 2 \times 3 + 1 \times 4} = \frac{4 \times 6469 + 6 \times 5845 + 4 \times 5440}{14} = 5908 \text{ W/m}^2 \cdot ^\circ\text{C} \end{aligned}$$

Let us check if the assumed value for the tube temperature was reasonable

$$h_i A_i \Delta T_i = h_o A_o \Delta T_o$$

$$(4000)\pi(0.030)L(T_{\text{tube}} - 20) = (5908)\pi(0.038)L(37 - T_{\text{tube}}) \longrightarrow T_{\text{tube}} = 31.1^\circ\text{C}$$

which is very close to the assumed value of 31°C. Therefore, the assumption was good. The overall heat transfer coefficient based on the outer surface is determined from

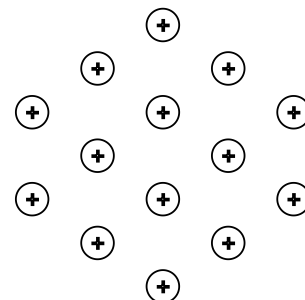
$$U_o = \left(\frac{D_o}{D_i h_i} + \frac{D_o \ln(D_o / D_i)}{2k} + \frac{1}{h_o} \right)^{-1} = \left(\frac{0.038}{0.030 \times 4000} + \frac{0.038 \ln(3.8 / 3.0)}{2(401)} + \frac{1}{5908} \right)^{-1} = \mathbf{2012 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

(b) The rate of heat transfer is

$$\dot{Q} = \dot{m}_{\text{condensation}} h_{fg}^* = (900 / 3600 \text{ kg/s})(1133 \times 10^3 \text{ J/kg}) = 2.833 \times 10^5 \text{ W}$$

Then the tube length may be determined from

$$\begin{aligned} \dot{Q} &= U_o A_o \Delta T \\ 2.833 \times 10^5 \text{ W} &= (2012 \text{ W/m}^2 \cdot ^\circ\text{C})(14)\pi(0.038 \text{ m})L(37 - 20) \longrightarrow L = \mathbf{4.96 \text{ m}} \end{aligned}$$



10-124 Saturated ammonia vapor at a saturation temperature of $T_{\text{sat}} = 25^\circ\text{C}$ condenses on the outer surfaces of a tube bank in which cooling water flows. The rate of condensation of ammonia, the overall heat transfer coefficient, and the tube length are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tubes are isothermal. 3 The thermal resistance of the tube walls is negligible.

Properties The properties of ammonia at the saturation temperature of 25°C are $h_{fg} = 1166 \times 10^3 \text{ J/kg}$ and $\rho_v = 7.809 \text{ kg/m}^3$ (Table A-11). We assume that the tube temperature is 20°C . Then, the properties of liquid ammonia at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (25 + 20) / 2 = 22.5^\circ\text{C}$ are (Table A-11)

$$\begin{aligned}\rho_l &= 606.5 \text{ kg/m}^3 \\ \mu_l &= 1.479 \times 10^{-4} \text{ kg/m} \cdot \text{s} \\ c_{pl} &= 4765 \text{ J/kg} \cdot ^\circ\text{C} \\ k_l &= 0.4869 \text{ W/m} \cdot ^\circ\text{C}\end{aligned}$$

The water properties at the average temperature of $(14+17)/2 = 15.5^\circ\text{C}$ are (Table A-9)

$$\begin{aligned}\rho &= 999.0 \text{ kg/m}^3 \\ c_p &= 4185 \text{ J/kg} \cdot ^\circ\text{C} \\ \mu &= 1.124 \times 10^{-3} \text{ kg/m} \cdot \text{s} \\ k &= 0.590 \text{ W/m} \cdot ^\circ\text{C} \\ \text{Pr} &= 7.98\end{aligned}$$

Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \\ &= 1166 \times 10^3 \text{ J/kg} + 0.68 \times 4765 \text{ J/kg} \cdot ^\circ\text{C}(25 - 20)^\circ\text{C} \\ &= 1182 \times 10^3 \text{ J/kg}\end{aligned}$$

The heat transfer coefficient for condensation on a single horizontal tube is

$$\begin{aligned}h &= h_{\text{horizontal}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(606.5 \text{ kg/m}^3)(606.5 - 7.809 \text{ kg/m}^3)(1182 \times 10^3 \text{ J/kg})(0.4869 \text{ W/m} \cdot ^\circ\text{C})^3}{(1.479 \times 10^{-4} \text{ kg/m} \cdot \text{s})(25 - 20)^\circ\text{C}(0.025 \text{ m})} \right]^{1/4} \\ &= 9280 \text{ W/m}^2 \cdot ^\circ\text{C}\end{aligned}$$

Then the average heat transfer coefficient for a 4-pipe high vertical tier becomes

$$h_o = h_{\text{horiz, N tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{4^{1/4}} (9280 \text{ W/m}^2 \cdot ^\circ\text{C}) = 6562 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The rate of heat transfer in the condenser is

$$\begin{aligned}\dot{m} &= 16\rho A_c \mathbf{V} = 16(999 \text{ kg/m}^3)\pi(0.25)(0.025 \text{ m})^2(2 \text{ m/s}) = 15.69 \text{ kg/s} \\ \dot{Q} &= \dot{m}c_p(T_{\text{out}} - T_{\text{in}}) = (15.69 \text{ kg/s})(4185 \text{ J/kg} \cdot ^\circ\text{C})(17 - 14) = 1.970 \times 10^5 \text{ W}\end{aligned}$$

Then the rate of condensation becomes

$$\dot{m}_{\text{cond}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{1.970 \times 10^5 \text{ W}}{1182 \times 10^3 \text{ J/kg}} = \mathbf{0.167 \text{ kg/s}}$$

(b) For the calculation of the heat transfer coefficient on the inner surfaces of the tubes, we first determine the Reynolds number

$$\text{Re} = \frac{VD\rho}{\mu} = \frac{(2 \text{ m/s})(0.025 \text{ m})(999.0 \text{ kg/m}^3)}{1.124 \times 10^{-3} \text{ kg/m} \cdot \text{s}} = 44,440$$

which is greater than 10,000. Therefore, the flow is turbulent. Assuming fully developed flow, the Nusselt number and the heat transfer coefficient are determined to be

$$Nu = 0.023 Re^{0.8} Pr^{0.4} = 0.023(44,440)^{0.8} (7.98)^{0.4} = 275.9$$

$$h_i = \frac{k}{D} Nu = \frac{(0.590 \text{ W/m} \cdot ^\circ\text{C})}{0.025 \text{ m}} (275.9) = 6511 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Let us check if the assumed value for the tube temperature was reasonable

$$\begin{aligned} h_i \Delta T_i &= h_o \Delta T_o \\ (6511)(T_{\text{tube}} - 15.5) &= (6562)(25 - T_{\text{tube}}) \\ T_{\text{tube}} &= 20.3^\circ\text{C} \end{aligned}$$

which is sufficiently close to the assumed value of 20°C . Disregarding thermal resistance of the tube walls, the overall heat transfer coefficient is determined from

$$U = \left(\frac{1}{h_i} + \frac{1}{h_o} \right)^{-1} = \left(\frac{1}{6511} + \frac{1}{6562} \right)^{-1} = \mathbf{3268 \text{ W/m}^2 \cdot ^\circ\text{C}}$$

(c) The tube length may be determined from

$$\begin{aligned} \dot{Q} &= UA\Delta T \\ 1.970 \times 10^5 \text{ W} &= (3268 \text{ W/m}^2 \cdot ^\circ\text{C})(16)\pi(0.025 \text{ m})L \left[25 - \frac{1}{2}(14 + 17) \right] \\ L &= \mathbf{5.05 \text{ m}} \end{aligned}$$

10-125 Saturated steam at 270.1 kPa pressure and thus at a saturation temperature of $T_{\text{sat}} = 130^\circ\text{C}$ (Table A-9) condenses inside a horizontal tube which is maintained at 110°C . The average heat transfer coefficient and the rate of condensation of steam are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal. 3 The vapor velocity is low so that $\text{Re}_{\text{vapor}} < 35,000$.

Properties The properties of water at the saturation temperature of 130°C are $h_{fg} = 2174 \times 10^3 \text{ J/kg}$ and $\rho_v = 1.50 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s) / 2 = (130 + 110) / 2 = 120^\circ\text{C}$ are (Table A-9),

$$\begin{aligned}\rho_l &= 943.4 \text{ kg/m}^3 \\ \mu_l &= 0.232 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \nu_l &= \mu_l / \rho_l = 0.246 \times 10^{-6} \text{ m}^2/\text{s} \\ c_{pl} &= 4244 \text{ J/kg}\cdot^\circ\text{C} \\ k_l &= 0.683 \text{ W/m}\cdot^\circ\text{C}\end{aligned}$$

Analysis The condensation heat transfer coefficient is determined from

$$\begin{aligned}h &= h_{\text{internal}} = 0.555 \left[\frac{g \rho_l (\rho_l - \rho_v) k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \left(h_{fg} + \frac{3}{8} c_{pl} (T_{\text{sat}} - T_s) \right) \right]^{1/4} \\ &= 0.555 \left[\frac{(9.81 \text{ m/s}^2)(943.4 \text{ kg/m}^3)(943.4 - 1.50) \text{ kg/m}^3 (0.683 \text{ W/m}\cdot^\circ\text{C})^3}{(0.232 \times 10^{-3} \text{ kg/m}\cdot\text{s})(130 - 110)^\circ\text{C}(0.025 \text{ m})} \right. \\ &\quad \left. \times \left(2174 \times 10^3 \text{ J/kg} + \frac{3}{8} (4244 \text{ J/kg}\cdot^\circ\text{C})(130 - 110)^\circ\text{C} \right) \right]^{1/4} \\ &= \mathbf{8413 \text{ W/m}^2\cdot^\circ\text{C}}\end{aligned}$$

The heat transfer surface area of the pipe is

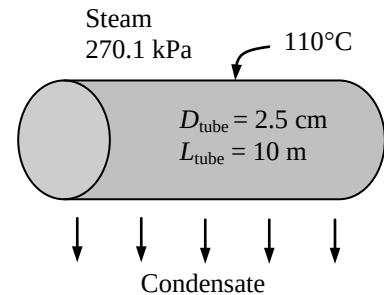
$$A_s = \pi DL = \pi(0.025 \text{ m})(10 \text{ m}) = 0.7854 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (8413 \text{ W/m}^2\cdot^\circ\text{C})(0.7854 \text{ m}^2)(130 - 110)^\circ\text{C} = 132,151 \text{ W}$$

The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}} = \frac{132,151 \text{ J/s}}{2174 \times 10^3 \text{ J/kg}} = \mathbf{0.0608 \text{ kg/s}}$$



Fundamentals of Engineering (FE) Exam Problems

10-126 When boiling a saturated liquid, one must be careful while increasing the heat flux to avoid “burnout.” Burnout occurs when the boiling transitions from _____ boiling.

- (a) convection to nucleate (b) convection to film (c) film to nucleate
(d) nucleate to film (e) none of them

Answer (d) nucleate to film

10-127 Heat transfer coefficients for a vapor condensing on a surface can be increased by promoting

- (a) film condensation (b) dropwise condensation (c) rolling action (d) none of them

Answer (b) dropwise condensation

10-128 At a distance x down a vertical, isothermal flat plate on which a saturated vapor is condensing in a continuous film, the thickness of the liquid condensate layer is δ . The heat transfer coefficient at this location on the plate is given by

- (a) k_l / δ (b) δh_f (c) δh_{fg} (d) δh_g (e) none of them

Answer (a) k_l / δ

10-129 When a saturated vapor condenses on a vertical, isothermal flat plate in a continuous film, the rate of heat transfer is proportional to

- (a) $(T_s - T_{\text{sat}})^{1/4}$ (b) $(T_s - T_{\text{sat}})^{1/2}$ (c) $(T_s - T_{\text{sat}})^{3/4}$ (d) $(T_s - T_{\text{sat}})$ (e) $(T_s - T_{\text{sat}})^{2/3}$

Answer (c) $(T_s - T_{\text{sat}})^{3/4}$

10-130 Saturated water vapor is condensing on a 0.5 m^2 vertical flat plate in a continuous film with an average heat transfer coefficient of $5 \text{ kW/m}^2\cdot\text{K}$. The temperature of the water is 80°C ($h_{fg} = 2309 \text{ kJ/kg}$) and the temperature of the plate is 60°C . The rate at which condensate is being formed is

- (a) 0.022 kg/s (b) 0.048 kg/s (c) 0.077 kg/s (d) 0.16 kg/s (e) 0.32 kg/s

Answer (a) 0.022 kg/s

Solution Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
hfg=2309 [kJ/kg]
dT=20 [C]
A=0.5 [m^2]
h=5 [kJ/m^2-K-s]
mdot=h*A*dT/hfg
```

10-131 Steam condenses at 50°C on a 1.2-m -high and 2.4-m -wide vertical plate that is maintained at 30°C . The condensation heat transfer coefficient is

- (a) $4260 \text{ W/m}^2\cdot\text{K}$ (b) $4780 \text{ W/m}^2\cdot\text{K}$ (c) $5510 \text{ W/m}^2\cdot\text{K}$ (d) $6260 \text{ W/m}^2\cdot\text{K}$ (e) $6940 \text{ W/m}^2\cdot\text{K}$

(For water, use $\rho_l = 992.1 \text{ kg/m}^3$, $\mu_l = 0.653 \times 10^{-3} \text{ kg/m}\cdot\text{s}$, $k_l = 0.631 \text{ W/m}\cdot^\circ\text{C}$, $c_{pl} = 4179 \text{ J/kg}\cdot^\circ\text{C}$, $h_{fg @ T_{sat}} = 2383 \text{ kJ/kg}$)

Answer (b) $4780 \text{ W/m}^2\cdot\text{K}$

Solution Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
T_sat=50 [C]
T_s=30 [C]
L=1.2 [m]
w=2.4 [m]
```

```
h_fg=2383E3 [J/kg] "at 50 C from Table A-9"
"The properties of water at (50+30)/2=40 C are (Table A-9)"
rho_l=992.1 [kg/m^3]
mu_l=0.653E-3 [kg/m-s]
nu_l=mu_l/rho_l
c_p_l=4179 [J/kg-C]
k_l=0.631 [W/m-C]
g=9.81 [m/s^2]
```

```
h_fg_star=h_fg+0.68*c_p_l*(T_sat-T_s)
Re=(4.81+(3.70*L*k_l*(T_sat-T_s))/(mu_l*h_fg_star)*(g/nu_l^2)^(1/3))^0.820
"Re is between 30 and 1800, and therefore there is wavy laminar flow"
h=(Re*k_l)/(1.08*Re^1.22-5.2)*(g/nu_l^2)^(1/3)
```

10-132 An air conditioner condenser in an automobile consists of 2 m^2 of tubular heat exchange area whose surface temperature is 30°C . Saturated refrigerant 134a vapor at 50°C ($h_{fg} = 152 \text{ kJ/kg}$) condenses on these tubes. What heat transfer coefficient must exist between the tube surface and condensing vapor to produce 1.5 kg/min of condensate?

- (a) $95 \text{ W/m}^2\cdot\text{K}$ (b) $640 \text{ W/m}^2\cdot\text{K}$ (c) $727 \text{ W/m}^2\cdot\text{K}$ (d) $799 \text{ W/m}^2\cdot\text{K}$ (e) $960 \text{ W/m}^2\cdot\text{K}$

Answer (a) $95 \text{ W/m}^2\cdot\text{K}$

Solution Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
hfg=152000 [J/kg]
dT=20 [C]
A=2 [m^2]
mdot=(1.5/60) [kg/s]
Q=mdot*hfg
Q=h*A*dT
```

10-133 Saturated water vapor at 40°C is to be condensed as it flows through a tube at a rate of 0.2 kg/s . The condensate leaves the tube as a saturated liquid at 40°C . The rate of heat transfer from the tube is

- (a) 34 kJ/s (b) 268 kJ/s (c) 453 kJ/s (d) 481 kJ/s (e) 515 kJ/s

Answer (d) 481 kJ/s

Solution Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
T1=40 [C]
m_dot=0.2 [kg/s]
h_f=ENTHALPY(Steam_IAPWS,T=T1,x=0)
h_g=ENTHALPY(Steam_IAPWS,T=T1,x=1)
h_fg=h_g-h_f
Q_dot=m_dot*h_fg
```

"Wrong Solutions:"

```
W1_Q=m_dot*h_f "Using hf"
W2_Q=m_dot*h_g "Using hg"
W3_Q=h_fg "not using mass flow rate"
W4_Q=m_dot*(h_f+h_g) "Adding hf and hg"
```

10-134 Steam condenses at 50°C on the outer surface of a horizontal tube with an outer diameter of 6 cm. The outer surface of the tube is maintained at 30°C. The condensation heat transfer coefficient is

- (a) 5493 W/m²·°C (b) 5921 W/m²·°C (c) 6796 W/m²·°C (d) 7040 W/m²·°C (e) 7350 W/m²·°C

(For water, use $\rho_l = 992.1 \text{ kg/m}^3$, $\mu_l = 0.653 \times 10^{-3} \text{ kg/m}\cdot\text{s}$, $k_l = 0.631 \text{ W/m}\cdot\text{°C}$, $c_{pl} = 4179 \text{ J/kg}\cdot\text{°C}$, $h_{fg @ T_{sat}} = 2383 \text{ kJ/kg}$)

Answer (c) 6796 W/m²·°C

Solution Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
T_sat=50 [C]
T_s=30 [C]
D=0.06 [m]
h_fg=2383E3 [J/kg] "at 50 C from Table A-9"
rho_v=0.0831 [kg/m^3]
"The properties of water at (50+30)/2=40 C are (Table A-9)"
rho_l=992.1 [kg/m^3]
mu_l=0.653E-3 [kg/m-s]
c_pl=4179 [J/kg-C]
k_l=0.631 [W/m-C]
g=9.81 [m/s^2]
h_fg_star=h_fg+0.68*c_pl*(T_sat-T_s)
h=0.729*((g*rho_l*(rho_l-rho_v)*h_fg_star*k_l^3)/(mu_l*(T_sat-T_s)*D))^0.25
```

10-135 Steam condenses at 50°C on a tube bank consisting of 20 tubes arranged in a rectangular array of 4 tubes high and 5 tubes wide. Each tube has a diameter of 3 cm and a length of 5 m and the outer surfaces of the tubes are maintained at 30°C. The rate of condensation of steam is

- (a) 0.12 kg/s (b) 0.28 kg/s (c) 0.31 kg/s (d) 0.45 kg/s (e) 0.62 kg/s

(For water, use $\rho_l = 992.1 \text{ kg/m}^3$, $\mu_l = 0.653 \times 10^{-3} \text{ kg/m}\cdot\text{s}$, $k_l = 0.631 \text{ W/m}\cdot^\circ\text{C}$, $c_{pl} = 4179 \text{ J/kg}\cdot^\circ\text{C}$, $h_{fg @ T_{sat}} = 2383 \text{ kJ/kg}$)

Answer (c) 0.31 kg/s

Solution Solved by EES Software. Solutions can be verified by copying-and-pasting the following lines on a blank EES screen.

```
T_sat=50 [C]
T_s=30 [C]
D=0.03 [m]
L=5 [m]
N=4
N_total=5*N

h_fg=2383E3 [J/kg] "at 50 C from Table A-9"
rho_v=0.0831 [kg/m^3] "at 50 C from Table A-9"
"The properties of water at (50+30)/2=40 C are (Table A-9)"
rho_l=992.1 [kg/m^3]
mu_l=0.653E-3 [kg/m-s]
c_pl=4179 [J/kg-C]
k_l=0.631 [W/m-C]
g=9.81 [m/s^2]

h_fg_star=h_fg+0.68*c_pl*(T_sat-T_s)
h_1tube=0.729*((g*rho_l*(rho_l-rho_v)*h_fg_star*k_l^3)/(mu_l*(T_sat-T_s)*D*N))^0.25
h_Ntubes=1/N^0.25*h_1tube
A_s=N_total*pi*D*L
Q_dot=h_Ntubes*A_s*(T_sat-T_s)
m_dot_cond=Q_dot/h_fg_star
```

10-136 ... 10-141 Design and Essay Problems

